II

VOLUME TWO OF THREE

A Candidate's Revision Handbook

The examination, section by section

Curriculum-mapped core revision notes
Thirty sections against the sixty-five curriculum items
The numbers, classifications and lists in one place
Diagnostic-case and case-presentation frameworks
The oral examination: set-pieces and technique
A dated countdown to 5 November

Written throughout to the standard of a general dental practitioner with a particular interest in orthodontics, and mapped to the five assessed components of the Diploma.

For candidates sitting the examination of 5 November 2026

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The set

Three volumes, in the order you should use them. Each is cross-referenced to the other two, so you can enter from whichever you happen to have open.

I	**Notes In Orthodontics** Revised edition · 275 pp	The book that teaches the material. A crash notebook of forty-two schematics and four number cards; the twenty essay topics with their skeletons; the sixteen chapters; model viva answers; and four new units on ethics and consent, radiation protection, infection control, and governance. **Reach for it** when you are learning something for the first time, planning an essay, or want the detail behind a fact.
II	**A Candidate's Revision Handbook** 161 pp	The book that maps the material onto the examination. Thirty sections against the sixty-five curriculum items, the numbers on six cards, the diagnostic cases, the case presentation, the oral, the literature by theme, and the dated countdown. **Reach for it** when you are revising against the syllabus, preparing the components that are not written, or covering what Volume I does not.
III	**The Question Bank** 300 pp	The book that tells you whether it has stuck. 1,016 single best answers in thirty blocks with explanations and references, a mark sheet, 207 viva questions with model answers, and a viva tracker. **Reach for it** at the end of every study session, and let the score decide what you re-read.

LEARN, REVISE, TEST

Volume I teaches the material. Volume II checks it against the sixty-five curriculum items and covers the four assessed components that are not knowledge — the diagnostic cases, the case presentation, the oral and examination technique. Volume III tells you whether any of it has stuck. Work in that order, and let a poor score in Volume III send you back to the section Volume II names.

The plan

Seven working weeks and a weekend, ending **Thursday 5 November 2026**. The written paper is the engine — revising for it is simultaneously revising for the oral and the diagnostic cases, because all three test the same knowledge. The case submissions depend on other people, so their *requests* go out first.

Before Monday — one session, about three hours

1. Pick your three write-up cases and your two presentation cases. For each, list what is missing: pre-treatment records, radiographs, photographs, models or scans, progress notes, debond records, PAR scoring. **2.** Send every request the same afternoon — patient recalls, radiograph retrieval, trainer sign-off. **3.** Ask the Programme Management Office when the three written-up cases are due. If that date falls before 5 November, it outranks everything below. **4.** Sit Question Bank Blocks **H** and **AD** cold. 55 questions, ethics and governance — the likeliest thin spot and the cheapest marks in the paper.

The spine is the curriculum, not the book. Volume II's thirty sections and Volume III's thirty blocks run one-to-one — §1 is Block A, §30 is Block AD — so the weeks below advance through them in order. The Volume I column then names the units whose *main* treatment sits in that week's sections. **Volume I has two separate series and they are not the same thing**: the twenty essay **topics** of Part II and the sixteen **chapters** of Part III. Chapter 1 is Evidence-Based Orthodontics; Topic 1 is the biological basis of tooth movement. Every chapter and every topic appears exactly once below.

Week	Dates	Curriculum	Read in Volume I	Alongside
1	14–20 Sep	§1–§5 · Blocks A–E	Chapters 2–6 · Topic 1	Viva V1–V20 aloud
2	21–27 Sep	§6–§10 · Blocks F–J	Chapters 1, 7 · Topic 2 · Part V.1	V21–V48. First diagnostic drill
3	28 Sep–4 Oct	§11–§15 · Blocks K–O	Chapters 8, 11 · Topics 3–5, 7	V49–V80. Write up case 1
4	5–11 Oct	§16–§20 · Blocks P–T	Chapters 9, 10, 12–14 · Topics 6, 8, 13–15, 20	V81–V111. Write up case 2
5	12–18 Oct	§21–§25 · Blocks U–Y	Topics 11, 12, 16, 17, 19	V112–V151. Write up case 3
6	19–25 Oct	§26–§30 · Blocks Z–AD	Chapters 15, 16 · Topics 9, 10, 18 · Part V.2–V.4	V152–V207
7	26 Oct–1 Nov	Re-sit every block scored under 70%	Only what the mark sheet sends you back to	Mock oral. Present both cases aloud
Final	2–4 Nov	Two or three blocks a session at examination pace	Crash notebook · number cards	Nothing new

Week 5 lists no chapters because §21–§25 are Chapter 13 territory, read in week 4; that week is topic-led. Part V has no chapter or topic of its own — it is new in this edition.

The daily loop — about two hours

One **Volume I** unit (45 min) → the **Volume II** section its teal strap names (20 min) → that **Volume III** block, timed at 60 seconds a question (30 min) → read the explanation for *every* question, including the ones you got right (25 min).

Log the score on the mark sheet. Anything under 70 per cent goes on the week 7 list. Close each session with one viva section, aloud, to the end, before reading the model.

THREE RULES THAT DECIDE THIS

1. **Speak from week one, not week seven.** The oral tests whether you can produce a structured spoken answer under pressure — a separate skill from knowing the material. Question blocks are measurable and satisfying, which is exactly why they expand to fill the time and crowd out the one thing you cannot cram.
2. **Protect the case deadline.** You cannot revise your way out of a missing pre-treatment photograph. A patient recall is a three-to-four week loop you do not control.
3. **Do not re-read Volume I cover to cover.** Use it as a textbook — when a Volume III explanation does not land, follow the strap back to that one unit.

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About this handbook

This is the consolidating volume. It brings together, into a single curriculum-mapped spine, the material previously issued as *Notes In Orthodontics — A Complete Revision Companion* and Supplementary Notes A, B, C, E, F and G. Nothing has been discarded; it has been re-ordered so that every one of the sixty-five curriculum items has a home, and so that the six source voices can be heard on the same page.

The six sources, and what each is for

Volume	Source text	What it contributes here
Companion	Cobourne & DiBiase, *Handbook of Orthodontics*, 3e (2025)	The spine. Twenty essay topics, sixteen chapters, the numbers, the landmark evidence. Where this handbook gives a page reference without qualification, it is to this text.
Supplement A	*Mosby's Orthodontic Review*	Breadth. American norms, prevalence figures, lasers, CBCT dosimetry, periodontal interface, the five ages.
Supplement B	Cobourne et al., *Clinical Cases in Orthodontics*	102 worked cases. The source for diagnostic-case and case-presentation practice.
Supplement C	Proffit et al., *Contemporary Orthodontics*, 6e	The differential voice. Where Proffit's position materially differs, it is flagged.
Supplement E	Bennett & McLaughlin (1993)	Lacebacks, bendbacks, tiebacks, sliding mechanics, light continuous force.
Supplement F	McLaughlin, Bennett & Trevisi (2001)	The MBT system: bracket selection, positioning, archwire sequence, force levels.
Supplement G	Nanda, *Biomechanics and Esthetic Strategies*	Force systems, vertical control, skeletal anchorage, smile aesthetics.

How this volume is organised

Part	What it is	When to use it
Front matter	Routing tables, a full two-way concordance with *Notes In Orthodontics*, and three revision routes	First, before anything else
1 — Core revision notes	Thirty curriculum-mapped sections	Six weeks out, then again in the final fortnight
2 — Numbers and lists	Every quotable figure, classification and list	Nightly from four weeks out
3 — The diagnostic case	A repeatable protocol for the two diagnostic exercises	Four weeks out; drill weekly
4 — Case presentation	The framework for the two treated cases	From the day you choose your cases
5 — The oral examination	Set-pieces, question flows, technique	Three weeks out

6 — The literature	Which book, which paper, what number	Throughout
7 — Examination technique	How the marks are actually awarded	The week before

Three points about level

The Diploma is not a specialist examination. It is designed, in the programme's own words, "to develop the interest and capability of family and primary-care dentists in orthodontics so that clinical practice is based on sound, evidence-based knowledge and fundamental clinical skills." Three consequences follow, and they are the commonest reasons candidates disappoint.

ONE — THE SAFE ANSWER BEATS THE SOPHISTICATED ANSWER

The examiners are looking for a candidate who knows the limits of what a general dentist with an interest in orthodontics should attempt, and who says so unprompted. A candidate who proposes segmental intrusion with miniscrews and a three-piece base arch for a deep bite in a 45-year-old with reduced periodontal support is answering a different examination.

TWO — GROWTH STATUS IS THE FIRST THING YOU SAY

Almost every planning question in orthodontics resolves differently in a grower and a non-grower. Say which you are dealing with in your first sentence and return to it in your last.

THREE — THE EVIDENCE BASE IS WEAK, AND SAYING SO EARNS MARKS

Orthodontics has a thin randomised evidence base and a long tradition of confident assertion. Candidates who can name the study that overturned a tradition — O'Brien on early treatment, Scott and Fleming on self-ligation, Behrents and White on TMD, Benson on interceptive canine extraction — demonstrate scholarship faster than any other single device.

The examination

What is assessed

Component	Format	What it is really testing
Written paper	Multiple choice, 2 hours	Recall and application across the whole 65-item curriculum. Breadth, not depth. Numbers, thresholds, classifications, and the single best management step.
Case presentation	Two treated or almost-completed cases	Whether you can defend decisions you actually made, including the ones that went wrong.
Diagnostic cases	Two diagnostic exercises	Whether you can move from records to a problem list to a defensible plan in a fixed, reproducible order.
Oral examination	Structured viva	Judgement, limits, and the evidence behind what you assert.
Clinical case log	20 supervised cases; 3 written up	Submitted before the examination; the written cases may be discussed.

The written paper — how to read the stem

Two hours of multiple choice across sixty-five curriculum items means roughly one item per minute and almost no time to reconsider. Three habits are worth building now.

- **Find the discriminating word before you read the options.** *Most likely*, *first*, *single best*, *least appropriate*, *initial* and *definitive* each change the answer. "What is the first step" and "what is the definitive treatment" have different answers in almost every impaction question.
- **Decide the answer, then look for it.** Reading five plausible options before forming a view is how examiners harvest marks from well-prepared candidates.
- **Age and growth status are almost never decoration.** If the stem gives an age, it is because the answer changes at that age.

The curriculum, and where each item is covered

Items	Curriculum topic	Section
1–2	Concepts of physical growth and development; biological, skeletal and dental age; sexual maturation; the adolescent growth spurt	§1
3	Normal growth and development of the face and jaws	§2
4	Normal development of the dentition; leeway and primate space; anomalies of number, size and position	§3
5–6	Tissue response to orthodontic force; bone remodelling; tooth eruption; root resorption	§4
7	Orthodontics: general principles and aetiology	§5
8	Treatment need: assessment, indices, application	§6
64	Growth and treatment analysis; appraising the evidence	§7
9–10	Ethics and informed consent; patient communication, motivation, cooperation, psychosocial impact	§8

11	Orthodontic clinical examination	§9
12–16	Supplementary investigation; lateral cephalometry and its development; 2-D analyses; 3-D imaging; 3-D cephalometric analysis	§10
17–18	Study model examination and analysis; digital set-up	§11
19–20	Classification and diagnosis; problem list; treatment planning; treatment alternatives	§12
20	Space creation and the choice of extraction	§13
21–22	Biomechanical basis of therapy; mechanical principles in force control	§14
23, 33–34	Mechanical properties and orthodontic materials; bonding agents, wires, instruments, expanders, lasers; impression materials and intra-oral scanning	§15
28–29	Appliance design and mechanical characteristics; functional appliances	§16
30–32, 61–62	Fixed labial; fixed lingual; "invisible" aligners; invisible orthodontics; individualised lingual	§17
35–37	Stage I, II and III treatment goals	§18
38	Retention; relapse of dental crowding	§19
—	Interceptive treatment and the developing dentition	§20
39–44	Class I in children, adolescents and adults; bimaxillary protrusion — orthodontic, surgical and periodontal	§21
45–52	Class II — interceptive, orthopaedic, high- and low-angle, surgical, and with TMD	§22
53–56	Class III — interceptive, orthopaedic, orthodontic and surgical	§23
57–58, 16	Asymmetry in children and adolescents; crossbite correction	§24
59–60	Open bite in children and adolescents; deep bite	§25
24–27, 51–52	Myofunctional therapy; TMD; condylar trauma; adenoid hypertrophy and the airway	§26
43, 50, 52, 56	Adult orthodontics; combined orthodontic and orthognathic treatment	§27
—	Cleft lip and palate; craniofacial syndromes	§28
65	Risks of orthodontic treatment and their management	§29
63	Orthodontic practice; infection control; governance	§30

Finding your way around

This is **Volume II** of three. It does not replace **Volume I**, *Notes In Orthodontics*; it re-orders it against the examination. If you have been revising from the Companion and the six supplements, everything you already know is still here — the twenty essay topics, the sixteen chapters, the crash notebook, the number cards — but arranged against the curriculum the examiners actually work from, with the gaps filled in.

Three navigation aids follow: a routing table for when you know what you want to revise, a full two-way concordance with *Notes In Orthodontics*, and three revision routes depending on how much time is left.

If you know what you want, start here

What you want to do	Where to go
Revise one curriculum topic properly	Part 1, the relevant §
Learn the numbers	Part 2, Cards 1–4
Learn the classifications and lists	Part 2, Card 5
Read the traps the morning of the examination	Part 2, Card 6
Practise reading a set of records	Part 3, the protocol, then the six drills
Prepare your two treated cases	Part 4
Rehearse spoken answers	Part 5, the four set-pieces
See what an oral actually feels like	Part 5, the four question flows
Check who said what, and when	Part 6
Work out how to spend the time left	The dated plan in the front matter; Part 7 for the full countdown
Test yourself on a topic	The Question Bank block named in that §'s companion strap
Find a term you half-remember	The index at the back of this volume

How each section is laid out

Element	What it is for
Section head	Each § begins on a fresh page, and its title runs along the top of every page in that section, so you can flip to it
Curriculum strap (grey)	Which of the sixty-five curriculum items this section serves, and which examination components test it
Notes In Orthodontics strap (rust)	Exactly where this material sits in the Companion — topic, chapter and crash-notebook panel — which supplements add to it, and which Question Bank block and viva range to practise on
In one minute panel (teal)	One headline and three core points. Read these alone on a revision sweep
Numbered subsections	The substance, in tables wherever a table will carry it
Exam phrase panel	A spoken answer, ready to use, in the register an examiner expects
Written-paper trap / **Viva trap** panel (rust)	The specific error this topic is set to catch

A WORD ABOUT THE RUST-COLOURED STRAPS

They are the bridge back to your existing notes. If a section looks unfamiliar, read the strap first: it will usually tell you that you already know this as Topic 7 or Chapter 11, and that only the arrangement has changed. The gaps the first edition of the Companion left have since been written as **Part V** of its revised edition — ethics and consent, radiation protection, infection control, and governance and audit — so a strap that once said "not covered" now names a unit you can go and read.

Concordance with *Notes In Orthodontics*

The twenty essay topics

	Essay topic	Where it lives here
1	The biological basis of tooth movement	**§4**; force levels also §14
2	Cephalometry: uses and limitations	**§10**
3	Factors in formulating a treatment plan	**§12**; space analysis §11
4	Factors affecting the choice of extraction	**§13**
5	Anchorage and its clinical significance	**§14**
6	First, second and third order movements, and the evolution of fixed appliances	**§17**
7	The ideal archwire; stainless steel versus nickel titanium	**§15**
8	Functional appliances	**§16**; Class II application §22
9	Benefits and complications of orthodontic treatment	**§29**; consent §8
10	Medical conditions relevant to orthodontics	**§29.3**
11	Overbite and methods of overbite reduction	**§25.1**
12	Reducing overjet in a growing child	**§22**; the appliances §16
13	Management of a missing maxillary central incisor	**§20.6**
14	Management of an impacted maxillary canine	**§20.4**
15	Management of an impacted lower second molar	**§20.5**
16	Methods of crossbite correction	**§24**; appliances §15.5
17	Correcting a Class III incisor relationship	**§23**
18	Malocclusion and temporomandibular disorder	**§26**
19	Mild facial asymmetry in a growing child	**§24.4**
20	Late lower incisor crowding	**§19.4**

The sixteen chapters

	Chapter	Where it lives here
1	Evidence-Based Orthodontics	**§7**
2	Occlusion and Malocclusion	**§5**; indices §6; TMD history §26.1
3	Prenatal Development of the Craniofacial Region	**§2.1**; clefts §28
4	Postnatal Growth of the Craniofacial Region	**§1** and **§2.2–2.4**
5	Development of the Dentition	**§3**; late crowding §19.4
6	Orthodontic Tooth Movement	**§4**; anchorage and headgear §14
7	The Orthodontic Patient: Examination and Diagnosis	**§9**; cephalometrics §10; medical history §29.3
8	The Orthodontic Patient: Treatment Planning	**§12**; space analysis §11; extraction §13
9	Functional Jaw Orthopaedics	**§16**; expansion §24.2; facemask §23.3
10	Contemporary Fixed Appliances	**§17**; materials §15; mechanics §18
11	Skeletal Anchorage	**§14.6**
12	Management of the Developing Dentition	**§20**; anomalies §3.3
13	Management of the Permanent Dentition	**§18**, **§21**, **§22**, **§23**, **§25**; retention §19
14	Clear Aligners	**§17.5**

15	Orthodontics and Orthognathic Surgery	**§27**; asymmetry §24.4
16	Cleft Lip and Palate, and Syndromes	**§28**

The crash notebook and the appendices

In the Companion	Here
Part I — the four number cards	**Part 2, Cards 1–4**, extended with the supplements' figures
Part I — the twelve traps	**Part 2, Card 6**
Part I — ten things the evidence overturned	**§7.3**, and repeated in Part 6
Part I — the 42 schematics	**Redrawn and shared**. Thirty-two of them now appear inline in the § they belong to, and all of them in the Companion's crash notebook
Part I — 56 figures to look up in the textbook	Not reproduced; use the Companion's list alongside §9 and §10
Part I — the last hour, six things to say	**Part 2, Card 6** and **Part 7**
Part II — cross-cutting themes	**Part 1 opener**, the seven themes
Appendix A and B — model viva answers	Extended by **Question Bank Part 3**, 207 questions. The Companion's own model answers are still there
Appendix C — self-test answers	Extended by **Question Bank Parts 1 and 2**, 1,016 questions
Appendix D — landmark evidence	**Part 6**, rearranged by theme
Appendix E — the numbers	**Part 2**, Cards 1–4
Appendix F — figure finder	Use the Companion's
Appendix G — "Where is the evidence?" boxes	Distributed through Part 1 and consolidated in **§7.3**
Appendix I and J — revision schedule and route	**The plan** in the front matter, **Part 7** for the dated countdown, and the three routes below
Appendix K — quick index	**The index** at the back of this volume
Appendix L — Proffit cross-reference	Supplement C, and the rust straps in each §

Which supplements feed which sections

Supplement	Sections it contributes to
A — Mosby	Throughout, and principally §1, §2, §5, §6, §9, §10 (imaging and lasers), §13 (nasolabial angle), §15, §22 (choose from the vertical), §23, §27 (molar uprighting), §30
B — Cobourne cases	§9, §11–§13, §20–§25 and the whole of Part 3. This is the volume to work through alongside the diagnostic drills
C — Proffit	The differential voice, flagged in §4 (force), §10 (natural head position), §12 (problem-oriented planning), §13 (thresholds), §14 (the anchorage ladder), §19 (three causes of relapse), §23 (bone-anchored protraction), §24 (rapid versus slow expansion), §26 (prevalence argument), §27 (hierarchy of stability)
E — Bennett & McLaughlin	§14 (lacebacks), §17, §18
F — McLaughlin, Bennett & Trevisi	§15 (wire sequence), §17, §18 (the four elements, arch form, finishing), §19 (retention protocol)
G — Nanda	§14 (force systems, miniplates), §18.4 (the six aesthetic rules), §22.3 (distalisation), §23.3 (chin cap), §25 (vertical control), §27 (interdisciplinary finishing)

Three revision routes

Pick the one that matches the time you actually have, not the one you wish you had.

Route A — six weeks or more, the full route

Week 1–2	Part 1, §1–§15, one section a day. After each section, sit the Question Bank block named in its rust strap. Choose your two presentation cases and audit their records against Part 4
Week 3–4	Part 1, §16–§30, same pattern. Begin the Part 3 drills, one a week, timed
Week 5	Part 2 cards nightly. Work the 207 viva questions out loud. Present your two cases to a colleague
Week 6	Re-read Part 1 in a week, skimming for the panels only. Repeat any Question Bank block scoring under 70%. One timed full-length paper

Route B — two weeks

Days 1–5	Part 1, reading only the *In one minute* panels, the tables and the trap panels — not the connecting prose. Six sections a day
Days 6–9	Question Bank, three blocks a day, marking each block before reading the answers. Re-read the § behind any block under 70%
Days 10–12	Part 2 cards; Part 5 set-pieces out loud; two Part 3 drills, timed
Days 13–14	Part 6, Part 7, Card 6. One timed paper. Sleep

Route C — forty-eight hours

Day 1 morning	Every *In one minute* panel in Part 1, in order. Nothing else. That is thirty headlines and ninety points
Day 1 afternoon	Part 2, Cards 1–4. Write them out rather than reading them
Day 1 evening	Part 5, the four set-pieces, out loud until they are automatic
Day 2 morning	Part 2, Card 5 (classifications and lists) and Part 3, the protocol
Day 2 afternoon	Part 7. One block of the Question Bank, timed, to calibrate pace
The night before	Cards 1–4 only. Nothing new
The morning	Card 6 — the twelve traps and the six things to say. Then stop

IF YOU HAVE ALREADY REVISED FROM THE COMPANION

You are not starting again. Read the rust strap at the head of each § first: it tells you which Companion topic or chapter this is. Where the strap names a topic you know well, read only the *In one minute* panel and the trap panel, then go straight to the Question Bank

block. Spend the time you save on the seven sections the Companion treats thinnest — §6 (indices), §8 (consent and communication), §11 (space analysis and the digital set-up), §18 (treatment mechanics), §26 (condylar trauma and the airway), §29 (risks) and §30 (practice and governance). Four of those now have full units in **Part V** of the revised Companion; the other three are covered here and nowhere else. They are the curriculum items most likely to appear in the written paper and least likely to be in your notes.

Part 1 in detail

Every section of the core notes, with its subsections. Use this rather than the contents page when you are looking for something specific.

§1 Concepts of growth and development
1.1 The vocabulary, used precisely
1.2 The five "ages", and what each is for
1.3 The adolescent growth spurt
1.4 Theories of growth control — and why they still matter
1.5 Numbers to hold

§2 Craniofacial growth — prenatal and postnatal
2.1 Prenatal development, in five steps
2.2 Postnatal growth — four regions, four mechanisms
2.3 Mandibular rotation — Björk and Skieller
2.4 Adult craniofacial change

§3 Development of the dentition
3.1 The arithmetic of the developing arch
3.2 Eruption — sequence, timing and consequence
3.3 Anomalies of number, size, form and position
3.4 The timing of the first permanent molar decision

§4 The biology of tooth movement, eruption and root resorption
4.1 The opening statement
4.2 Bone and the periodontal ligament
4.3 Pressure–tension and the three phases
4.4 The molecular sequence, in order
4.5 Mechanotransduction and the discarded theories
4.6 Optimal force — the honest answer
4.7 Root resorption (OIIRR)
4.8 Eruption
4.9 Accelerating tooth movement — what to say

§5 Occlusion, malocclusion and aetiology
5.1 Definitions worth having verbatim
5.2 The classifications
5.3 Aetiology of malocclusion
5.4 Prevalence — the numbers

§6 Treatment need, indices and epidemiology
6.1 IOTN — the Index of Orthodontic Treatment Need
6.2 PAR — the Peer Assessment Rating
6.3 The other indices worth naming

§7 Evidence-based orthodontics
7.1 The hierarchy, top to bottom
7.2 Appraisal vocabulary you must be able to use
7.3 Ten things the evidence overturned

§8 Ethics, consent, communication and the psychosocial patient
8.1 What must be disclosed
8.2 Consent in the young patient
8.3 Motivation and cooperation

§9 Clinical examination and records
9.1 The sequence
9.2 Extra-oral examination
9.3 Intra-oral examination
9.4 The records, and the justification for each

§10 Imaging and cephalometrics
10.1 Technique
10.2 Landmarks and planes
10.3 The four limitations
10.4 Growth prediction — a use the evidence undermined
10.5 Three-dimensional imaging

§11 Study models, space analysis and the digital set-up
11.1 What study casts are examined for
11.2 The space analysis — six areas
11.3 Mixed dentition space analysis
11.4 The digital set-up

§12 Diagnosis, the problem list and treatment planning
12.1 The nine-step sequence
12.2 The arbiter: the skeletal pattern
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§13 Extraction decisions and space management
13.1 The four ways to create space
13.2 The four factors in the choice
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13.6 The extraction debate — the evidence

§14 Biomechanics, force control and anchorage
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§15 Orthodontic materials, wires, bonding and instruments
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§16 Removable and functional appliances
16.1 Removable appliances — design in four parts
16.2 Functional appliances — what they actually do
16.3 The appliances
16.4 The evidence — the numbers to quote
16.5 Myofunctional therapy

§17 Fixed appliances — labial, lingual and aligners
17.1 The three orders
17.2 The evolution — Angle to MBT
17.3 Self-ligation and customisation — the evidence
17.4 Lingual appliances

17.5 Clear aligners

§18 Treatment mechanics: Stage I, II and III

18.1 Stage I — alignment and levelling
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18.3 Stage III — finishing and detailing
18.4 The six aesthetic rules worth checking before debond (Nanda)
18.5 Arch form

§19 Retention, relapse and stability

19.1 Why teeth relapse — three mechanisms, three timescales
19.2 The retainers
19.3 Regimen, and what to tell the patient
19.4 Late lower incisor crowding

§20 Interceptive orthodontics and the developing dentition

20.1 The five-step triage
20.2 What is worth intercepting — graded by evidence
20.3 Serial extraction
20.4 The impacted maxillary canine
20.5 The impacted mandibular second molar
20.6 The missing or lost maxillary central incisor
20.7 Dental trauma: the rest period before orthodontic force

§21 Class I malocclusion and bimaxillary protrusion

21.1 Class I: the four presentations
21.2 Bimaxillary protrusion

§22 Class II malocclusion

22.1 Diagnosis
22.2 The interceptive question, answered by the trials
22.3 The orthopaedic phase in adolescents
22.4 Camouflage in the permanent dentition
22.5 Surgery
22.6 Class II with TMD

§23 Class III malocclusion

23.1 Diagnosis
23.2 The favourable/unfavourable split in the mixed dentition
23.3 Early intervention
23.4 Camouflage or surgery

§24 Transverse problems and facial asymmetry

24.1 Classifying a crossbite
24.2 Choosing the expansion approach
24.3 Anterior crossbite
24.4 Facial asymmetry

§25 Vertical problems: deep bite and open bite

25.1 Deep overbite — the four situations
25.2 Anterior open bite
25.3 Vertical control — the rules

§26 TMD, condylar trauma, the airway and myofunctional therapy

26.1 The history, in five dates
26.2 The evidence, and the position to state
26.3 Screening, classification and investigation
26.4 Condylar trauma and orthodontic treatment
26.5 Adenoid hypertrophy, the airway and sleep-disordered breathing
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§27 Adult orthodontics and orthognathic surgery

27.1 What is different about the adult
27.2 The orthognathic pathway
27.3 The operations
27.4 The hierarchy of stability
27.5 Consent for orthognathic surgery

§28 Cleft lip and palate, and craniofacial syndromes

28.1 Epidemiology
28.2 The care timeline
28.3 Dental features in the cleft patient
28.4 Craniofacial syndromes worth recognising

§29 Risks of orthodontic treatment and their management

29.1 The benefits — with the right level of confidence
29.2 The risks, quantified
29.3 The medical history — what it changes

§30 Orthodontic practice, infection control and governance

30.1 Infection control in the orthodontic setting
30.2 Radiation protection
30.3 Record keeping and governance
30.4 Knowing your limits

Part 1 — Core revision notes

Thirty sections, each mapped to the curriculum items it serves. Every section opens with an *In one minute* panel — one headline and three core points — and closes with the traps that cost marks. The material is written to be read twice: once for coverage, once for the numbers.

SEVEN THEMES THAT RUN THROUGH EVERYTHING

1. **Growth status determines everything.** Growth permits growth modification, makes overbite reduction by buccal segment extrusion stable, allows facemask protraction and sutural expansion, and permits spontaneous alignment after well-timed extraction. Its absence converts each of those into dentoalveolar compromise or a surgical problem.
2. **Most "skeletal" correction is dentoalveolar.** A few millimetres at best, and essentially transient. Functional appliances probably work by unlocking the occlusion and letting normal mandibular growth express itself.
3. **Anchorage is Newton's third law, not an optional extra.** It cannot be avoided, only planned. Unsupported, around two-thirds of the space created is lost.
4. **Evidence frequently contradicts tradition.** Name the study that overturned it.
5. **Retention is planned at the start, not the end.** Agreed before active treatment begins.
6. **Consent, radiation and risk are examinable, not decorative.** Candidates routinely omit them.
7. **The mandibular arch is the template.** Intercanine width and lower incisor inclination are the least amenable to change; the lower arch therefore forms the template for the whole plan.

§1 Concepts of growth and development

Curriculum: Items 1–2 — early and later stages of growth and development; biological, skeletal and dental age; stages of sexual development; the adolescent growth spurt and its relevance to craniofacial growth. Tested in the written paper and the oral.

Key reading: Companion Ch 4; Proffit Ch 2 and 4; Mosby ch. on craniofacial growth.

Notes In Orthodontics: Chapter 4 (Postnatal Growth of the Craniofacial Region); crash notebook *Growth and development*
Supplements: A: growth chapter · C: Proffit Ch 2 and 4
Practise this: Question Bank Block A, Q1–25 · Viva V1–V10

IN ONE MINUTE

Chronological age is the least useful age you have. Everything clinical runs on maturational age.

- Growth has three properties: pattern, variability and timing. Timing is what you manipulate
- Peak height velocity: girls ~12 years, boys ~14; CVM stage CS3 means the peak is within a year — start now
- Widths mature before depths, which mature before heights. There is no anteroposterior growth spurt of the chin

1.1 The vocabulary, used precisely

Term	Meaning, stated as an examiner would want it
Growth	Anatomical. An increase in size, by hyperplasia, hypertrophy and secretion of extracellular matrix.
Development	Physiological and behavioural. Progression towards maturity — a change in complexity, not merely in size.
Pattern	The proportional relationships between parts at a point in time, and their orderly change over time. Cephalocaudal gradient of growth: structures nearer the head mature earlier.
Variability	Normal biological range. The clinical question is never "is this child different" but "how different, and does it matter".
Timing	When growth occurs relative to the individual's own maturation. The variable that decides whether growth modification will work.
Site vs centre of growth	A **site** is a location where growth occurs; a **centre** is a location where independent, genetically controlled growth occurs. The distinction is the whole nasal-septum-versus-sutures debate.

1.2 The five "ages", and what each is for

Age	How it is assessed	What it is used for
Chronological	Date of birth	Almost nothing clinical. Correlates poorly with everything that matters

Dental	Stage of tooth formation and eruption (Demirjian; Nolla)	Interceptive timing; the decision to extract; predicting eruption
Skeletal	Hand–wrist radiograph; cervical vertebral maturation (CVM) on the lateral cephalogram	Timing of functional appliance therapy, facemask, RME, orthognathic surgery
Sexual	Tanner stages; menarche in girls, voice change and facial hair in boys	A cheap chairside proxy for the growth spurt; menarche indicates the peak has *passed*
Morphological	Height and weight against standard charts; serial height velocity	Confirming the spurt has begun or ended

1.3 The adolescent growth spurt

- There is a **juvenile spurt** at about 6–8 years, then a **prepubertal minimum**, then the adolescent spurt.
- **Peak height velocity** occurs at approximately 12 years in girls and 14 years in boys, with wide individual variation; girls both start and finish earlier.
- In girls, **menarche** follows peak height velocity by around a year — so a girl who has begun menstruating has largely passed the window for orthopaedic change.
- Statural growth precedes mandibular growth slightly; mandibular growth continues after statural growth appears complete, and in Class III patients may continue into the twenties.
- **Puberty begins in the brain**, with pulsatile GnRH release; the peripheral changes follow.

Cervical vertebral maturation (Baccetti, Franchi & McNamara)

Stage	Mean age	Interpretation and clinical action
CS1	9.7 y	Peak is 2 years away. Too early for a functional appliance
CS2	10.5 y	Peak within 1 year. Prepare; consider starting late in the stage
CS3	12.0 y	Peak occurs during this year. **This is the window** — start now
CS4	13.4 y	Peak was 1–2 years ago. Growth modification is losing its power
CS5–CS6	14–16 y	Growth largely complete; camouflage or surgery

THE CAVEAT YOU MUST ADD

None of the underlying principles of maturational staging is universally accepted, and the correlation between any single index and the amount of remaining growth is moderate at best. CVM tells you *where in the curve the patient is*, not *how much growth is left*.

1.4 Theories of growth control — and why they still matter

Theory	Claim	Clinical consequence
Genetic (Brodie)	Growth is entirely predetermined	If true, growth modification is impossible. Not sustained
Sutural (Sicher)	Sutures are growth centres pushing bones apart	Would make sutural expansion an orthopaedic certainty. Overstated

Cartilaginous / nasal septum (Scott)	Cartilage is the primary determinant; sutures respond	Explains why condylar cartilage behaves differently from epiphyseal cartilage
Functional matrix (Moss)	Bone grows in response to the soft tissue capsules and functional spaces it serves	The intellectual basis of functional appliances, and of the effect of airway obstruction on facial form
Servosystem (Petrovic)	Growth is a cybernetic system with local and systemic feedback; condylar cartilage is a secondary, adaptive cartilage	The most defensible modern position: mandibular growth can be modified within limits, and the limits are individual

EXAM PHRASE TO HAVE READY

"I would place myself with Petrovic and the modern synthesis: the condyle is a secondary, adaptive cartilage, so mandibular growth is modifiable — but within genetically determined limits, and the evidence is that the modification is a few millimetres and largely transient."

WRITTEN-PAPER TRAP

"The condylar cartilage is a primary growth centre" → false. It is a *secondary*, adaptive cartilage; it grows in response to the functional environment, which is precisely why functional appliances can influence it at all — and why the influence is modest.

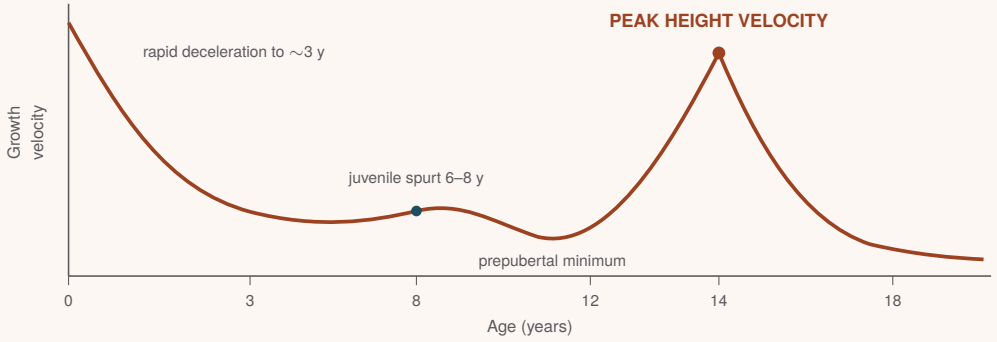
1.5 Numbers to hold

- Peak height velocity: girls ∼12 y, boys ∼14 y; juvenile spurt 6–8 y
- CVM: CS1 9.7 y · CS2 10.5 y · CS3 12.0 y · CS4 13.4 y
- Menarche follows peak height velocity by approximately 1 year
- Order of maturation: **widths** before **depths** before **heights**
- Growth never entirely stops; the cumulative adult effect is "surprisingly large"

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

GROWTH VELOCITY AND CERVICAL VERTEBRAL MATURATION



CVM staging (Baccetti, Franchi & McNamara)

CS1 9.7 y → peak in 2 y CS2 10.5 y → peak in 1 y CS3 12.0 y → peak within 1 y (start now) CS4 13.4 y → peak was 1–2 y ago

Caveat: none of the underlying principles is universally accepted.

Compare Figs 4.3, p. 171 and 4.47, p. 240.

§2 Craniofacial growth — prenatal and postnatal

Curriculum: Item 3 — normal growth and development of the face and jaws; the prenatal basis of clefting. Tested in the written paper and the oral.

Key reading: Companion Ch 3 and Ch 4; Proffit Ch 2; Mosby ch. on prenatal development.

Notes In Orthodontics: Chapters 3 and 4 (Prenatal Development; Postnatal Growth); crash notebook *Face and palate*
Supplements: A: prenatal development · B: prenatal chapter · C: Proffit Ch 2 and 5

Practise this: Question Bank Block B, Q26–55 · Viva V1–V10

IN ONE MINUTE

Learn prenatal development as the basis of clefting; learn postnatal growth as the basis of everything done to a growing patient.

- Five facial processes appear at week 4; the face is complete by week 8, the palate by week 10
- Cleft lip ± primary palate is a week-8 fusion failure; isolated cleft palate is a weeks 6–10 elevation or fusion failure
- The cranial vault grows by sutural apposition, the cranial base by synchondroses, the maxilla by sutures and remodelling, the mandible by the condyle and by surface remodelling

2.1 Prenatal development, in five steps

1. **Week 4** — five facial processes appear around the stomodeum: the single **frontonasal** process, the paired **maxillary** processes and the paired **mandibular** processes. Neural crest cells form practically all of the facial mesenchyme.
2. **Week 5** — nasal placodes develop; the **medial** and **lateral nasal processes** form around each placode.
3. **Week 6** — the palatine shelves appear *vertically*, one either side of the tongue.
4. **Week 8** — the face is complete. The upper lip is formed by fusion of the maxillary processes with the medial nasal processes.
5. **Week 10** — the palate is complete. The shelves elevate above the tongue as the mandible grows forwards and the head extends, then fuse in the midline, and with the primary palate anteriorly at the incisive foramen.

Where clefts arise — learn the pair as a contrast

	Cleft lip ± palate	Isolated cleft palate
Developmental failure	Maxillary process fails to fuse with the medial nasal process	Palatine shelves fail to elevate or to fuse
Timing	By week 8	Weeks 6–10
Incidence	1 : 700	1 : 2000
Sex ratio	Male 2 : 1	Female 4 : 1
Non-syndromic	~70%	~50%
Laterality	~80% unilateral, left commoner	—

WRITTEN-PAPER TRAP

The *opposite* sex ratios are the giveaway that these are different developmental failures at different times, not two grades of the same thing. A stem that gives you "female, isolated cleft palate, other anomalies present" is steering you towards a syndrome — 50% of isolated cleft palates are syndromic.

2.2 Postnatal growth — four regions, four mechanisms

Region	Mechanism	Clinical consequence
Cranial vault	Sutural apposition driven by the expanding brain, plus surface remodelling	Neural growth curve: ~90% complete by age 5. Craniosynostosis therefore presents early
Cranial base	Synchondroses (spheno-occipital, spheno-ethmoidal, intersphenoid) — endochondral	The spheno-occipital synchondrosis fuses at 13–15 y (girls earlier). Cranial base angle influences apparent jaw relationship
Maxilla	Displacement forwards and downwards by the surrounding sutures, plus surface remodelling (resorption anteriorly, apposition posteriorly at the tuberosity)	Explains why the maxilla can be protracted before sutural interdigitation, and why arch length increases posteriorly
Mandible	Endochondral growth at the condyle plus extensive surface remodelling; ramus remodels posteriorly	The condyle is a *secondary* adaptive cartilage — the basis for functional appliances

2.3 Mandibular rotation — Björk and Skieller

Björk's metallic implant studies revealed that the mandible rotates internally during growth, and that this rotation is largely masked by surface remodelling of the lower border.

Rotation	Definition
Total (true)	Rotation of the mandibular *body*, relative to the anterior cranial base, measured on implants
Matrix (apparent)	Rotation of the soft tissue matrix relative to the cranial base — what a cephalogram shows without implants
Intramatrix (angular remodelling)	The difference between the two: remodelling of the lower border. **Total = matrix + intramatrix**

	Forward (anterior) rotation	Backward (posterior) rotation
Which face height grows more	Posterior	Anterior
Occlusal consequence	Deep overbite; reduced lower face height	Anterior open bite; increased lower face height
Frequency	Commoner	Less common, and the harder cases

Björk's structural signs	Forward-inclined condyle, curved mandibular canal, thick symphysis, shallow antegonial notch, interincisal angle increases	Backward-inclined condyle, straight canal, thin symphysis, deep antegonial notch, anterior symphyseal inclination
Lower incisors	Upright as the rotation continues — the engine of late lower incisor crowding	Procline

EXAM PHRASE TO HAVE READY

"Growth direction matters more than growth amount. A backward-rotating, high-angle face is the one that will defeat me: overbite reduction by extrusion is unstable, anchorage is harder, and open bite recurs. I would want that pattern identified before I chose the mechanics, not after."

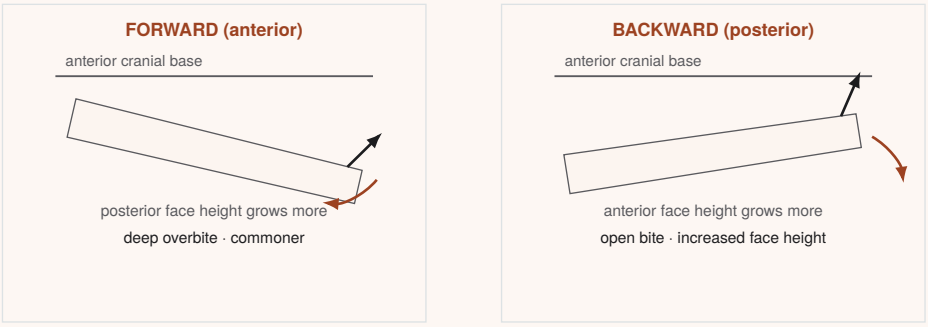
2.4 Adult craniofacial change

Growth does not stop at eighteen. Behrents' longitudinal work showed continued, slow, rotational change throughout adult life, in the individual's original pattern. Practically this means: continued uprighting of the lower labial segment (late lower incisor crowding); continued vertical facial growth; soft tissue changes — thinning of the vermilion, lengthening of the nose and upper lip, reducing incisor show at rest. It is why implants in a young adult can end infra-occluded, and why long-term retention is the honest answer.

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

MANDIBULAR ROTATION — FORWARD AND BACKWARD



The three rotations (Björk & Skieller; simplified by Solow & Houston)

Total (true) = mandibular body versus cranial base, measured on implants.

Matrix (apparent) = soft tissue matrix versus cranial base — what a cephalogram shows without implants.

Intramatrix (angular remodelling) = the difference; remodelling of the lower border. Total = matrix + intramatrix.

Compare Fig. 4.37, p. 223; extremes Fig. 4.40, p. 229.

§3 Development of the dentition

Curriculum: Item 4 — normal development of the deciduous and permanent dentitions; leeway and primate space; abnormalities of number, size and position, and their orthodontic consequences. Tested in the written paper and the oral.

Key reading: Companion Ch 5 and Ch 12; Proffit Ch 3; Mosby ch. on the developing dentition.

Notes In Orthodontics: Chapter 5 (Development of the Dentition); crash notebook *The arithmetic of the developing arch*
Supplements: A: developing dentition · B: development chapter · C: Proffit Ch 3

Practise this: Question Bank Block C, Q56–95 · Viva V11–V20

IN ONE MINUTE

Two numbers do most of the work: incisor liability and leeway space.

- Incisor liability ≈ 7 mm maxilla, 5 mm mandible — a deficit to be met
- Leeway space ≈ 1.5 mm per quadrant maxilla, 2.5 mm mandible — an excess to be used
- The larger liability is maxillary; the larger leeway is mandibular. Note the asymmetry

3.1 The arithmetic of the developing arch

	Incisor liability	Leeway space
What it is	Permanent incisors are wider than their predecessors	Primary canine + molars are wider than permanent canine + premolars
Maxilla	≈ 7 mm	≈ 1.5 mm per quadrant
Mandible	≈ 5 mm	≈ 2.5 mm per quadrant
Sign	A deficit to be met	An excess to be used
How it is met / used	Primate and developmental spacing; increase in intercanine width (∼3 mm mandible, ∼4.5 mm maxilla); labial positioning of the permanent incisors	Mesial drift of the first permanent molars into a Class I relationship; or preserved with a lingual arch to relieve crowding

- **Primate spaces**: mesial to the maxillary canine, distal to the mandibular canine. Present from eruption; their absence predicts crowding.
- **Developmental spacing**: generalised spacing in the primary dentition. Normal and necessary — a primary dentition without spacing is very likely to produce a crowded permanent dentition.
- Space for the mandibular incisors is **negative for about two years** after they erupt. Transient early crowding of lower incisors is normal and self-correcting.

3.2 Eruption — sequence, timing and consequence

Item	Detail
When a tooth erupts	When approximately two-thirds to three-quarters of the root is formed

Primary dentition	Erupts 6 months to 2.5 years; complete by ~3 years
First permanent molars	~6 years — the "six-year molars"
Mixed dentition	~6 to ~12 years
Mandibular eruption sequence	Canine → 1st premolar → 2nd premolar. **The second premolar is the one left crowded**
Maxillary eruption sequence	1st premolar → 2nd premolar → canine. **The canine is the one left crowded**
Governing principle	The last tooth to erupt in each arch is the one that ends up crowded or displaced
Terminal plane	Flush terminal plane in ~60–75%: a coin toss, correcting to Class I using leeway space, or remaining end-to-end. **Mesial step** → Class I. **Distal step** → Class II, and never self-corrects

WRITTEN-PAPER TRAP

The distal step is the one to remember: a distal step terminal plane in the primary dentition *will* become a Class II molar relationship. A flush terminal plane may go either way. A mesial step becomes Class I. Stems love this.

3.3 Anomalies of number, size, form and position

Anomaly	Prevalence	Orthodontic consequence
Hypodontia (excl. third molars)	6.4%	Commonest: mandibular second premolar, maxillary lateral incisor, maxillary second premolar. Spacing, retained primary teeth, infra-occlusion; a space-opening versus space-closing decision
Third molar agenesis	Up to 30%	Not a reason for anything on its own
Oligodontia (≥6 missing)	~0.1–0.3%	Multidisciplinary; often syndromic (ectodermal dysplasia)
Supernumerary teeth	Up to 4%; ~2% in permanent dentition	90% maxillary, most in the premaxilla. Conical, tuberculate, supplemental, odontome. **Tuberculate** typically prevents eruption; **conical** typically causes displacement or diastema
Megadontia / microdontia	—	Peg lateral incisor ~2%; tooth-size discrepancy, Bolton analysis
Double teeth (fusion / gemination)	~0.5% permanent	Aesthetics, space, sometimes division and restoration
Ectopic first permanent molar	0.75–4.3%; 25× commoner in the maxilla	Resorbs the distal of the second primary molar. Separator, brass wire or distalising spring; may reverse spontaneously up to ~8 y
Transposition	~0.3%	True transposition involves crowns *and* roots; pseudo-transposition crowns only. Commonest: maxillary canine–first premolar
Infra-occlusion (ankylosed primary molar)	Up to 9%	Monitor; extract if the permanent successor is present and eruption is obstructed, or if infra-occlusion is severe
Molar–incisor hypomineralisation	~15%	Changes the extraction calculus: a compromised first permanent molar may become the planned extraction

ANOMALIES TRAVEL IN PACKS

One dental anomaly should prompt a search for the others. Palatally displaced canines,

hypodontia, peg lateral incisors, infra-occluded primary molars, enamel hypoplasia and transposition co-occur far more often than chance — a shared genetic field effect. Finding a peg lateral is a reason to palpate for canines and to look for missing premolars.

3.4 The timing of the first permanent molar decision

If a first permanent molar is of poor long-term prognosis, the classic guidance is to time its removal so that the second permanent molar erupts into its place.

- **Ideal timing**: bifurcation of the second permanent molar just calcifying — radiographically about **8–10 years** of dental age.
- Better spontaneous space closure in the **mandible** than the maxilla; the maxillary second molar drifts mesially readily at almost any age, the mandibular one does not.
- Confirm the presence and position of **all** permanent teeth, including third molars, before extracting.
- Balancing and compensating extractions: consider a **compensating** extraction of the opposing maxillary molar to prevent over-eruption; **balancing** extraction of the contralateral tooth is rarely indicated for molars (it is a centre-line consideration, and centre-line shift from molar loss is limited).

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

THE ARITHMETIC OF THE DEVELOPING ARCH

Incisor liability

Permanent incisors are WIDER than their predecessors.

Maxilla	≈ 7 mm
Mandible	≈ 5 mm

a DEFICIT to be met

Leeway space

Primary canine + molars are WIDER than permanent canine + premolars.

Maxilla	≈ 1.5 mm/quadrant
Mandible	≈ 2.5 mm/quadrant

an EXCESS to be used

Note the asymmetry: the larger liability is MAXILLARY; the larger leeway is MANDIBULAR.

Eruption sequence and its consequence

Mandible:	canine → 1st premolar → 2nd premolar	→ **2nd PREMOLAR crowded**
Maxilla:	1st premolar → 2nd premolar → canine	→ **CANINE crowded**

The last tooth to erupt in each arch is the one that ends up crowded.

Compare Figs 5.23–5.26, pp. 268–280.

§4 The biology of tooth movement, eruption and root resorption

Curriculum: Items 5–6 — tissue response to orthodontic force; bone remodelling; biological responses to force application; tooth eruption and root resorption. Tested in the written paper and the oral.

Key reading: Companion Ch 6, pp. 287–307; Proffit Ch 8; Nanda ch. 1; Mosby ch. 1.

Notes In Orthodontics: **Topic 1** (The biological basis of tooth movement); Chapter 6
Supplements: A: Topic 1 · B: Topic 1 (root resorption) · C: Topic 1 and Proffit Ch 8 · G: Topic 1
Practise this: Question Bank Block D, Q96–140 · Viva V21–V30

IN ONE MINUTE

Force is mechanical; movement is biological. An intact periodontal ligament is an absolute prerequisite.

- Pressure–tension: resorption on the leading side, deposition on the trailing side
- Hyalinisation produces a lag phase of 20–30 days; RANKL drives resorption, OPG opposes it
- No universal optimal force exists: 50–100 cN for bodily movement, and lighter is safer

4.1 The opening statement

Orthodontic tooth movement is a **biological** phenomenon, not a mechanical one. The clinician applies force; the movement is produced by remodelling of the periodontal ligament and the alveolar bone. The cleanest proof is clinical: neither an implant nor an ankylosed tooth will move under orthodontic force, which localises the response to the ligament rather than to bone alone. The histological foundation was laid independently by **Carl Sandstedt** and **Albin Oppenheim** at the turn of the twentieth century, giving the classic pressure–tension theory.

4.2 Bone and the periodontal ligament

Item	Detail
Bone composition	Extracellular matrix ~30% (type I collagen, proteoglycans); mineral ~60% (hydroxyapatite); cells ~10%
Osteocytes	85–90% of bone cells. Osteoblasts enveloped during mineralisation; linked by canaliculi; the principal mechanosensors
Osteoblasts	Bone-forming; derived from mesenchymal stem cells; express RANKL and OPG
Osteoclasts	Multinucleate; from haematopoietic stem cells; circulate as monocytes; resorb via cathepsin K, MMPs and H^+, forming Howship's lacunae
Basic multicellular unit (BMU)	The remodelling unit; apposition and resorption are tightly coupled
Sclerostin	Secreted by osteocytes; inhibits osteoblast differentiation and canonical Wnt signalling — a brake against excess bone formation
PDL width	~0.2 mm on average; thinnest around the middle third of the root
PDL contents	Type I collagen bundles (Sharpey's fibres), transeptal fibres, elastic oxytalan fibres, fibroblasts, cementoblasts, immune cells, epithelial rests of Malassez, pluripotent stem cells, GAG-rich ground substance

4.3 Pressure–tension and the three phases

Leading (pressure) side	Trailing (tension) side
Compression; blood flow reduced	Tension; fibres stretched
RANKL and M-CSF increased	OPG increased, RANKL reduced
OPG decreased	TGF-β, PGE_2 raised
Osteoclast recruitment $\rightarrow$ resorption	Osteoblast activity $\rightarrow$ bone deposition
Hyalinisation if the force is excessive	VEGF and FGF2 support angiogenesis

Phase	Duration	What is happening
1 — Initial	$\sim$2 days	Displacement of the tooth within the PDL space; fluid expressed; no true bone remodelling yet
2 — Lag	20–30 days	Hyalinisation: a cell-free, glass-like necrotic zone forms. Osteoclasts cannot resorb beneath it, so **undermining resorption** proceeds from the adjacent marrow spaces until the necrotic tissue is cleared
3 — Post-lag	Sustained	Direct **frontal** resorption resumes; movement proceeds at $\sim$1 mm/month

- Hyalinisation occurs even at **30 g** (Reitan 1967) and throughout treatment, not only at the start. It is greater with **tipping** than with bodily movement, because tipping concentrates stress at the alveolar crest and apex.
- **Frontal** resorption is what you want; **undermining** resorption is what heavy force forces you to accept, and it is slower.

4.4 The molecular sequence, in order

1. Matrix strain and altered interstitial fluid flow; distortion of nerve endings releases substance P and CGRP.
2. Reduced blood flow causes hypoxia, apoptosis and denaturation of extracellular matrix.
3. Necrotic tissue triggers **sterile inflammation** — interleukins and prostaglandins.
4. **M-CSF** drives monocyte differentiation; **RANKL**, expressed by osteoblasts, osteocytes and PDL cells, binds **RANK** on pre-osteoclasts and is *essential* for osteoclastic resorption.
5. **OPG**, a soluble decoy receptor, opposes this. OPG is increased on the tension side and reduced under compression.
6. Osteoclasts resorb through cathepsins, MMPs and hydrogen ions.
7. RANKL peaks at **3 days** and returns to baseline between days 6 and 58 — implying other pathways sustain the post-lag phase.
8. On the tension side, IL-1β, IL-10 and PGE_2 raise OPG and lower RANKL; TGF-β drives fibroblast and osteoblast proliferation; VEGF and FGF2 support angiogenesis.

> **WRITTEN-PAPER TRAP**
>
> **RANKL up on compression; OPG up on tension.** Reversing this is the single commonest and costliest slip in this topic. A useful mnemonic: RANKL *rankles* the bone away where the tooth is pressing.

4.5 Mechanotransduction and the discarded theories

- **Henneman's four-step model**: extracellular matrix strain and fluid flow → strain transferred to cells → cell recruitment and differentiation → tissue remodelling.
- **Bone-bending** (Baumrind, 1969) was proposed to explain how differential force could arise if the PDL behaved as a continuous hydrostatic system under Pascal's law. There is little evidence that orthodontic forces bend bone to any clinically meaningful degree.
- **Piezoelectricity**: currents are too small and too short-lived to matter.
- **Streaming potentials**: of unclear significance.

4.6 Optimal force — the honest answer

Source	What to be able to quote
Storey & Smith (1952)	Historical optimal force range 150–200 g
Quinn & Yoshikawa (1985)	Positive dose–response at low forces until a plateau is reached
Ren et al. (2003, 2004)	No universal ideal force can be defined; modelled maximum canine retraction at 272 cN (1 cN ≈ 1 g)
Theodorou et al. (2019)	50–250 cN gives similar rates of 0.1–0.47 mm/week; **50–100 cN optimal for bodily movement**; >250 cN marginally faster with greater adverse effects
Proffit's formulation	Optimum force is *the lightest force producing a near-maximum biological response*. Different movements need different forces because the loaded PDL area differs
Force duration	Threshold ∼4–8 h/day for a clinically useful effect; continuous beats intermittent

Movement	Force, and why
Tipping	35–60 g. Small PDL area loaded at crest and apex
Bodily (translation)	50–100 cN (Theodorou); traditionally 100–150 g. Whole root surface loaded
Root uprighting / torque	50–100 g. Concentrated apically
Rotation	35–60 g
Extrusion	35–60 g
Intrusion	**10–20 g per tooth**; ∼60 g total for a four-incisor segment. The lightest movement there is, because force is dissipated through the small apical surface area

4.7 Root resorption (OIIRR)

Item	Detail
Radiographic incidence	Up to 73%
Histological incidence	Up to 90%; near-universal on CBCT
Severe ($\geq \frac{1}{3}$ root length)	1–5% of teeth; ∼5% of patients lose >5 mm
Distribution	Maxillary incisors > canines > mandibular incisors > first permanent molars

Proven risk factor	**Heavy force** (Weltman 2010) is the only one with substantive evidence
Suggested only	Continuous rather than intermittent force; intrusion; total amount of movement; Class II elastics; abnormal root morphology; treatment duration; roots driven into cortical bone; allergy or asthma; teeth without occlusal contact; previous trauma; previous resorption
Management	Where significant resorption is identified, **pause force application for 3–6 months** (Levander 1994)
Long-term significance	Severe resorption does not necessarily cause greater mobility or discolouration long term
Consent	Mention it explicitly, quantify it, and record that you did

EXAM PHRASE TO HAVE READY

"I would tell the patient that some root shortening happens in most people treated with fixed appliances, that it is usually a fraction of a millimetre and of no consequence, but that in about one to five per cent of teeth it exceeds a third of the root — and that I reduce the risk by using light forces, keeping treatment short, and taking a radiograph part-way through so that I can pause if I see anything."

4.8 Eruption

- Eruption occurs when about **two-thirds to three-quarters** of the root is formed, and continues until occlusal contact, then slowly thereafter to compensate for wear.
- The **dental follicle** is essential: it coordinates the osteoclastic resorption that forms the eruption pathway. Remove the follicle and the tooth does not erupt; remove the tooth and leave the follicle, and the pathway still forms.
- **Primary failure of eruption** — a loss-of-function mutation in **PTH1R**. Affected teeth do not respond to orthodontic traction; attempting it causes ankylosis. Distinguish it from mechanical obstruction, which *does* respond.
- **Alveolar bone exists only where a tooth erupted** (Proffit's argument). Erupting and extruding teeth bring alveolar bone with them; a site left edentulous early develops a permanent alveolar defect. Hence: erupt a tooth through a deficient site, then move it back, and bone remains.

4.9 Accelerating tooth movement — what to say

Corticotomy, micro-osteoperforation and piezocision can increase the local rate through the **regional acceleratory phenomenon**, but the effect appears short-lived and evidence that any of them shortens overall treatment time is sparse. Photobiomodulation may accelerate discrete phases but does not influence overall treatment time (El-Angbawi 2023). **Do not claim acceleration techniques work.**

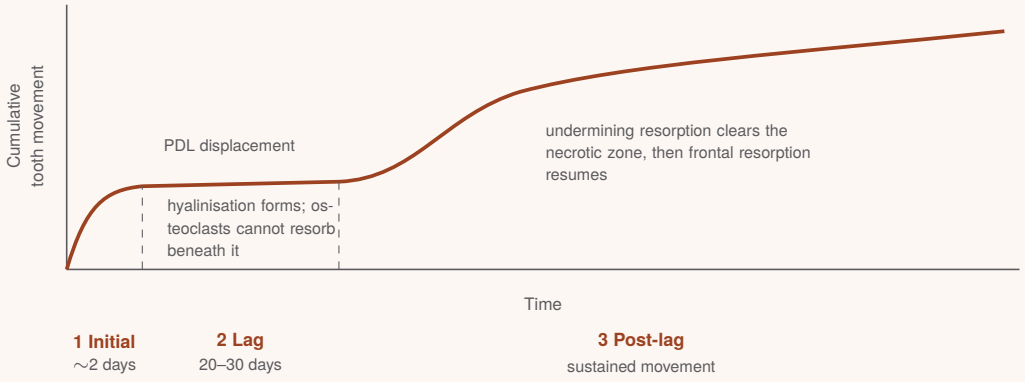
VIVA TRAP

"How would you speed up treatment for this patient?" The examinable answer is: good bracket positioning, an efficient wire sequence, light continuous force, avoiding breakages, and attending appointments. Not a device.

Schematics

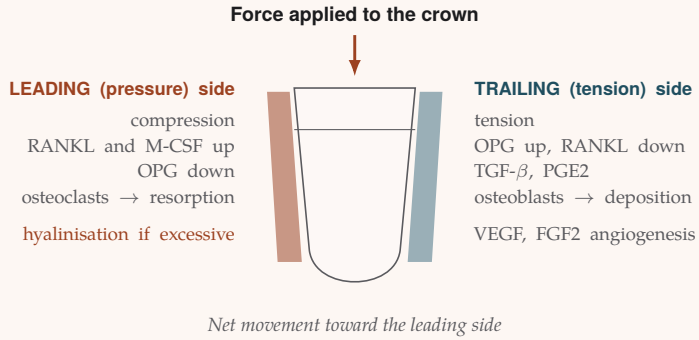
The same figures appear in the crash notebook of Notes In Orthodontics.

THE THREE PHASES OF TOOTH MOVEMENT



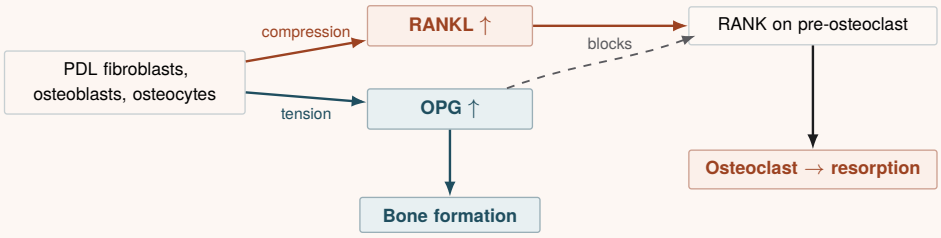
Three phases. The lag is the rate-limiting step, and it is why heavier force does not move teeth faster. Compare Fig. 6.3, p. 295.

PRESSURE AND TENSION



Pressure–tension. Compare Fig. 6.2, p. 294 — the histology is worth seeing.

THE RANKL / OPG AXIS



RANKL is essential for osteoclastic resorption; OPG is a soluble decoy receptor. RANKL peaks at 3 days and returns to baseline from day 6–58 (Danz et al., 2021).

Compare Fig. 6.6, p. 302.

§5 Occlusion, malocclusion and aetiology

Curriculum: Items 7 and 19 — orthodontics: general principles and aetiology; classification of malocclusion. Tested in the written paper and the oral.

Key reading: Companion Ch 2; Proffit Ch 1 and 5; Mosby ch. on classification.

Notes In Orthodontics: Chapter 2 (Occlusion and Malocclusion); Topic 18 background
Supplements: A: classification chapter · B: occlusion chapter · C: Proffit Ch 1 and 5
Practise this: Question Bank Block E, Q141–170 · Viva V31–V38

IN ONE MINUTE

Angle classifies molars; the British Standard classifies incisors. Ideal occlusion is a description, not a treatment goal you can always reach.

- An acceptable occlusion develops naturally in only 30–40% of people
- Only $\leq$5% of malocclusion has a specific identifiable cause
- RCP and ICP coincide in only $\sim$10%; a 0–2 mm discrepancy is normal (Posselt)

5.1 Definitions worth having verbatim

Term	Definition
Ideal occlusion	A theoretical concept: the anatomically perfect relationship of the dental arches, rarely found
Normal occlusion	An occlusion within an accepted range of variation, functioning without pathology
Malocclusion	A variation from ideal occlusion of sufficient degree to be, or to have the potential to be, of aesthetic or functional significance to the individual
Centric relation (RCP)	The condyle in its most superior–anterior position against the articular eminence, independent of tooth contact
Intercuspal position (ICP)	Maximum intercuspation of the teeth
Line of occlusion	The smooth curve through the central fossae and cingula of the upper arch, and the buccal cusps and incisal edges of the lower — Angle's defining construct

5.2 The classifications

Angle (1899) — molar-based

- **Class I**: the mesiobuccal cusp of the upper first permanent molar occludes with the buccal groove of the lower first permanent molar.
- **Class II**: the lower molar is distal to that position. Division 1 — upper incisors proclined or of average inclination with an increased overjet. Division 2 — upper central incisors retroclined, overjet usually reduced, overbite increased.
- **Class III**: the lower molar is mesial to that position.
- **Limitations**: assumes the first molar is in a stable, correct anteroposterior position; ignores the vertical and transverse planes; ignores the skeletal pattern; unusable if the molar is missing or displaced.

British Standards Institute — incisor-based

- **Class I**: the lower incisor edges occlude with or lie immediately below the cingulum plateau of the upper central incisors.
- **Class II**: the lower incisor edges lie posterior to the cingulum plateau. **Division 1** — upper central incisors proclined or of average inclination, overjet increased. **Division 2** — upper central incisors retroclined, overjet usually minimal but may be increased.
- **Class III**: the lower incisor edges lie anterior to the cingulum plateau; overjet reduced or reversed.

Andrews' six keys of occlusion (1972)

Derived from 120 non-orthodontic normal occlusions.

1. Correct molar relationship (including the distal marginal ridge of the upper 6 contacting the mesial marginal ridge of the lower 7).
2. Correct crown angulation (**tip** — second order).
3. Correct crown inclination (**torque** — third order).
4. No rotations.
5. Tight contact points, no spaces.
6. Flat or minimal curve of Spee.

A seventh — correct crown size and shape / tooth-size harmony — is often added.

5.3 Aetiology of malocclusion

Level	Content
Specific causes ($\leq$5%)	Genetic syndromes and single-gene disorders; embryological defects (clefting); trauma (condylar fracture, incisor avulsion); pathology (cysts, tumours); functional deviations; dental disease (early loss, caries)
Genetic influence	Strong for skeletal pattern and tooth size/number; the Habsburg mandible is the standard illustration. Twin and family studies show high heritability for facial form, lower for occlusal detail
Environmental influence	Explained by **equilibrium theory**: teeth sit where resting soft tissue forces balance. Light, sustained forces (resting tongue and lip posture, a lip trap) move teeth; heavy, brief forces (chewing, swallowing) do not
Digit sucking	Proclines uppers, retroclines lowers, asymmetric anterior open bite, posterior crossbite from a lowered tongue position and cheek pressure. Usually resolves if the habit stops before the permanent incisors are in place
Soft tissue	Lip competence, lower lip line, tongue posture. A lip trap perpetuates a Class II division 1
Airway	Chronic nasal obstruction is associated with — not proved to cause — increased face height and a narrow maxilla. State it carefully

5.4 Prevalence — the numbers

Trait	Figure
Acceptable natural occlusion	30–40%
Class II (US, 12–17 y)	15–22%
Class III (US, 12–17 y)	5–6%

Class III worldwide	~6.8%, with striking regional variation (much higher in East Asian populations)
Class II division 2 (Caucasian)	~5–10%
Overjet ≥5 mm	23% of children falling to 13% of adults — the mandible catches up
RCP = ICP	~10% (Posselt); 0–2 mm discrepancy normal
Normal overbite	2–4 mm, or one-third to one-half of the lower incisor crown
Ideal overjet	2–4 mm

§6 Treatment need, indices and epidemiology

Curriculum: Item 8 — treatment need: assessment, indices and their application. Tested in the written paper and the oral.

Key reading: Companion Ch 2, pp. 106–112; Proffit Ch 1.

Notes In Orthodontics: Chapter 2, pp. 106–112 (IOTN and PAR); crash notebook *IOTN and PAR*

Supplements: A and B: treatment need · C: Proffit Ch 1

Practise this: Question Bank Block F, Q171–195 · Viva V39–V48

IN ONE MINUTE

IOTN measures need before treatment; PAR measures what treatment achieved. Do not confuse them.

- IOTN is hierarchical: record only the **single worst** trait, using MOCDO
- PAR is accumulative, has no maximum cut-off, and is best used to audit a clinician or a service
- ≥22 PAR points, or >70% reduction, is "greatly improved"

6.1 IOTN — the Index of Orthodontic Treatment Need

Two independent components, recorded separately.

Dental Health Component — grades 1 to 5

Grade	Need	Typical traits
1	None	Minor irregularity <1 mm
2	Little	Overjet >3.5 but ≤6 mm with competent lips; overbite ≥3.5 mm without contact; crowding 1–2 mm; prenormal or postnormal occlusion with no other anomaly
3	Moderate / borderline	Overjet >3.5 but ≤6 mm with *incompetent* lips (3a); reverse overjet >1 but ≤3.5 mm (3b); crossbite with 1–2 mm displacement (3c); crowding 2–4 mm (3d); deep overbite complete to gingiva without trauma (3f)
4	Great	Overjet >6 but ≤9 mm (4a); reverse overjet >3.5 mm without masticatory or speech difficulty (4b); crossbite with >2 mm displacement (4c); severe displacement >4 mm (4d); extreme lateral or anterior open bite >4 mm (4e); increased and complete overbite with gingival or palatal trauma (4f); less extensive hypodontia (4h); posterior lingual crossbite with no functional occlusal contact (4l); reverse overjet >1 but ≤3.5 mm with recorded difficulties (4m); partially erupted, tipped and impacted teeth (4t); existing supernumerary (4x)
5	Very great	Impeded eruption (5i); extensive hypodontia with restorative implications (5h); overjet >9 mm (5a); reverse overjet >3.5 mm with reported difficulties (5m); defects of cleft lip and palate (5p); submerged deciduous teeth (5s)

MOCDO — THE HIERARCHY, IN ORDER

Missing teeth · **O**verjet · **C**rossbite · **D**isplacement of contact points · **O**verbite. Work down the list, record the first (worst) trait you meet, and stop. IOTN is *not* cumulative.

Aesthetic Component

Ten colour photographs of anterior occlusions in rank order of attractiveness. The clinician (or the patient) selects the photograph closest in *aesthetic impairment* — not in morphological similarity. Grades 1–4 no need, 5–7 borderline, 8–10 need.

6.2 PAR — the Peer Assessment Rating

- Scored from study casts, before and after. Components: upper and lower anterior segment alignment, buccal occlusion (in three planes), overjet, overbite, centre lines — each weighted.
- Accumulative; no maximum score.
- Outcome bands: **<30% reduction** — worse or no different; **>30% reduction** — improved; **≥22 points** (or >70%) — **greatly improved**.
- Best used to audit an operator or a service over many cases, not to judge an individual case.

6.3 The other indices worth naming

Index	Purpose
ICON	Index of Complexity, Outcome and Need — combines need, complexity and outcome in one instrument
ABO-OGS	American Board of Orthodontics Objective Grading System. Finishing quality: alignment, marginal ridges, buccolingual inclination, occlusal relationship, occlusal contacts, overjet, interproximal contacts, root angulation. **The measure on which aligners perform significantly worse than fixed appliances**
Little's Irregularity Index	Sum of the linear displacements of the anatomical contact points of the mandibular incisors. Used for relapse research; <3.5 mm proposed as an acceptable level of relapse
Bolton analysis	Tooth-size discrepancy: anterior ratio 77.2%, overall ratio 91.3%

WRITTEN-PAPER TRAP

PAR shows *no* difference between aligners and fixed appliances; ABO-OGS shows aligners significantly *worse*. PAR is the coarser instrument and the two do not correlate. If a stem asks which index detected the difference, it is ABO-OGS.

§7 Evidence-based orthodontics

Curriculum: Item 64 — growth and treatment analysis; and the appraisal skills that underpin every other item. Tested in the written paper and, heavily, in the oral.

Key reading: Companion Ch 1; Proffit Ch 1.

Notes In Orthodontics: Chapter 1 (Evidence-Based Orthodontics); Appendix D and the *Where is the evidence?* boxes; crash notebook *Ten things the evidence overturned*

Supplements: B: the overturned-claims unit · C: Proffit Ch 1

Practise this: Question Bank Block G, Q196–225 · Viva V39–V48

IN ONE MINUTE

Orthodontics has a weak evidence base and a strong tradition of assertion. Knowing which is which is the fastest way to demonstrate scholarship.

- Evidence-based practice = best available evidence + clinical expertise + patient values
- Learn the hierarchy, and learn where orthodontic claims usually sit on it
- Name the study that overturned the tradition

7.1 The hierarchy, top to bottom

Clinical practice guidelines from high-quality meta-analysis → meta-analysis → systematic review (*filtered / secondary research*) → randomised controlled trial → cohort study → case–control study → cross-sectional and ecological studies → case series and case reports (*unfiltered / primary research*) → expert opinion.

7.2 Appraisal vocabulary you must be able to use

Term	Meaning
PICO	Patients / Population · Intervention · Comparison · Outcome. The structure of an answerable question
Randomisation	Removes selection bias by making groups comparable in known *and unknown* confounders
Allocation concealment	Prevents the recruiter from knowing the next allocation. Distinct from blinding
Blinding	Rarely possible for the operator in orthodontics; possible for the outcome assessor, which is what matters most
Intention to treat	Analyse participants in the group to which they were randomised, whatever they actually received. Preserves the benefit of randomisation
Type I error (α)	False positive. Conventionally set at 0.05
Type II error (β)	False negative. Power = $1 - \beta$; conventionally 80–90%
Confidence interval	The range within which the true value probably lies. A wide CI crossing 1 (for RR/OR) or 0 (for a difference) means the result is compatible with no effect
I^2	Proportion of variability due to heterogeneity rather than chance: **25% low · 50% moderate · 75% high**
Relative risk	Risk in the exposed divided by risk in the unexposed. RR 0.58 = a 42% reduction

Number needed to treat	1 / absolute risk reduction. The number the patient actually understands
Publication bias	Positive results are published preferentially; assessed by a funnel plot

7.3 Ten things the evidence overturned

The tradition	What the evidence shows	Name
Early Class II treatment improves the result	No long-term difference; the late group finished slightly better	O'Brien 2009
Third molars cause late lower incisor crowding	Crowding develops without them; NICE advises against removal for this reason	Richardson 2002; NICE
Orthodontics causes or cures TMD	It does neither	Behrents & White 1992
Self-ligation is faster	No difference in rate, treatment time, visits or occlusal outcome	Scott 2008; Fleming 2010
Customised appliances shorten treatment	Comparable outcome, more breakages, more planning time	Penning 2017
Extractions damage the face	Change is small and usually judged favourable	Bowman & Johnston 2000
Extractions harm the airway	No evidence; "scaremongering and marketing"	Larsen 2015; Orabi 2021
Interceptive canine extraction prevents surgery	Very weak evidence for eruption; none for reducing surgery	Benson 2021 (Cochrane)
Growth can be predicted cephalometrically	At best unreliable	Baumrind 1984
An ideal functional occlusion improves stability	No evidence for stability or TMD prevention	Luther 2007

EXAM PHRASE TO HAVE READY

"I'd want to separate what we know from what we do. The randomised evidence in orthodontics is thin, and several things I was taught as fact have not survived it — early Class II treatment and self-ligation are the two I'd name. Where I have no good evidence, I say so and choose the option that is easiest to reverse."

§8 Ethics, consent, communication and the psychosocial patient

Curriculum: Items 9–10 — ethics and informed consent; patient communication in orthodontic treatment, motivation, cooperation and psychosocial impact (4 hours of the curriculum). Tested in the written paper and heavily in the oral.

Key reading: Companion Ch 2 and Ch 7; Mosby ch. on motivation; GDC Standards.

Notes In Orthodontics: **Topic 9** (Benefits and complications) in part; Chapter 7, pp. 356–366
Supplements: A: motivation and the five ages · G: the chapter on why the patient came
Practise this: Question Bank Block H, Q226–255 · Viva V49–V58

IN ONE MINUTE

Orthodontics is elective. That single fact raises the consent standard, not lowers it.

- Valid consent: capacity, sufficient information, voluntariness — and it is a process, not a form
- Montgomery: disclose the risks a *reasonable patient in this position* would attach significance to
- Body dysmorphic disorder affects 5–13% of orthodontic patients and up to 17% of orthognathic patients

8.1 What must be disclosed

- The nature and purpose of the proposed treatment, in plain language.
- **All reasonable alternatives**, including the alternative of no treatment and what will happen if nothing is done.
- Material risks — **demineralisation** (up to half of fixed appliance patients), **root resorption** (severe in 1–5% of teeth), **relapse** (near-universal), **recession**, **loss of vitality**, **pain** (peaks at 48 hours, gone by 7 days), **headgear injury**, **failure to achieve the aim**, and the possibility of **needing surgery** if growth is unfavourable.
- The **time commitment**, appointment frequency, and what happens if appointments are missed.
- **Retention** — that it is lifelong if the result is to be held, and what the patient's part in that is. Retention is agreed *before* treatment starts.
- Cost, and the cost of replacing lost or broken retainers.

8.2 Consent in the young patient

- Most orthodontic patients are minors. Consent is obtained from a person with parental responsibility, but the child's **assent** is clinically decisive: a technically ideal plan the child will not wear is not a plan.
- **Gillick competence** / **Fraser guidelines**: a child under 16 may consent if they have sufficient understanding and intelligence to understand fully what is proposed.
- Capacity is **decision-specific and time-specific**, assessed by the four-part functional test: understand, retain, weigh, communicate.
- Involve the child in the discussion, not only the parents. Nanda makes this point explicitly.

8.3 Motivation and cooperation

Factor	What to say about it
Internal vs external motivation	Internally motivated patients — those seeking treatment for themselves — cooperate better than those brought by a parent or a dentist
Predicting cooperation	No reliable predictor exists. The best available proxies are attendance history, oral hygiene at the assessment visit, and whether the patient can articulate their own concern
Discontinuation	High rates in children and adolescents, especially with removable appliances (Klaus 2017)
Psychosocial benefit	The best-supported benefit of treatment. Increased overjet makes a child more susceptible to teasing and bullying (Seehra 2011); treatment improves the emotional and social domains of OHRQoL (Javidi 2017)
Long-term psychological health	Little objective evidence of benefit (Kenealy 2007). Do not overclaim
Body dysmorphic disorder	0.7–2.5% general population; **5–13% of orthodontic patients**; up to 17% of those seeking orthognathic surgery. Red flags: a defect invisible or trivial to the clinician, serial dissatisfaction with previous treatment, extensive photographic evidence, refusal to accept reassurance

THE FOUR ETHICAL PRINCIPLES, APPLIED TO AN ORTHODONTIC DECISION

Autonomy — the patient's own concern is the outcome you will be judged against, so establish it first. **Beneficence** — the benefit of orthodontics is narrower than commonly claimed; be honest about which benefits are strong. **Non-maleficence** — this is elective treatment on a healthy dentition; the risks of demineralisation and resorption are yours to minimise. **Justice** — the allocation of a scarce, largely publicly-funded resource is why indices of need exist at all.

VIVA TRAP

"The patient is 14 and desperate for treatment but her oral hygiene is poor. What do you do?" The wrong answer is to start and hope. The examinable answer: defer active treatment, put a defined oral hygiene programme in place with review, explain in writing that appliances will not be placed until plaque control is adequate, and re-assess. Demineralisation in an elective treatment you chose to start is indefensible.

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

BENEFITS AND RISKS — WITH THE RIGHT LEVEL OF CONFIDENCE

Claimed benefit	Confidence	What can honestly be said
Dental aesthetics	STRONG	consistently improved, and it is the commonest reason patients seek treatment
Trauma reduction (large overjet)	STRONG	RR 0.58 — a 42% reduction
Plaque scores	MODERATE	small and inconsistent effect
Caries reduction	WEAK	not demonstrated; appliances increase risk during treatment
Speech	WEAK	no reliable effect
TMD prevention or cure	NONE	no evidence of causation or prevention
Airway or sleep apnoea	WEAK	extraction does not damage the airway (Ch 2, p. 91)

Orthodontics is **elective**. Stating the benefits at the confidence the evidence supports — rather than overclaiming — is itself a markable act of professionalism.

Compare Ch 2, pp. 85–105.

§9 Clinical examination and records

Curriculum: Item 11 — orthodontic clinical examination. Tested in the written paper, the diagnostic cases and the oral.

Key reading: Companion Ch 7, pp. 356–410; Proffit Ch 6; Mosby ch. on diagnosis.

Notes In Orthodontics: Chapter 7 (Examination and Diagnosis)
Supplements: A and B: examination chapters · C: Proffit Ch 6
Practise this: Question Bank Block I, Q256–285 · Viva V59–V68

IN ONE MINUTE

A fixed order, recorded reproducibly. The clinical examination is the most important diagnostic tool; everything else supplements it.

- Complaint and motivation first — they set the outcome you will be judged against
- Extra-oral before intra-oral; three planes of space, every time
- One aim of the clinical examination is to decide which records you actually need

9.1 The sequence

1. **Presenting complaint**, in the patient's own words, and what they expect treatment to achieve.
2. **History** — dental (attendance, caries, trauma, previous orthodontics, parafunction), medical, social, family (skeletal pattern, hypodontia, impacted canines), habits.
3. **Extra-oral examination**.
4. **Intra-oral examination**.
5. **Provisional problem list**, then decide what records are needed.

9.2 Extra-oral examination

Plane / feature	What to assess, and how
Position	Patient seated upright, Frankfort plane horizontal, in **natural head position**, teeth in centric occlusion, lips at rest
Anteroposterior	Skeletal Class I, II or III, assessed by palpating A and B points or by the relationship of the soft tissue profile. Class I: mandible 2–3 mm behind maxilla
Vertical	Upper face height (glabella–subnasale) vs lower face height (subnasale–menton) — should be approximately equal. **Frankfort–mandibular plane angle** ~27°; **maxillary–mandibular plane angle** ~27°. Lower anterior face height ~55% of total
Transverse	Assess from in front and from above and behind. Look for chin point deviation, occlusal cant, nasal deviation. Rule of fifths across the face
Rule of thirds	Trichion–glabella–subnasale–menton, approximately equal thirds. Lower third divides one-third upper lip, two-thirds lower lip and chin
Lips	**Competent** (meet at rest with no muscular effort), **potentially competent** (would meet but for proclined incisors), **incompetent** (do not meet at rest). Interlabial gap: >3–4 mm is the key soft tissue sign of vertical excess
Lower lip line	Where the lower lip rests relative to the upper incisor. A lip trap behind the upper incisors perpetuates a Class II division 1

Incisor show	At rest ~2–4 mm (less with age); on smiling, an average smile reveals **75–100%** of the upper incisor crown plus up to 2 mm of gingiva
Smile analysis	Smile arc parallel to the lower lip in ~85%; buccal corridors; the terminal tooth in the smile is a premolar in ~90%; midline and occlusal cant
Nasolabial angle	~90–110°. **Acute** — extraction and retraction will help. **Obtuse** — retraction will harm
TMJ	Range of opening, deviation, joint sounds, muscle tenderness, and **path of closure** from RCP to ICP

9.3 Intra-oral examination

- **Oral hygiene, gingival health, caries and restorative status** — recorded before anything else, because they determine whether treatment can be started at all.
- **Teeth present**, teeth absent, teeth of poor prognosis, tooth size and form anomalies.
- **Lower arch**: crowding or spacing quantified in millimetres; incisor inclination; curve of Spee depth; intercanine width; rotations.
- **Upper arch**: crowding or spacing; incisor inclination; canine position — **palpate the buccal sulcus from age 9–10 in every patient**; palatal vault; midline.
- **In occlusion**: incisor relationship (BSI class), overjet (mm), overbite (increased, average or reduced; complete or incomplete; traumatic or not), centre lines and their relationship to the facial midline, molar and canine relationship in halves and thirds of a unit, crossbites with or without displacement, open bites.
- **Path of closure**: an anterior or lateral displacement from RCP to ICP is one of the most important findings you can make, because it changes urgency, prognosis and sometimes the whole diagnosis (pseudo-Class III).

EXAM PHRASE TO HAVE READY

"I would record the malocclusion in three planes of space, and I would say explicitly whether there is a displacement on closing — because a crossbite with a displacement is treated on diagnosis, and a pseudo-Class III that manipulates back to edge-to-edge is a completely different prognosis from a true skeletal Class III."

9.4 The records, and the justification for each

Record	Indication	Note
Study casts / intra-oral scan	Almost every case	Permits space analysis, measurement, and a medico-legal record of the starting point
Extra- and intra-oral photographs	Almost every case	Frontal at rest, frontal smiling, profile, three-quarter; five intra-oral views
Panoramic radiograph	Where a survey of the developing dentition, pathology or third molars is needed	The default screening film in orthodontics. Not for fine detail
Periapical / occlusal	Localisation, root morphology, resorption, pathology	**Parallax** — horizontal (two periapicals) or vertical (panoramic + occlusal). "SLOB": the object that moves with the tube is lingual/palatal
Lateral cephalogram	When incisors will be moved significantly, growth will be modified, or surgery is planned	**Not justified routinely**, and specifically not justified for minor tooth movement

Posteroanterior cephalogram	Significant transverse asymmetry	Identifying the midsagittal plane is the hard part
CBCT	Impacted teeth where conventional imaging is inadequate, suspected root resorption of adjacent incisors, clefts, complex asymmetry, orthognathic planning	Highest dose. Must be individually justified; a large proportion of scans carry incidental findings
Vitality testing	Any traumatised or discoloured tooth before force is applied	Record the baseline

WRITTEN-PAPER TRAP

"No threshold dose" — ionising radiation risk is stochastic, so every exposure must be individually **justified**, **optimised** and **limited**. A radiograph taken because it would be "interesting to see" is not justified. This is easy marks and candidates routinely omit it.

§10 Imaging and cephalometrics

Curriculum: Items 12–16 — supplementary investigation and imaging modalities; lateral cephalometry and its development; 2-D analyses and clinical application; 3-D imaging in clinical orthodontics; 3-D cephalometric analyses. Tested in the written paper, the diagnostic cases and the oral.

Key reading: Companion Ch 7, pp. 411–482; Proffit Ch 6; Mosby ch. on imaging.

Notes In Orthodontics: **Topic 2** (Cephalometry: uses and limitations); Chapter 7, pp. 411–482; crash notebook *Why cephalometric numbers are not exact*

Supplements: A: Topic 2 and the imaging chapter · B: Topic 2 · C: Topic 2 (natural head position, superimposition)

Practise this: Question Bank Block J, Q286–325 · Viva V59–V68

IN ONE MINUTE

Reproducible numbers, but not exact ones. Cephalometric analysis supplements a thorough clinical examination; it does not replace it.

- Uses: diagnosis, monitoring, research, pre-surgical planning — and *not* growth prediction
- Four sources of error: reproducibility, validity, projection, superimposition
- Houston (1983): "cephalometrics is definitely not an exact science"

10.1 Technique

Cephalometric radiography images the craniofacial region in a standardised, reproducible manner. Standardisation is achieved with a **cephalostat**, which holds the mid-sagittal plane at a fixed distance from source and film so that magnification is constant and films are comparable between visits and between subjects.

- **Lateral**: mid-sagittal plane perpendicular to the beam and parallel to the film. Subject in natural head position or with Frankfort horizontal, teeth in **retruded contact position**, lips at rest.
- **Posteroanterior**: mid-sagittal plane parallel to the beam and perpendicular to the film.
- Modern solid-state cephalostats scan horizontally over an approximately 10-second exposure.

10.2 Landmarks and planes

Landmark	Definition
S (sella)	Centre of the pituitary fossa
N (nasion)	Most anterior point of the frontonasal suture
A point	Deepest concavity on the anterior maxilla between ANS and the alveolar crest
B point	Deepest concavity on the anterior mandibular symphysis
Po (porion)	Uppermost point of the external auditory meatus
Or (orbitale)	Lowest point of the orbital margin
Go (gonion)	Most posterior inferior point on the angle of the mandible
Me (menton)	Lowest point of the symphysis
Gn (gnathion)	Most anteroinferior point of the symphysis

Pog (pogonion)	Most anterior point of the symphysis
ANS / PNS	Anterior / posterior nasal spine
Ar (articulare)	Intersection of the posterior ramus and the inferior cranial base

Measurement	Caucasian norm	What it tells you
SNA	$81 \pm 3°$	Maxillary position relative to the cranial base
SNB	$78 \pm 3°$	Mandibular position relative to the cranial base
ANB	$3 \pm 2°$	Skeletal pattern. Class I 2–4°; Class II >4°; Class III <2°
Wits appraisal	Male −1 mm, female 0 mm (±2)	Perpendiculars from A and B to the functional occlusal plane. Sidesteps cranial base error altogether
MMPA	$27 \pm 4°$	Maxillary–mandibular plane angle; vertical proportion
FMPA	$27 \pm 4°$	Frankfort–mandibular plane angle
SN–MP	$32 \pm 5°$	Mandibular plane to sella–nasion
LAFH	$55 \pm 2\%$	Lower anterior face height as a proportion of total
UI to maxillary plane	$109 \pm 6°$	Upper incisor inclination
LI to mandibular plane	$93 \pm 6°$	Lower incisor inclination
Interincisal angle	$135 \pm 10°$	The determinant of overbite stability
LI to A–Pog	0 ± 2 mm	Lower incisor position; the classic stability reference
Holdaway angle	7–15°	Soft tissue profile
Nasolabial angle	90–110°	Whether incisor retraction will help or harm

TWO CORRECTIONS EXAMINERS LIKE

Eastman correction — ANB is distorted by the anteroposterior position of nasion and by the inclination of the SN plane. If SN–maxillary plane $\neq 8°$, adjust ANB by 0.5° for every degree of difference; if SNA >81°, reduce ANB by 0.5° per degree above 81, and vice versa. **Proffit's version** — record SN to true horizontal in natural head position; if it differs materially from 6°, correct every SN-based measurement by that difference.

10.3 The four limitations

Limitation	Detail
Reproducibility	Landmark identification error. Each landmark has a characteristic *non-circular* envelope of error in x and y (Baumrind & Frantz 1971a). Error is **cumulative**, because the same landmarks are reused across an analysis, and can produce artificial correlation between measurements (Baumrind & Frantz 1971b)
Validity	Points A and B represent the anterior limit of the apical bases, but no specific anatomical structure exists there — and both **remodel as the incisors move**. Frankfort plane requires porion and orbitale, both hard to locate, both bilateral, and it does not lie in the mid-sagittal plane, so it is sensitive to head positioning

Projection	A three-dimensional object rendered in two dimensions. Every machine has its own enlargement factor, which must be known before linear values are compared between machines. Landmarks not parallel to the film distort, and correction is "often ignored because of the difficulty in calculating correction values"
Superimposition	Assumes stable reference structures. The evidence base is limited and no method has been proved to deliver completely accurate results (Graf 2022); all methods carry appreciable error and none is significantly more reliable than the others

Superimposition — the three, in order

1. **Cranial base** — on SN at sella, or on Bolton–nasion; shows total change of both jaws.
2. **Maxilla** — on the anterior surface of the zygomatic process and stable palatal structures; shows maxillary dental change.
3. **Mandible** — on the internal cortical outline of the symphysis, the mandibular canal and the tooth germ crypts (Björk's stable structures); shows mandibular dental change.

Superimpose on **structures**, never on outlines.

10.4 Growth prediction — a use the evidence undermined

Ricketts' visualised treatment objective (1957, 1972) and Björk's implant studies (1969) underpinned cephalometric growth prediction, and it has since been shown to be **at best unreliable**. A radiograph cannot be justified on the basis of growth prediction alone. The constructive alternative is **serial films about a year apart**, which give a more accurate assessment of the direction of change — particularly valuable in Class III, where treatment decisions are deliberately deferred.

10.5 Three-dimensional imaging

Modality	Role and caveats
CBCT	Impacted and ectopic teeth where conventional imaging is inadequate; suspected resorption of adjacent incisor roots; clefts and alveolar bone graft planning; complex asymmetry; orthognathic planning; TAD site assessment. **Dose is far higher than a panoramic**; incidental findings are reported in a quarter to nearly all scans; **CBCT cannot image the disc or muscle — that needs MRI**
3-D facial surface imaging (stereophotogrammetry)	No ionising radiation. Soft tissue outcome assessment, serial monitoring of asymmetry (Häner 2021 — good reproducibility using forehead, mid-nose and infraorbital regions), research
Intra-oral scanning	Replaces impressions; digital study models; digital set-up; aligner and indirect bonding workflows; monitoring
3-D cephalometrics	Landmarks defined in three planes, removing projection and superimposition error in principle. In practice: no agreed normative database, landmark definitions differ from 2-D, dose is higher, and **2-D norms cannot be applied to 3-D measurements**
MRI	The disc, the retrodiscal tissues, effusion, marrow oedema. The investigation of choice for internal derangement of the TMJ

VIVA TRAP

> "Would you take a CBCT for this palatally displaced canine?" Do not answer yes reflexively. The examinable answer names the specific question the scan will answer — the relationship of the canine to the lateral incisor root, and whether that root is resorbing — states that the answer would change management (whether the lateral is retained), and confirms that conventional films have not answered it.

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

WHY CEPHALOMETRIC NUMBERS ARE NOT EXACT

The four errors

1 Projection — magnification and the two sides not superimposing. **2 Landmark identification** — the largest error; each landmark has its own envelope, and several (Ba, Or, Go, Gn) are inherently unreliable. **3 Measurement** — reference planes are themselves variable; Frankfort is hard to locate, sella–nasion easier but not anatomically meaningful. **4 Interpretation** — norms are population means with wide standard deviations, derived largely from white populations, and a single value cannot separate which jaw is at fault.

What follows from this

Cephalometry **supplements** clinical examination; it does not replace it. Report trends over serial films rather than absolute values, use the Eastman correction where the cranial base is atypical, and never plan from a single number. Growth prediction from cephalometry has been discredited.

Compare Fig. 7.60, p. 465 — the envelope of error in landmark identification.

§11 Study models, space analysis and the digital set-up

Curriculum: Items 17–18 — study model examination and analyses; digital set-up in orthodontic examination. Tested in the written paper and, decisively, in the diagnostic cases.

Key reading: Companion Ch 7 and Ch 8, Box 8.6, pp. 507–509; Proffit Ch 6.

Notes In Orthodontics: Chapter 7 (study models); Chapter 8, Box 8.6 (space analysis)
Supplements: A: the space-analysis unit · B: Topic 3 (Royal London space plan) · C: Proffit Ch 6
Practise this: Question Bank Block K, Q326–350 · Viva V69–V80

IN ONE MINUTE

Space analysis converts findings into millimetres. Anchorage follows from the number, not the other way round.

- Six areas of space requirement; two-part chart — required (negative) and available (positive)
- The difference in space requirement between the arches should reflect the molar relationship you intend to finish to
- Formal analysis gives "a more disciplined and less varied approach" — Royal London space analysis

11.1 What study casts are examined for

- Arch form and symmetry, arch width, arch length, palatal depth.
- Crowding or spacing, measured: sum of mesiodistal tooth widths minus available arch perimeter.
- Curve of Spee depth — measured from the deepest cusp tip to a plane joining the incisal edges and the distal cusp of the last molar. As a rule of thumb, **levelling a curve of Spee consumes approximately 1 mm of arch length per 1 mm of depth** (commonly taught as ~1 mm per side for a 2–3 mm curve).
- Tooth size discrepancy — **Bolton**: anterior ratio 77.2%, overall ratio 91.3%. A ratio above the norm means mandibular excess; below, maxillary excess.
- Occlusal relationships measured on the casts, and casts mounted where centric relation matters.

11.2 The space analysis — six areas

Area of requirement	How it is quantified
1. Crowding / spacing	Direct measurement; negative if space is required
2. Levelling the curve of Spee	~1 mm of arch length per 1 mm of curve depth
3. Arch width change	Expansion yields as usable space **approximately half** the amount of expansion
4. Incisor anteroposterior change	**2 mm of space per 1 mm of incisor retraction** (1 mm each side); conversely, 1 mm of proclination gains 2 mm
5. Angulation (tip) change	Uprighting distally-angulated canines consumes space

6. Inclination (torque) change	Palatal root torque of the upper labial segment consumes space — approximately **1 mm per 10°**

Source of space	Note
Tooth reduction (IPR)	Up to ~0.5 mm per contact anteriorly; **up to 1 mm per contact** in the buccal segments. Total available is modest
Extraction	The most predictable source
Space opening for a prosthesis	Adds to the requirement, not the supply
Molar distal movement	Headgear, distalising appliances, TADs. Reciprocal mesial movement of premolars is the cost
Molar mesial movement	Where space is surplus
Differential growth	Only in the grower
Residue	What is left, and where it will be

THE INTERPRETATION STEP CANDIDATES SKIP

The difference in space requirement *between the arches* should reflect the molar relationship you intend to finish to. Equal requirement in both arches finishes Class I molars. A greater requirement in the maxilla finishes Class II molars. Get this backwards and the extraction pattern is wrong before you start.

11.3 Mixed dentition space analysis

- **Moyers' probability tables** — predict the combined mesiodistal width of the unerupted canine and premolars from the measured mandibular incisors, at a chosen probability level (usually 75%).
- **Tanaka–Johnston** — half the sum of the mandibular incisor widths, plus 10.5 mm for the mandibular arch or 11.0 mm for the maxillary arch, per quadrant.
- **Radiographic methods** — direct measurement from a periapical, corrected for magnification using a tooth measurable both clinically and radiographically.
- Remember to add the leeway space to the space available.

11.4 The digital set-up

- A diagnostic **Kesling set-up** rendered digitally: teeth segmented from an intra-oral scan and repositioned to the planned finish.
- **Uses**: testing whether a plan is geometrically possible; demonstrating space requirement and the consequence of extracting versus expanding; showing the patient the likely outcome; designing indirect bonding trays and custom appliances; generating aligner staging.
- **Limitations**: it is a *geometric* prediction only. It contains no biology, no root position unless a CBCT is merged, no soft tissue response and no anchorage. It shows what the teeth would look like if they went where you asked, not whether they will.
- **Garbage in, garbage out** — the errors enter at the scan and, in surgical planning, at the bite registration.

WRITTEN-PAPER TRAP

A stem describing a beautiful digital set-up in a case with 9 mm of crowding and a thin mandibular symphysis is testing whether you know that the set-up cannot tell you the

lower incisors will be outside the alveolar envelope. The answer is the space analysis and the biological limit, not the software.

§12 Diagnosis, the problem list and treatment planning

Curriculum: Items 19–20 — classification of malocclusion and diagnosis; problem list; diagnosis and treatment planning; consideration of treatment alternatives (4 hours of the curriculum). Tested in every component of the examination.

Key reading: Companion Ch 8, pp. 487–530; Proffit Ch 6 and 7; Cobourne cases; Mosby ch. on treatment planning.

Notes In Orthodontics: **Topic 3** (Factors in formulating a treatment plan); Chapter 8; crash notebook *The treatment planning sequence*

Supplements: A: Topic 3 (the nine boxes) · B: Topic 3 · C: Topic 3 (problem-oriented sequence, soft tissue paradigm)

Practise this: Question Bank Block L, Q351–390 · Viva V69–V80

IN ONE MINUTE

A fixed sequence, not a list. Each stage constrains the next.

- Diagnosis → problem list → aims → patient factors → space analysis → anchorage → appliance → phasing → retention
- Facial aims and occlusal aims must be reconciled against the skeletal pattern
- Planning starts with the maxillary incisors within the face, not with the occlusion

12.1 The nine-step sequence

	Stage	Content
1	Diagnosis	Examination, records, and the patient's own concerns
2	Problem list	Crowding, overjet, overbite, crossbite, skeletal pattern in three planes — **ranked in order of importance to the patient**
3	Aims	**Facial** aims and **occlusal** aims, reconciled against the skeletal pattern
4	Patient factors	Age and growth, dental and periodontal health, medical history, compliance, expectations
5	Space analysis	The six areas; required versus available
6	Anchorage	Follows directly from the space analysis and the extraction pattern
7	Appliance	Aims, patient preference, operator expertise
8	Phasing	Align and level → overjet and overbite → space closure → detailing and finishing
9	Retention	Planned and **agreed** before treatment begins

THE SENTENCE TO QUOTE

"Planning begins with the position of the dentition, particularly the *maxillary incisors*, within the *face*." This is a deliberate shift from the classical approach of planning around mandibular incisor position and isolated static occlusal goals. The soft tissues define the limits of orthodontic treatment in each individual (Ackerman & Proffit; Holdaway).

12.2 The arbiter: the skeletal pattern

Situation	Route
No significant skeletal discrepancy	Occlusal correction — orthodontics alone
Growing, with a discrepancy	**Growth modification** — functional appliance, headgear, facemask, expansion. Relies on compliance and favourable growth; results are variable; **much more effective for Class II than Class III**
Non-growing, mild–moderate discrepancy	**Camouflage** — accept the skeletal pattern, position the teeth to disguise it
Non-growing, severe discrepancy, or facial aims unattainable	**Orthognathic surgery**

Camouflage limits

- **Class II** (Proffit 1992): surgery indicated if overjet >10 mm, pogonion to nasion perpendicular ≥18 mm, mandibular body length <70 mm, or anterior face height >125 mm.
- **Class III**: surgery more likely if Wits ≤ -11 mm, ANB $\leq -4°$, reverse overjet ≤ -5 mm, Holdaway angle $\leq 3.5°$, gonial angle $\leq 120°$, LAFH $\geq 66\%$, lower incisor inclination $\leq 80°$. Discriminant analysis on these predicts the need for surgery with >90% accuracy.
- **Soft tissue response**: for every 1 mm of maxillary incisor retraction the lips retract 0.7 mm and the nasolabial angle increases 1.6° (Konstantonis 2018).

12.3 The envelope of discrepancy

The range of movement achievable by orthodontics alone, by orthodontics with growth modification, and by surgery, in each direction. Three features to state:

- The envelope is **asymmetric**: more movement is possible in some directions than others — maxillary incisors can be retracted further than they can be advanced.
- Growth modification extends the envelope only in the growing patient, and only modestly.
- **Soft tissue limits are frequently tighter than hard tissue limits.** You may be able to move a tooth there; the lip may not tolerate it.

12.4 Writing the problem list

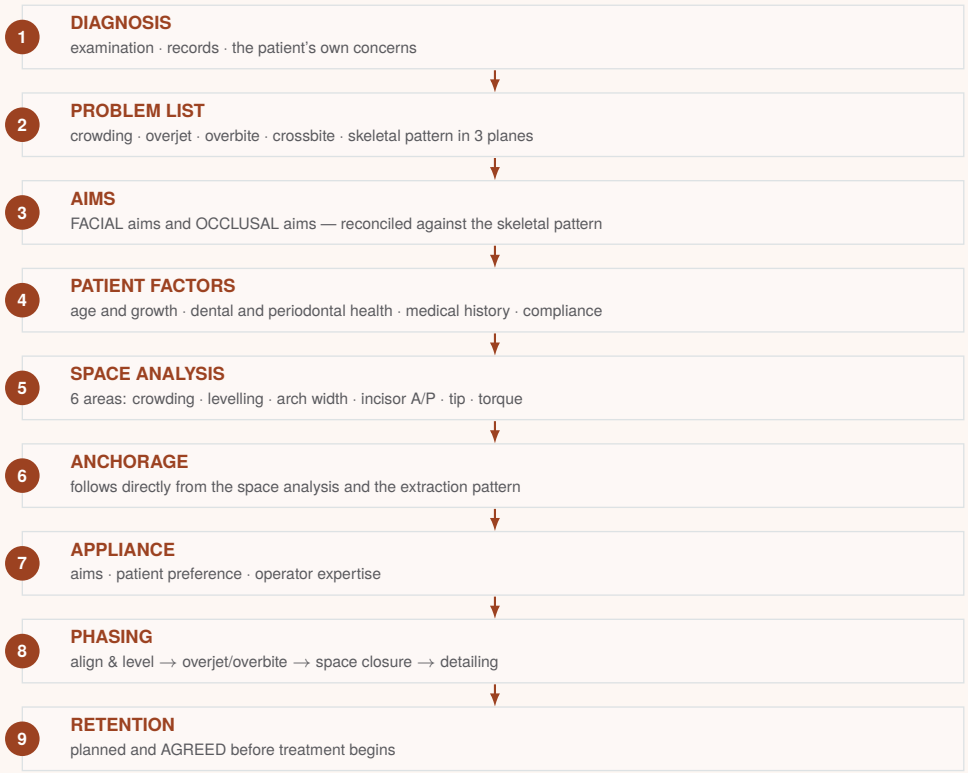
Proffit's discipline is worth adopting whole: build a **database** (interview, clinical examination, records), derive a **problem list**, **rank** it in order of importance *to the patient*, list **possible solutions** for each, then reconcile them into a single plan. The chief complaint always ranks high. "Rubber-stamp" objectives — "achieve Class I canines and a good functional occlusion" — are not objectives, because no colleague could measure you against them.

> **VIVA TRAP**
>
> The commonest failure in the diagnostic case is producing an unranked list of ten findings and no argument. Rank them, say which two you would treat if you could only treat two, and say why.

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

THE TREATMENT PLANNING SEQUENCE — EACH STAGE CONSTRAINS THE NEXT



Planning begins with the position of the dentition — particularly the MAXILLARY INCISORS — within the FACE.

The soft tissues define the limits of orthodontic treatment in each individual (Ch 8, p. 490).

Compare the envelope of discrepancy, Fig. 8.3, p. 494.

§13 Extraction decisions and space management

Curriculum: Item 20 — consideration of treatment alternatives; space creation and the choice of extraction. Tested in the written paper, the diagnostic cases and the oral.

Key reading: Companion Ch 8, pp. 505–530; Proffit Ch 7; Cobourne cases 4.x; MBT ch. 3.

Notes In Orthodontics: **Topic 4** (Factors affecting the choice of extraction); Chapter 8, pp. 505–530; crash notebook *Extraction pattern by class*

Supplements: A: Topic 4 (nasolabial angle) · B: Topic 4 (four factors) · C: Topic 4 (Proffit's thresholds) · F: Topic 3

Practise this: Question Bank Block M, Q391–430 · Viva V81–V90

IN ONE MINUTE

Extraction is one of four ways to make space, and the most predictable. Prognosis beats symmetry.

- Crowding: mild 1–4 mm · moderate 5–8 mm · severe $\geq$9 mm
- Class II $\rightarrow$ extract further *back* in the mandible; Class III $\rightarrow$ further *forward*
- Remove the compromised tooth, not the convenient one

13.1 The four ways to create space

Method	Yield	Caveat
Extraction	The full mesiodistal width of the tooth	Irreversible; anchorage demand follows
Expansion	**About half** the amount of expansion becomes usable space	Relapse; the lower arch is the template and should not be expanded
Interproximal reduction	~0.5 mm per anterior contact; up to 1 mm per buccal contact	Enamel is finite; needs good technique and fluoride
Distalisation / proclination	2 mm per 1 mm of incisor advance; molar distalisation limited	Proclination is unstable beyond the alveolar envelope; distalisation buys reciprocal mesial premolar movement

13.2 The four factors in the choice

1. **Condition of the teeth.** Prognosis beats symmetry. A compromised tooth should always be removed in preference to a healthy one, even if it sits in a less convenient place — a heavily restored or hypomineralised first permanent molar, a peg lateral, a tooth with a poor root.
2. **Site and degree of crowding.** Space is most useful where it is needed. Crowding at the front is relieved most efficiently by extracting near the front.
3. **Type of malocclusion.** Determines the differential space requirement between the arches.
4. **Age.** Well-timed extraction in the mixed dentition permits spontaneous alignment; in the permanent dentition it does not, and appliances become obligatory.

13.3 Extraction pattern by class

Class	Pattern	Rationale
Class I	Corresponding premolar in both arches (usually first premolars)	Maintains the Class I buccal segment relationship; equal space requirement
Class II	Extract further **back** in the mandible — maxillary first premolars with mandibular second premolars	Greater maxillary space requirement to retract the upper labial segment; the mandibular molars are allowed to come forward
Class III	Extract further **forward** in the mandible — mandibular first premolars with maxillary second premolars	Greater mandibular space requirement to retract the lower labial segment

WRITTEN-PAPER TRAP

Reversing the Class II and Class III patterns is one of the twelve most costly slips. The rule: **put the space where the movement is needed.** In Class II you need to move the upper teeth back, so put the upper space at the front (first premolars) and the lower space further back (second premolars).

13.4 Which tooth, and when

Tooth	When it is the right choice
First premolar	Anterior crowding, a large overjet, or where maximum anterior space is needed. Best spontaneous improvement, especially in the mixed dentition
Second premolar	Mild to moderate crowding sited more posteriorly; where anchorage should favour the anterior segment; where the second premolar is absent, hypoplastic or displaced. Eases Class II mechanics in the lower arch
First permanent molar	Only when of poor prognosis. Time it to the calcification of the second molar bifurcation (~8–10 y dental age). Better spontaneous closure in the mandible
Second permanent molar	To relieve crowding at the back, to ease Class II mechanics, or to allow the third molar to erupt. Requires a third molar present, of good size, correctly angulated ($\leq 30^\circ$ to the second molar long axis) and at the right stage of development
Lower incisor	Rarely — severe lower labial segment crowding, a lower incisor of poor prognosis, mild Class III camouflage in a non-grower. Costs: tooth-size discrepancy, residual spacing, deepened overbite
Third molar	Not prophylactically; NICE advises against removal to prevent late lower incisor crowding
Primary canine	Interceptive, for a palatally displaced permanent canine, or to relieve incisor crowding (with the balancing extraction considered)

13.5 Balancing and compensating

- **Balancing** — extraction of the contralateral tooth in the same arch, to prevent a centre-line shift. Most relevant for **canines** and, to a lesser extent, incisors. Rarely needed for molars.
- **Compensating** — extraction of the opposing tooth in the same quadrant, to preserve the buccal occlusion and prevent over-eruption. Considered for first permanent molars.
- Neither is automatic. Both are decisions justified by the centre line and the occlusion in the individual case.

13.6 The extraction debate — the evidence

Claim	The evidence
Extraction flattens the profile	Soft tissue differences between extraction and non-extraction groups are minimal; both are perceived to benefit (Rushing 1995; Paquette 1992; Iared 2017). Lip retraction must be considerable before a profile is judged unattractive (Bowman & Johnston 2000)
Extraction widens the buccal corridors and spoils the smile	It does not (Meyer 2014; Christou 2019; Cheng 2016)
Extraction narrows the airway and causes sleep apnoea	No evidence; described as "scaremongering and marketing" (Larsen 2015; Orabi 2021)
Extraction causes TMD	No evidence of causation (Behrents & White 1992)
Proffit's thresholds	<4 mm rarely extract · 5–9 mm a judgement · ≥10 mm almost always extract
Arch width indices (Pont)	Not acceptable predictors of arch width

EXAM PHRASE TO HAVE READY

"I don't regard extraction as damaging in itself. The comparative evidence shows very little facial difference between comparable cases treated either way, and none for the airway claims. What decides it for me is the space requirement in millimetres, the soft tissue profile — particularly the nasolabial angle — and the prognosis of the individual teeth."

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

EXTRACTION PATTERN BY CLASS OF MALOCCLUSION



Rust = extraction site. **Rationale:** put the space where the movement is needed, and let the differential in space requirement between the arches reflect the molar relationship you intend to finish to.

Compare Figs 8.10–8.12, pp. 517–518.

§14 Biomechanics, force control and anchorage

Curriculum: Items 21–22 — biomechanical basis of orthodontic therapy; mechanical principles in orthodontic force control (3 hours of the curriculum). Tested in the written paper and heavily in the oral.

Key reading: Companion Ch 6, pp. 307–351 and Ch 11; Nanda ch. 1 and 2; Proffit Ch 9; MBT ch. 1.

Notes In Orthodontics: **Topics 5 and 6**; Chapters 6 and 11; crash notebook *Anchorage demand*, *Headgear*, *Miniscrew failure by site*

Supplements: A: Topic 5 (TADs) · B: Topic 5 · C: Topic 5 (the anchorage ladder) · E: Topic 2 · G: Topics on force systems and miniplates

Practise this: Question Bank Block N, Q431–481 · Viva V91–V102

IN ONE MINUTE

You cannot apply a force at the centre of resistance. Every appliance is a workaround, and the moment-to-force ratio is how you control the result.

- Centre of resistance is fixed by the anatomy; centre of rotation moves with the force system
- Anchorage is Newton's third law. Unsupported, about two-thirds of the space created is lost
- Skeletal anchorage changes the biological response, not the force system

14.1 The definitions

Term	Definition
Force	A vector: magnitude, direction, point of application, line of action. Measured in cN or g
Centre of resistance (C_{res})	The point through which a single force produces pure translation. For a single-rooted tooth, **one-third to one-half down the root** from the alveolar crest (commonly quoted as $\frac{1}{3}$–$\frac{1}{4}$ of root length apical to the CEJ); for a multirooted tooth, **1–2 mm apical to the furcation**
Centre of rotation (C_{rot})	The point about which the tooth actually rotates. It *moves* with the force system
Moment	Force × perpendicular distance to C_{res}. Units g mm or cN mm. The greater the distance, the greater the rotation
Couple	Two equal, opposite, parallel, non-collinear forces. **No net force**; produces pure rotation about C_{res} *wherever it is applied*
Moment-to-force ratio (M/F)	The single number that decides the type of movement

14.2 The fundamental problem, and its four solutions

C_{res} is inside bone and inaccessible. Force must be applied at the crown, so a moment is *always* produced. Varying the M/F ratio is how the movement is controlled.

Movement	M/F ratio	Where C_{rot} sits, and what it looks like

Uncontrolled tipping	0	Near C_{res}. Crown and root move in opposite directions. What a removable appliance does
Controlled tipping	~7:1	Near the apex. Crown moves, root is held
Bodily movement (translation)	~10:1	At infinity. The force acts effectively through C_{res}
Root torque (root movement)	~12:1 and above	At the incisal edge. The root moves further than the crown

THE CONSEQUENCE YOU MUST BE ABLE TO STATE

Periodontal attachment loss moves C_{res} **apically**. The same force applied at the same bracket now acts at a greater distance from C_{res}, so it produces a *larger* moment and more tipping. This is why the reduced periodontium changes the mechanics, not merely the risk: lighter forces, higher M/F ratios, and often a different appliance altogether.

14.3 Determinate and indeterminate systems

- **Equilibrium**: the sum of forces = 0 and the sum of moments = 0. Every force you apply produces an equal and opposite reaction somewhere. The only question is where you put it.
- **One-couple systems** (a cantilever or utility arch engaged in one bracket and tied as a point contact at the other end) are **statically determinate**: the force system can be calculated and measured.
- **Two-couple systems** (a continuous archwire engaged in two or more brackets) are **statically indeterminate**: the force system cannot be calculated, only estimated. Most of clinical orthodontics is indeterminate, which is why experience matters.

14.4 Anchorage

Classification	Meaning
Minimum (Group C)	The anchorage unit may move mesially freely; up to two-thirds of the space is used by the posterior segment
Moderate (Group B)	Both units move similar amounts; roughly half the space each
Maximum (Group A)	No, or almost no, mesial movement of the anchorage unit; the space is used entirely for anterior retraction
Reciprocal	Equal and opposite movement of two units is acceptable and intended — midline diastema closure, expansion, crossbite correction with a fixed appliance
Simple / stationary	Simple: tipping against tipping. Stationary: the anchor unit is being moved bodily while the active unit tips — a more efficient use of the same absolute anchorage
Intramaxillary / intermaxillary	Within one arch; or between arches (Class II and Class III elastics)
Extraoral	Headgear
Skeletal	Miniscrews, miniplates, palatal implants, onplants

What determines how much anchorage a unit has

- **Root surface area** — Jepsen's measurements of the permanent dentition are the basis of relative anchorage values. Number of teeth matters less than the summed root area.

- **Root form and length**, bone quality, and whether roots lie in cortical bone (cortical anchorage).
- **Type of movement demanded** of the anchor unit — bodily movement resists more than tipping.
- **Force magnitude** — a heavier force is more likely to exceed the anchor unit's threshold and move it too.
- **Occlusal interdigitation** and the presence of the second molars in the system.

Proffit's anchorage ladder — learn the ratios

Arrangement	Anterior retraction : posterior protraction
Reciprocal closure without reinforcement	≈ 1 : 1, determined by PDL area, not tooth number
Stabilising lingual or transpalatal arch	≈ 2 : 1
Headgear plus interarch elastics	3 : 1 or 4 : 1
Skeletal anchorage	Effectively absolute

The evidence on anchorage aids

Aid	What the evidence shows
Unsupported anchorage	**Around two-thirds of the space created is lost** (Haque 2017). Anchorage loss is the same for first and second premolar extraction
Transpalatal arch	Provides very little additional anchorage in the sagittal plane (Diar-Bakirly 2017)
Lingual arch	Provides little anchorage support and can procline the lower incisors (Rebellato 1997)
Anchor teeth in the initial aligning wire	In maximum anchorage cases, anchor teeth should *not* be subject to initial aligning forces (Melsen & Verna 1999)
The maxillary arch	Particularly susceptible to anchorage loss (Giannopoulou 2016)
Lacebacks	Without them the lower incisor advances 1.4 mm during levelling; with them it retracts 1.0 mm (McLaughlin & Bennett). Cochrane evidence is weaker than the popularity of the technique

14.5 Headgear

Direction of pull	Vertical effect	Indicated when
High pull / occipital	Intrusive on the molar	Increased vertical proportions, high angle, anterior open bite tendency
Straight pull / combination	Neutral	Average vertical proportions; pure molar distalisation
Cervical / low pull	Extrusive on the molar	Reduced vertical proportions, low angle, deep bite

Purpose	Duration	Force per side
Anchorage support	10–12 h/day	250–350 g

Distalisation / orthopaedic effect	14 h/day	450–500 g

HEADGEAR SAFETY — NEVER OMIT THIS

Two independent safety mechanisms are mandatory: a self-releasing (snap-away) neck strap and a locking facebow. Two injury mechanisms were described by Samuels (1996): the **catapult** injury, where the bow disengages under tension and recoils into the face, and **nocturnal disengagement**, where the bow comes out during sleep. Consequences include facial and ocular laceration, penetrating eye injury with loss of vision, and intra-orbital infection. Warn the patient in writing, record the warning, and instruct them to stop wearing it and attend if the bow ever comes out.

14.6 Skeletal anchorage

Item	Detail
Principle	A TAD changes the *biological response* to the force system, not the force system itself. Newton's third law still applies; the reaction is simply placed in bone, where nothing moves
Direct anchorage	The active force acts on or through the screw. **The line of force must pass through the screw**
Indirect anchorage	The screw stabilises an anchor tooth or unit; ligate to it with a *large rectangular stainless steel* wire, rigid enough not to flex
Insertion torque	**5–10 Ncm**. Too low → poor primary stability. Too high → microfracture and poor secondary stability
Dimensions	Higher failure below **8 mm** length and **1.2 mm** diameter
Principal cause of failure	**Root contact**. Also thin cortical bone, mobile mucosa, inflammation, immediate heavy loading
Cortical bone	TAD success is governed by cortical bone thickness
Palatal implants (Nanda)	Minimum 4 mm of bone thickness needed; the hard palate generally does not exceed 3 mm outside the paramedian region; osseointegration period 3 months unloaded
Miniplates (Nanda)	Sit outside the dentition so they obstruct nothing; need 2–3 mm of cortical bone depth; load at about **three weeks**, not immediately

Miniscrew failure by site — learn the extremes

Site	Failure rate
Median palate	1.3%
Paramedian palate	4.8%
Maxillary buccal interradicular	7.5%
Overall (pooled)	**12.3%**
Maxillary between first and second molars	25.9%
Zygomatic buttress	28.1%
Mandibular buccal shelf	65.0%

EXAM PHRASE TO HAVE READY

"Placement in the palate is not restricted by the dentition, which is why it is by far the most reliable

site — around 1% failure in the median palate against 65% on the mandibular buccal shelf. If I were placing a screw as a general dentist with an interest in orthodontics, that is where I would want it, and I would be planning the case so that root contact was designed out rather than avoided by luck."

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

THE MECHANICS IN ONE PAGE

Centre of resistance

1/3–1/2 down the root (single-rooted)

1–2 mm apical to the furcation (multirooted)

Moment

force × perpendicular distance to CRes

greater distance = greater rotation

Couple

two equal, opposite forces

NO net force; acts about CRes wherever applied

Centre of rotation

force nearer CRes → CRot further away

force THROUGH CRes → CRot at infinity = bodily

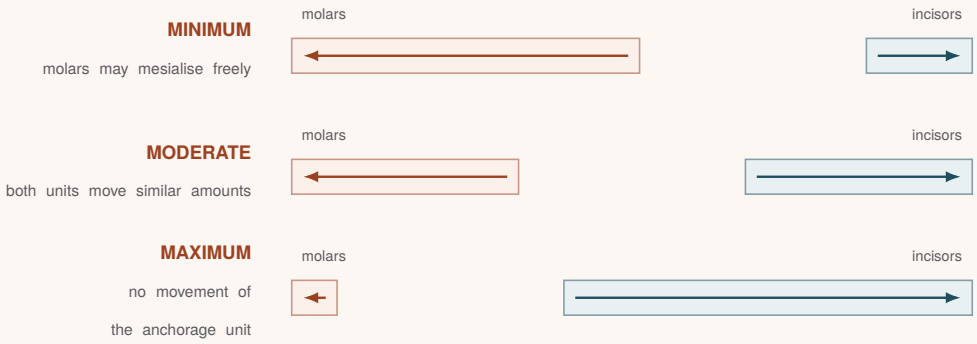
The fundamental problem

CRes is not accessible — force must be applied at the crown, so a moment is **always** produced. **Varying the moment-to-force ratio is how you control the type of movement.**

Periodontal bone loss moves CRes apically = same force, bigger moment, more tipping. Ch 6, pp. 307–312.

Compare Figs 6.11–6.15, pp. 310–314.

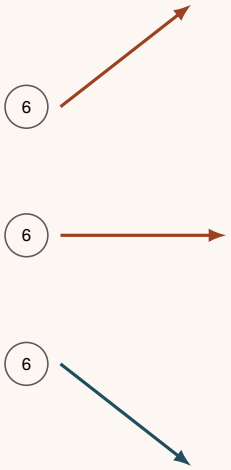
ANCHORAGE DEMAND



Reciprocal anchorage = equal movement of both units is acceptable (expansion, midline diastema). **Unless anchorage is supported, around two-thirds of the space created is lost** (Haque et al., 2017).

Compare Fig. 6.35, p. 339.

HEADGEAR — DIRECTION OF PULL



HIGH / occipital

increased vertical proportions — intrusive

STRAIGHT / combi

normal proportions — molar distalisation

LOW / cervical

reduced vertical proportions — extrusive

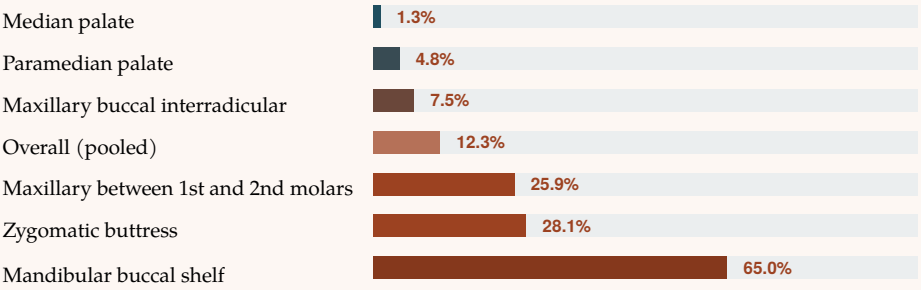
Force and duration

Anchorage support	10–12 h/day	250–350 g per side
Distalisation / orthopaedic	14 h/day	450–500 g per side

Two independent safety devices are mandatory: a self-releasing strap and a locking facebow.

Compare Figs 6.42–6.43, pp. 346–347; injury Fig. 6.45, p. 349.

MINISCREW FAILURE RATE BY SITE



Placement in the palate is not restricted by the dentition — which is why it is by far the most reliable site. **Insertion torque 5–10 Ncm**: too low gives poor primary stability, too high causes microfracture and poor secondary stability. Failure is higher below 8 mm length and 1.2 mm diameter, and root contact is a major cause.

Compare Fig. 11.5, p. 716.

§15 Orthodontic materials, wires, bonding and instruments

Curriculum: Items 23, 33–34 — mechanical properties and orthodontic materials; bonding agents, orthodontic wires and sequencing, instruments, expanders, intra-oral laser applications; dental impression materials and intra-oral scanning. Tested in the written paper.

Key reading: Companion Ch 10, pp. 645–665; Proffit Ch 9; Mosby ch. on materials; MBT ch. 6.

Notes In Orthodontics: Topic 7 (The ideal archwire; stainless steel versus nickel titanium); Chapter 10, pp. 645–665

Supplements: A: Topic 7 (etching, modulus) · B: Topic 7 · C: Topic 7 (strength = stiffness × range) · F: the archwire-sequence chapter

Practise this: Question Bank Block O, Q482–525 · Viva V103–V111

IN ONE MINUTE

No wire has every property, and the properties you need change through treatment. That is the whole reason a sequence exists.

- The ideal wire: high strength, low stiffness, high range, high formability — plus biocompatibility, weldability, low friction and cost
- **Strength = stiffness × range**. You cannot raise one without paying elsewhere
- Nickel titanium modulus is ∼0.26 that of stainless steel; beta titanium $\sim\frac{1}{3}$

15.1 The mechanical properties, defined on the stress–strain curve

Property	Definition
Stiffness	The gradient of the elastic portion — Young's modulus. **Low stiffness is desirable** in an aligning wire
Strength	The maximum load the wire can bear. **Yield strength** is the stress at 0.1% permanent deformation
Range	The distance the wire can be deflected without permanent deformation
Springback	Yield strength ÷ modulus of elasticity — effectively the working range
Resilience	Area under the elastic portion of the curve; the energy stored
Formability	The amount of permanent deformation the wire tolerates before fracture — how far you can bend it
Not on the curve	Biocompatibility, environmental stability, joinability (weldability/solderability), friction, ductility, cost

15.2 The wires

Wire	Composition / properties	Where it is used
Stainless steel	"18–8" austenitic: Fe ∼70%, Cr 17–20%, Ni 8–12%, C max 0.08%. High stiffness, high strength, excellent formability, low friction, cheap, weldable	Working and finishing wires; arch form; sliding mechanics; the reference against which everything else is compared

Multistrand / coaxial steel	Several thin strands; low stiffness with steel's other virtues	Initial alignment; bonded retainers. Coaxial superelastic NiTi produced greater movement over 12 weeks than single strand (Wang 2018, Cochrane)
Nickel titanium	~55% Ni, 45% Ti. Low stiffness, very large range, shape memory (thermoelastic) and superelasticity (a load-deflection plateau with hysteresis)	Initial alignment. **Cannot be bent, welded or soldered.** Austenitic-active slightly better than martensitic-stabilised; copper NiTi slightly slower to a working wire and more painful (Papageorgiou 2014)
Beta titanium (TMA)	Ti–Mo. Stiffness $\approx \frac{1}{3}$ of stainless steel; good formability; weldable; higher friction	Intermediate and finishing wires; auxiliaries, intrusion arches, loops
Cobalt–chromium (Elgiloy)	Supplied soft, formable, then heat-treated to raise stiffness	Where complex bending is needed before hardening
Aesthetic (composite, coated)	Poor mechanical properties; coating degrades	Limited role

Stage	Typical wire
Alignment	0.012–0.018 in round NiTi (or multistrand)
Levelling	0.016–0.020 in round stainless steel
Working	0.019×0.025 in stainless steel in a 0.022 slot — the MBT working wire, giving overbite control and free sliding
Finishing	0.017×0.025 in an 0.018 slot; 0.021×0.025 in a 0.022 slot; or beta titanium for detailing bends

TWO NUMBERS FROM THE MBT LITERATURE

A 0.016×0.022 wire deflects **47% more** than 0.019×0.025 in the same slot. Popular superelastic NiTi archwires expand the lower intercanine width by up to **5.9 mm** if allowed to express fully — which is why arch form must be a deliberate decision, and why buccal uprighting relapses in about 70% of cases.

15.3 Bonding

Step	Detail
Clean	Pumice prophylaxis, no fluoride
Etch	**37% phosphoric acid, 15–30 seconds**; rinse and dry to a frosted appearance. Self-etching primers combine etch and prime, are faster, produce a shallower etch pattern, and give comparable clinical bond failure rates
Prime and bond	Resin, light- or chemically-cured. Bond strength required is modest — it must be strong enough to resist function and weak enough to allow debonding without enamel damage
Bands	Where bonding is unreliable — molars carrying headgear, transpalatal arches, quadhelices, RME; heavily restored teeth. Cemented with glass ionomer, which releases fluoride
Ceramic brackets	Bond by **silane** coupling (chemical) or mechanical retention. **They bond too well**: the risk at debond is enamel fracture. More brittle, higher friction, and they abrade opposing enamel
Enamel loss	Debonding and clean-up removes a small but real thickness of enamel. Use a tungsten carbide bur at low speed, then polish

White spot lesions	Up to **half** of fixed appliance patients; visible within **6 months**; commonest on first molars, upper lateral incisors and lower canines; more prevalent in males
Prevention	Daily high-fluoride toothpaste (2800 or 5000 ppm) plus professional high-concentration fluoride foam or varnish every **6–8 weeks** (Benson 2019); glass ionomer cement for bands; steel ligatures rather than elastomeric modules in a poor brusher

15.4 Impression materials and scanning

- **Alginate** (irreversible hydrocolloid): cheap, well tolerated, but dimensionally unstable — cast within 10 minutes or store in 100% humidity. Adequate for study models.
- **Addition silicone (PVS)**: dimensionally stable, accurate, more expensive; used where accuracy matters — bonded retainers, appliances, indirect bonding.
- **Intra-oral scanning**: no material, better tolerated, immediately transmissible, permits digital storage and set-up, and avoids the pouring and storage of casts. Accuracy is comparable to PVS for full arches in most studies; deep undercuts, subgingival margins and edentulous spans remain harder.

15.5 Expanders, auxiliaries and lasers

Device	Note
Removable expansion plate	Midline screw, 0.25 mm per turn, **twice weekly**. Slow expansion, largely dental
Quadhelix	Fixed, adjustable, delivers a light continuous force; **about 3 months faster than a plate** and preferred on Cochrane evidence in the 7–11 year group
Rapid maxillary expander (hyrax/haas)	**1–4 turns per day** (0.25–1 mm/day) over 2–4 weeks; forces up to **10 kg**; up to 4 mm at the suture ≈ 50% of screw expansion; retain in situ 3–6 months
Transpalatal arch	Anchorage, molar derotation, expansion or constriction; little sagittal anchorage value
Lingual arch	Space maintenance and leeway preservation; little anchorage value
Nance button	Palatal anchorage; risk of ulceration under the acrylic button
Diode laser (810–980 nm)	Absorbed by melanin and haemoglobin; cuts soft tissue and spares enamel and metal. Exposure of partially erupted teeth, gingival recontouring, frenectomy, operculectomy. No sutures, no dressings, controlled bleeding. **A finishing instrument** — diagnose the cause of the gingival disproportion before reaching for it

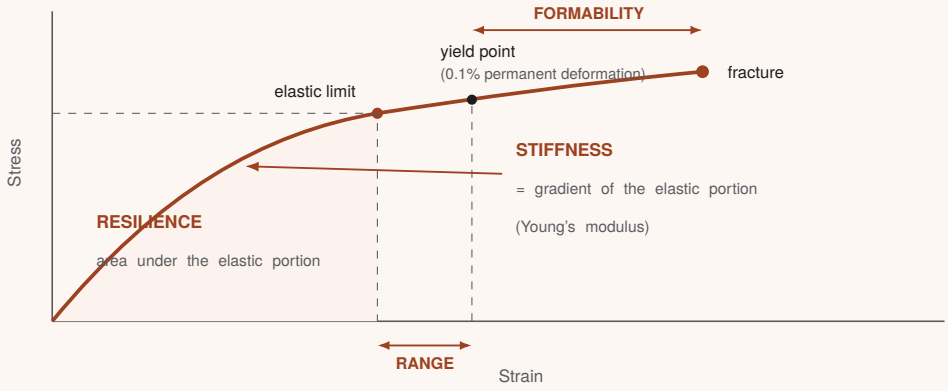
15.6 Nickel and latex

- **Nickel allergy**: approximately **10–30% of females** and **1–3% of males**. Type IV delayed hypersensitivity. Intra-oral reactions are rare; extra-oral (headgear, facial) reactions are commoner.
- **A history of nickel allergy does not contraindicate orthodontics.** Stainless steel is safe — the nickel is bound within the austenitic lattice. **Avoid nickel titanium**; use steel, beta titanium or fibre-reinforced alternatives, and avoid extra-oral nickel-containing components.
- **Latex**: prevalence ∼4.3% in the general population, 9.7% in healthcare workers. Use latex-free gloves, dam and elastics.

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

ARCHWIRE PROPERTIES — THE STRESS–STRAIN CURVE



Springback = yield strength ÷ modulus of elasticity — effectively the working range. Not on the curve: biocompatibility · environmental stability · joinability · friction · ductility.

Compare Box 10.2, p. 652; Fig. 10.28, p. 654.

§16 Removable and functional appliances

Curriculum: Items 24, 28–29 — myofunctional therapy; design of orthodontic appliances and their mechanical characteristics; functional appliances. Tested in the written paper and the oral.

Key reading: Companion Ch 9; Proffit Ch 10 and 11; Cobourne cases 8.x; Mosby ch. on growth modification.

Notes In Orthodontics: **Topic 8** (Functional appliances); Chapter 9; crash notebook *What a functional appliance actually does*
Supplements: A: Topic 8 · B: Topic 8 · C: Topic 8 (the three trials) · G: Topic on growth modification
Practise this: Question Bank Block P, Q526–566 · Viva V112–V121

IN ONE MINUTE

They posture the mandible forwards. Most of what follows is dentoalveolar.

- Skeletal gain is a few millimetres at best, and essentially transient
- The sophisticated explanation: they *unlock* the occlusion and let normal mandibular growth express itself
- Best timed to the circumpubertal period — CVM stage CS3

16.1 Removable appliances — design in four parts

Design every removable appliance under **ARAB**: **A**ctive components, **R**etention, **A**nchorage, **B**aseplate.

Component	Detail
Active	Springs (palatal finger 0.5 mm, buccal canine retractor 0.7 mm, Z-spring 0.5 mm, T-spring), screws (0.25 mm per quarter turn), bows (Roberts retractor 0.5 mm in tubing), elastics
Retention	Adams clasp on molars (0.7 mm) and premolars (0.6 mm); Southend clasp on incisors; ball-ended clasps; labial bow contributes
Anchorage	Derived from the baseplate, the clasped teeth and the palatal coverage
Baseplate	Self-cure or heat-cure acrylic. May incorporate a flat anterior bite plane (overbite reduction), a posterior bite plane (crossbite correction), or buccal capping

- Removable appliances deliver **tipping** movements only (uncontrolled tipping), because they achieve only **two-point contact** on a tooth. Bodily movement, rotation and torque are beyond them.
- Their two fatal limitations are **compliance** and **two-point contact**.
- Force from a spring: 25–40 g for tipping a single tooth. Activate to about half the spring's diameter for a palatal finger spring.

16.2 Functional appliances — what they actually do

Effect	Magnitude and character

Skeletal	A few millimetres at best; **essentially transient**. A "headgear effect" restraining the maxilla is present in most functional appliance treatment
Dentoalveolar	The dominant effect. Upper incisors retrocline, lower incisors procline, buccal segment eruption is guided
Soft tissue	The lip trap is eliminated; lip competence is gained. Functional appliance therapy has a positive effect on the soft tissue profile in Class II subjects (Koretsi 2015)
Muscular	Voudouris (2003), using implanted electrodes, showed a *reduction* in lateral pterygoid activity on protrusion — contradicting earlier surface EMG work

THE SOPHISTICATED EXPLANATION

A Class II relationship persists in part because the occlusion is *locked* by intercuspation. The appliance unlocks it and permits normal mandibular growth to express itself, alongside genuine dentoalveolar change. This reads far better than "it grows the mandible".

16.3 The appliances

Appliance	Type	Features
Andresen activator	Tooth-borne, passive, one piece	The original. Bulky, worn at night, wholly compliance-dependent
Harvold activator	Tooth-borne, passive	Larger construction bite opening
Bionator	Tooth-borne, passive	Less bulky, can be worn more of the day
Frankel (FR-2, FR-3)	Tissue-borne	Buccal shields and lip pads hold the soft tissues away from the dentition, allowing transverse and sagittal dentoalveolar development. Fragile
Twin block (Clark)	Tooth-borne, active, two pieces	**Bite blocks at ∼70°** to the occlusal plane. Two-piece design means it can be worn full time and eaten in; the most widely used. Posterior open bites during treatment are expected and are managed by trimming the blocks
Medium opening activator	Tooth-borne	
Herbst	**Fixed** functional	Bilateral telescopic rod-and-tube. Not compliance-dependent. Overjet reduction in **6–8 months**
Twin Force / Forsus and similar	Fixed, semi-rigid	Attach to a fixed appliance; used mid-treatment
Headgear-activator combinations		For vertical control in the high-angle Class II

16.4 The evidence — the numbers to quote

Study	Finding
Tulloch (1997, 2004); Keeling (1998); Dolce (2007)	The Florida and North Carolina trials: effective early overjet reduction, wide individual variation in skeletal effect, no long-term difference

O'Brien (2003)	Early twin block reduces the overjet *dentoalveolarly* and gives a transient increase in self-esteem
O'Brien (2009)	**No long-term difference.** Early-treated children had more attendances and longer treatment; the late-treated group had a slightly better final occlusal result. Early treatment is not normally justified
Cobourne (2022); Kallunki (2021)	Relative risk of new incisor trauma with early overjet reduction **RR 0.58 — a 42% reduction** — but with wide confidence intervals; headgear was marginally more effective than a functional appliance
Pancherz (1982)	Herbst reduces the overjet in 6–8 months
Pancherz (2014)	At **32 years**, 64% of molar correction and 86% of overjet correction remained stable; recommend at least 2 years of activator retention
Pacha (2023, 2024)	Randomised trial: **37% failure with twin block versus 17% with Herbst**; twin block 1.5 months longer; smaller mean overjet reduction (**5.8 vs 7.1 mm**); but more visits and emergencies with the Herbst
Santana (2020)	Little clinical difference between maximal and incremental mandibular advancement

16.5 Myofunctional therapy

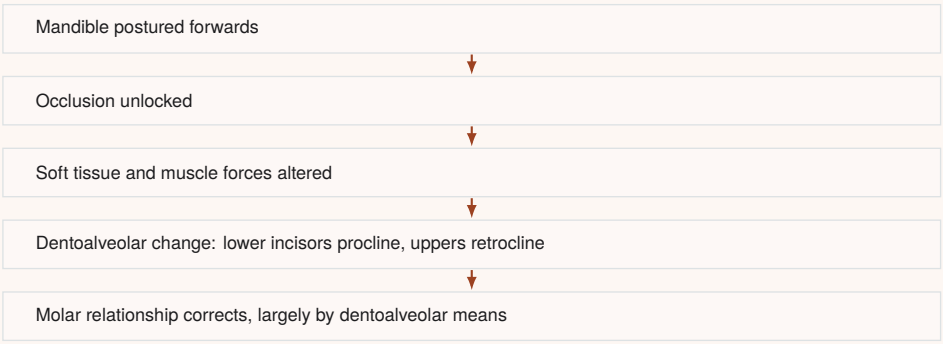
- Exercises intended to retrain lip, tongue and masticatory muscle behaviour: lip seal exercises, tongue posture training, habit interception.
- **The evidence is weak.** There is reasonable evidence that removing a digit habit before the permanent incisors are established allows spontaneous correction of an anterior open bite, and reasonable evidence that a lip trap perpetuates a Class II division 1. There is not good evidence that muscle exercises alone correct a malocclusion.
- **Habit interception**: positive reinforcement and reminder therapy first; a habit-breaking appliance (palatal crib or rake) if that fails, and only in a motivated child. Warn that the appliance is a reminder, not a punishment.
- Say clearly: "an anterior open bite from a digit habit will usually close spontaneously if the habit stops before the permanent incisors are in occlusion; a *skeletal* anterior open bite will not."

> **WRITTEN-PAPER TRAP**
>
> "A functional appliance grows the mandible" → overclaiming. The examinable statement is that any extra growth is "a matter of only a few millimetres at best, and these effects are essentially transient", and that the mechanism is probably unlocking the occlusion plus dentoalveolar change.

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

WHAT A FUNCTIONAL APPLIANCE ACTUALLY DOES



The sentence to write

Any extra mandibular growth is "a matter of only a few millimetres at best, and these effects are essentially transient" (Ch 9, p. 560). The likely mechanism is that the appliance *unlocks the occlusion* and permits normal mandibular growth to express itself, combined with dentoalveolar change (Ch 9, p. 562).

Compare Ch 9, pp. 560–562.

§17 Fixed appliances — labial, lingual and aligners

Curriculum: Items 6, 30–32, 61–62 — first, second and third order movements and the evolution of fixed appliances; fixed labial appliances; fixed lingual appliances; "invisible" aligners; invisible orthodontics; individualised lingual orthodontics (7 hours of the curriculum). Tested in the written paper and the oral.

Key reading: Companion Ch 10 and Ch 14; Proffit Ch 10; MBT ch. 4 and 5; Bennett & McLaughlin.

Notes In Orthodontics: **Topic 6** (The three orders and the evolution of fixed appliances); Chapters 10 and 14; crash notebook *Clear aligners versus fixed appliances*

Supplements: A: Topic 6 and the bracket chapter · B: Topic 6 and the aligner unit · C: Topic 6 (Angle's four appliances) · E and F: the MBT chapters

Practise this: Question Bank Block Q, Q567–606 · Viva V112–V121

IN ONE MINUTE

The three orders have not changed since Angle. Only their delivery has.

- First order = in–out · second order = tip (angulation) · third order = torque (inclination)
- Standard edgewise builds all three into the *wire*; pre-adjusted edgewise builds them into the *bracket*
- Pre-adjustment moved the error from the wire to the *bracket position* — which is now where cases are lost

17.1 The three orders

Order	Movement	Built into	Clinical expression
First	In–out	Slot-to-base distance	Arch form and crown thickness compensation
Second	Tip (angulation)	Mesiodistal slot angulation	Canine and premolar root angulation; parallel roots at extraction sites
Third	Torque (inclination)	Buccolingual slot inclination	Incisor root position; buccal root torque of upper molars

17.2 The evolution — Angle to MBT

Appliance	Idea, and why it was superseded
E-arch (Angle)	A heavy labial arch with ligatures. Tipping only
Pin and tube	Individually soldered pins. Full control in theory, unworkable in practice
Ribbon arch	Vertical slot; excellent alignment, poor root control
Edgewise (Angle, 1928)	The slot turned **horizontal**, giving greater control of tooth movement. All three orders bent into the wire, tooth by tooth. Results traditionally left 10–20% undertreated
Begg (1956)	Ribbon arch slot inverted; **differential force**; three stages; deliberate tipping then uprighting with auxiliary springs. An anchorage philosophy

Tip-Edge (Kesling, 1988)	Limited tipping, then full prescription expressed in rectangular wire at finishing
Pre-adjusted edgewise (Andrews, 1972, 1979)	The **straight-wire appliance**: tip, torque and in–out built into each bracket, derived from 120 non-orthodontic normals. The prescription cannot fix a misplaced bracket
Roth	One appliance rather than an inventory; over-corrected values built in
MBT (McLaughlin, Bennett & Trevisi, 2001)	Proven mechanics determined the bracket design, not the reverse. Reduced anterior tip; more palatal root torque on upper incisors; three canine torque options; light continuous force; lacebacks, bendbacks and tiebacks

SLOT SIZE

0.018 in and 0.022 in are both in wide use. **There is no evidence that either is superior.** The 0.018 slot expresses torque earlier in a smaller wire; the 0.022 slot allows more robust sliding mechanics on a 0.019×0.025 working wire. Finishing wires: 0.017×0.025 in an 0.018 slot, 0.021×0.025 in an 0.022 slot. The MBT principle is to *vary the stiffness by changing material, not dimension*.

17.3 Self-ligation and customisation — the evidence

Claim	Evidence
Self-ligation is faster	**No difference** in rate of movement, treatment time, number of visits or occlusal outcome (Scott 2008; Fleming 2009, 2010; DiBiase 2011; Johansson & Lundström 2012; Pringle 2009). Pain benefit marginal; expansion claims unsupported
Active vs passive clips	Active clips store energy against the wire; passive slides do not. A design difference, not a clinical advantage
Customised appliances (Insignia and similar)	Randomised trial: comparable outcomes, **no reduction in treatment time**, more breakages, more complaints and more planning time. Treatment time was dictated by case difficulty and by the treating orthodontist (Penning 2017)
Archwire superiority	Insufficient evidence that any archwire material is superior, apart from coaxial superelastic NiTi over single-strand at 12 weeks (Wang 2018, Cochrane)

17.4 Lingual appliances

Aspect	Detail
Indication	An adult whose sole objection is visibility, and who will tolerate the trade-offs
Advantage	Genuinely invisible; unlike aligners, it delivers full fixed-appliance biomechanics
Mechanical consequences	The bracket is much closer to C_{res} in the vertical dimension and lies palatal to it: intrusive and torquing effects differ; interbracket span is shorter anteriorly, raising stiffness; retraction tends to produce more bodily movement of incisors but also more crown lingual tipping if uncontrolled
Individualised / customised lingual	CAD-CAM brackets and robotically-bent wires (Incognito and similar). Compensates for the very variable lingual anatomy, which is the principal obstacle to a stock lingual appliance

Drawbacks	Speech disturbance (transient, usually 1–3 weeks), tongue discomfort and ulceration, difficult access and hygiene, longer chairside time, cost, steep learning curve
Evidence	Limited; comparative trials are few and small. Do not claim superiority of outcome

17.5 Clear aligners

Item	Detail
Mechanism	A viscoelastic plastic shell: a spring that relaxes. Force decays through the wear period; faster when warm and moist. **Plastic persuades; wire overpowers**
Movements it does well	Tipping, mild crowding relief, arch expansion by tipping, anterior intrusion, IPR-based space closure, mild rotations of round-sectioned teeth
Movements it does badly	**Root torque**, extrusion, rotation of round teeth (canines, premolars), bodily space closure at extraction sites, molar uprighting, large anteroposterior correction
Attachments	Bonded composite features that give the plastic something to push against — essential for extrusion and rotation
Predictability	Invisalign produces the predicted outcome with about **41%** accuracy (Kravitz); rotation is the least accurate movement
Refinement	Only **6%** of cases complete without a refinement; **over 17%** convert to fixed appliances to finish
PAR vs ABO-OGS	**PAR shows no difference**; **ABO-OGS shows aligners significantly worse**, particularly for buccolingual inclination, occlusal contacts and relationship, overjet and root angulation. PAR is the coarser measure and the two do not correlate
Where aligners are better	Less pain in the first few days; better periodontal index scores; less eating and chewing disturbance; far less visible
Where fixed appliances are better	Occlusal outcome; buccolingual inclination; occlusal contacts; overjet and root angulation; severe crowding and complex cases; less post-treatment change
Compliance	20–22 hours per day. **Compliance is the weakest element in the system**
The unsolved problem	The influence of adjacent teeth: the aligner is one appliance acting on every tooth simultaneously, and reaction forces are not designed, only accepted

EXAM PHRASE TO HAVE READY

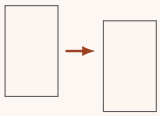
"I'd offer aligners to an adult with mild to moderate crowding, a Class I skeletal pattern and good motivation, and I'd be explicit that fewer than one in ten cases finishes without a refinement and that around one in six converts to fixed appliances. What I would not do is take on a case needing torque, extrusion or extraction space closure, because those are exactly the movements the evidence says the plastic does badly."

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

THE THREE ORDERS

FIRST ORDER
in–out



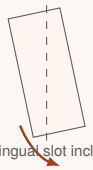
slot-to-base distance
arch form / crown thickness

SECOND ORDER
tip (angulation)



mesiodistal slot angulation
canine and premolar root angulation

THIRD ORDER
torque (inclination)



buccolingual slot inclination
incisor root position

Standard edgewise: all three built into the **wire**, tooth by tooth. Pre-adjusted edgewise: all three built into the **bracket** (Andrews, 1972, 1979).

Compare Fig. 10.9, p. 623.

CLEAR ALIGNERS VERSUS FIXED APPLIANCES

Aligners better

✓ Less pain in the first few days ✓ Better periodontal index scores ✓ Less eating and chewing disturbance ✓ Far less visible ✓ Shorter chairside appointments

Fixed appliances better

✓ Occlusal outcome (ABO-OGS) ✓ Buccolingual inclination ✓ Occlusal contacts and relationship ✓ Overjet and root angulation ✓ Severe crowding and complex cases ✓ Less relapse

PAR shows NO difference; ABO-OGS shows aligners significantly worse. PAR is the coarser measure — the two do not correlate.

The two numbers to quote

Only **6%** complete treatment without a refinement; over **17%** convert to fixed appliances to finish. Ch 14, pp. 1029–1031.

Compare Fig. 14.21, p. 1031.

§18 Treatment mechanics: Stage I, II and III

Curriculum: Items 35–37 — Stage I goals: alignment, correction of reverse bite, assisted eruption of impacted teeth, overjet and overbite reduction. Stage II: correction of buccal segment relationships, space closure of extraction sites. Stage III: finishing adjustments, bracket repositioning, vertical control of canines, development of the final occlusion and aesthetics. Tested in the written paper, the case presentation and the oral.

Key reading: MBT ch. 5–10; Bennett & McLaughlin; Companion Ch 10 and Ch 13; Proffit Ch 14–16.

Notes In Orthodontics: Chapter 10 (mechanics) and Chapter 13 (definitive treatment)

Supplements: E: lacebacks, bendbacks, tiebacks · F: the four elements, the working wire, arch form, finishing · G: optimal finishing

Practise this: Question Bank Block R, Q607–641 · Viva V122–V129

IN ONE MINUTE

Alignment → levelling → space closure → finishing, each with its own wire and its own anchorage cost.

- Anchorage is spent or saved during levelling and aligning, *before* any active mechanics begin
- Space closure at ~150 g; 200 g is no better
- Allow a month of passive rectangular wire before applying any active force

18.1 Stage I — alignment and levelling

Goal	How it is achieved
Alignment	Light, flexible, large-range wires: 0.012–0.018 in NiTi or multistrand. Progress by material and dimension. Bond second molars *as early as possible* — they are needed in the system for bite opening and for anchorage
Control of incisor position during alignment	**Lacebacks** — a passive 0.010 in ligature from the molar to the canine — distalise the canine without tipping it, and are the single most effective way of preventing incisor advance during levelling. **Bendbacks** — the wire bent distal to the molar tube — minimise labial tipping of the incisors as the wire straightens. Without lacebacks the lower incisor advances 1.4 mm; with them it retracts 1.0 mm
Correction of a reverse bite (anterior crossbite)	If there is room and a positive overbite will result: procline the upper incisor. Fixed appliance with an opening coil, or a removable appliance with a Z-spring and posterior bite planes. **Correct posterior crossbite before anterior** — correct the anterior first and you lose it when you expand
Assisted eruption of impacted teeth	Create space *first*, then expose, then apply light traction (30–50 g) along a controlled path, against adequate anchorage — usually a heavy stabilising base arch, and often a TAD
Overbite reduction	See §25
Overjet reduction	Begins here only in the sense that the incisors must be aligned and the arch levelled before retraction can start

18.2 Stage II — buccal segment correction and space closure

Element	Detail
Working wire	**0.019×0.025 in stainless steel** in a 0.022 slot: enough to control overbite and torque, loose enough in the slot to slide
Passive interval	Allow about **one month** of passive rectangular wire before any active force, so that the slots line up. Level slot line-up is what makes sliding mechanics possible at all
Space closure force	∼**150 g**. Attempts to transfer traditional edgewise forces of 500–600 g produced the "roller coaster" effect. 200 g is no better than 150 g
Sliding mechanics	Tiebacks: a 0.010 in wire ligature with a module stretched to about twice its resting length, from the molar hook to the canine hook. Reactivate at **4–6 weeks**
Loop mechanics	Closing loops (T-loop, teardrop) in beta titanium or stainless steel. No friction, but the force system is harder to control and wire bending is required
Coil springs	More consistent and slightly faster translation than modules in comparative work
Class II elastics	Correct the molar relationship at the cost of lower incisor proclination, upper incisor extrusion, and an increase in the occlusal plane cant. **Barred if the lower incisors are already forward**
Class III elastics	The mirror image; extrude the lower molars and rotate the mandible down and back — occasionally useful in a low-angle Class III, dangerous in a high-angle one
Anchorage check	Ask before every activation: is the anchorage holding? Measure it on the casts or with a superimposed photograph, not by impression

THE OVERJET-REDUCTION QUESTION, AS BENNETT PUTS IT

Ask first: *is lower arch anchorage available?* Every 1 mm the lower incisor is allowed to advance yields 2 mm of space. If the lower incisors are already forward, Class II elastics are barred and the correction must come from the upper arch, from growth, or from surgery. The lower incisor proclination limit conventionally quoted is **100° to the mandibular plane**.

18.3 Stage III — finishing and detailing

Task	Detail
Bracket repositioning	The commonest single reason a case does not finish. Reassess every bracket at the working-wire stage against marginal ridge levels and gingival contour, and reposition before finishing wires
Vertical control of canines	Canines are frequently left slightly extruded or under-torqued; check the gingival margin level against the central incisor and the cusp tip position in the smile
Root parallelism	Confirm on a panoramic radiograph at the finishing stage, particularly at extraction sites
Finishing wire	Fill or nearly fill the slot with an appropriately flexible wire: 0.021×0.025 steel or beta titanium in a 0.022 slot
Detailing bends	First order in–out; second order tip; third order torque; artistic positioning bends
Settling	**Never go straight from rectangular wires to retainers.** Settle in light round wires with vertical elastics for about **six weeks**, keeping every space closed

Interproximal reduction	For tooth-size discrepancy, black triangles and marginal ridge discrepancies
Aesthetic check	The appearance of the teeth, not merely the occlusion, is the target: midline, smile arc, gingival levels, incisal embrasures, connector areas, buccal corridors
Occlusal check in five planes	Horizontal, vertical, transverse, dynamic, and cephalometric/aesthetic. Finish to ABO requirements deliberately, not by luck

18.4 The six aesthetic rules worth checking before debond (Nanda)

- An average smile reveals **75–100%** of the upper incisor crowns.
- The smile curve is parallel to the lower lip in about **85%** of people.
- The terminal tooth in the smile is a **premolar** in about 90%.
- Connector areas descend in size from central–central to central–lateral to lateral–canine (the "50–40–30 rule").
- Incisal embrasures increase in size from the midline distally.
- Gingival zenith: the upper central and canine zeniths sit on a level line, the lateral incisor zenith 0.5–1 mm below it.

18.5 Arch form

- **The mandibular arch is the template.** Preserve the patient's original lower arch form; do not manufacture a new one.
- Distribution in a Caucasian practice: **tapered 50%, ovoid 42%, square 8%**; Japanese samples are almost the reverse. A recommended stock ratio of 45/45/10 follows.
- Change the arch form and it **returns in up to 70% of cases**. Buccal uprighting relapses in about 70% — which also means 30% is stable, so the manoeuvre is not worthless, merely unpredictable.
- Retain the *original* arch form, not the treated one.

§19 Retention, relapse and stability

Curriculum: Item 38 — retention: indications and principles; removable, fixed and active retainers; orthodontic relapse of dental crowding — current understanding and research. Tested in every component of the examination.

Key reading: Companion Ch 13, pp. 975–992 and Ch 5, pp. 283–284; Proffit Ch 17.

Notes In Orthodontics: **Topic 20** (Late lower incisor crowding); Chapter 13, pp. 975–992; Chapter 5, pp. 283–284
Supplements: A: Topic 20 · B: Topic 20 · C: Topic 20 (three causes of relapse) · F: the retention chapter
Practise this: Question Bank Block S, Q642–676 · Viva V130–V139

IN ONE MINUTE

Relapse is near-universal, and late lower incisor crowding is ageing rather than relapse. Retention is agreed before treatment starts.

- Three causes of relapse, on three different timescales: gingival fibre recoil, soft tissue pressures, continued growth
- An occlusal stop must be established or the overbite relapses (Houston 1989)
- By 4–5 years post-treatment the majority of patients have stopped wearing their retainers

19.1 Why teeth relapse — three mechanisms, three timescales

Mechanism	Timescale	Consequence and countermeasure
Elastic recoil of the gingival and periodontal fibres	Weeks to months (PDL ~3–4 months; supracrestal collagen and elastic fibres up to **6–12 months**)	Rotations relapse most. **Circumferential supracrestal fiberotomy** before debond; bonded retention
Soft tissue pressures	Continuous, lifelong	Teeth return towards the position of soft tissue equilibrium. Any tooth moved outside the neutral zone — lower incisor proclination, lower arch expansion — needs **permanent** retention
Continued growth	Lifelong, slow, in the original pattern	Late lower incisor crowding; recurrence of Class III and of open bite. Retention cannot be abandoned until growth is essentially complete

THE THREE RULES THAT PREDICT STABILITY

1. **Do not expand the lower intercanine width** — it relapses. 2. **Do not advance the lower incisors** more than about 2 mm; beyond that, permanent retention is very likely to be needed. 3. **Establish an occlusal stop** for any corrected overbite, and correct the interincisal angle; otherwise the overbite relapses (Houston 1989; Millett 2012).

19.2 The retainers

Retainer	Description	Comment
Vacuum-formed (Essix)	0.75–1 mm thermoplastic, full coverage	**More effective than a Hawley in the lower labial segment** (Rowland 2007). Cheap, well accepted, but occludes the posterior teeth apart and wears out; must be replaced
Hawley	Acrylic plate, labial bow, Adams clasps	Robust, adjustable, allows posterior settling; can incorporate a bite plane for deep bite retention
Begg / wrap-around	Labial bow passing distal to the last molar	Permits settling, does not obstruct the occlusion
Bonded (fixed) retainer	Multistrand wire bonded 3–3 (or 2–2)	**Marginally more effective in the lower arch for the first 12 months, especially for rotations; equivalent thereafter** (Forde 2017, 2018; Martin 2023). Requires lifelong monitoring: bond failure, wire fracture, unwanted torque and "wire syndrome", calculus, caries
Positioner	Elastomeric full-coverage appliance	Settles the occlusion and retains simultaneously; bulky and poorly tolerated
Active retainers	Spring retainer, Barrer retainer, aligner as an active retainer	For minor relapse of lower incisor alignment

19.3 Regimen, and what to tell the patient

- There is no universally accepted regimen. A common protocol is **full-time for the first 3 months** (or immediately to nights only with a vacuum-formed retainer), then **nights indefinitely**.
- The honest sentence: **"If you want the teeth to stay where they are, you wear a retainer for as long as you want that. There is no endpoint."**
- Retention should be **discussed prior to the commencement of active treatment**, and the patient must be told about relapse and about normal age-related change such as late lower incisor crowding, and must *agree* to the plan.
- Compliance is poor: by **4–5 years** post-treatment the majority of patients have stopped wearing their retainers (Al-Moghrabi 2018; Krämer 2022).
- Cost of replacement retainers and the consequences of loss should be stated at the outset.

19.4 Late lower incisor crowding

Type of crowding	Basis
Primary	Tooth size versus jaw size. Largely genetic
Secondary	Environmental — tongue, lips, cheeks
Tertiary	Adolescence onwards. **This is "late crowding"**

Proposed driver	Status
Mandibular growth rotation uprighting the lower labial segment	**The major factor** (Behrents; Solow)

Soft tissue maturation	Contributory
Physiological mesial drift	Contributory
Reduced interproximal wear (modern diet)	Contributory
Periodontal attachment loss	Contributory
Anterior component of occlusal force	**Doubted** — the forces are too brief (Southard 1990)
Third molars	**Not a primary cause.** NICE advises against prophylactic removal to prevent it (Richardson 2002; NICE)

- It happens in **untreated** people too: mandibular arch width and length decrease with age, with increasing anterior crowding, in untreated subjects (Sinclair & Little 1983; Thilander 2009).
- **It is ageing, not relapse**, and no extraction pattern or archwire sequence prevents it.
- Little (1975) proposed **under 3.5 mm** of lower incisor irregularity as an acceptable level.
- The answer is long-term bonded retention, and saying so honestly at consent.

WRITTEN-PAPER TRAP

Do not conflate relapse with age-related change. A stem describing 2 mm of lower incisor irregularity in a 27-year-old ten years after treatment is testing whether you will call it relapse (and blame the treatment) or normal ageing (and explain it).

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

LATE LOWER INCISOR CROWDING

What the evidence says

It occurs in treated *and untreated* dentitions alike. It cannot be prevented by any extraction pattern or archwire sequence. Third molars are not the cause (Ch 5, p. 284). It is a manifestation of continued late facial growth and the forward component of occlusal force — that is, **it is ageing, not relapse**.

What follows

The only reliable answer is **indefinite bonded retention**, and that must be said at the consent conversation before treatment begins — not discovered by the patient at debond.

Compare Ch 5, pp. 283–284; Ch 13, p. 979.

§20 Interceptive orthodontics and the developing dentition

Curriculum: Curriculum items 4, 35 and 39, 45, 53, 57, 59 as they apply to the mixed dentition — interceptive procedures, particularly in the mixed dentition stage, are an explicit objective of the Diploma. Tested in the written paper, the diagnostic cases and the oral.

Key reading: Companion Ch 12; Proffit Ch 12 and 13; Cobourne cases; Mosby ch. on the developing dentition.

Notes In Orthodontics: Topics 13, 14 and 15 (missing central incisor; impacted canine; impacted lower second molar); Chapter 12; crash notebook *Canine prognosis*, *Lower second molar*, *Trauma rest periods*

Supplements: A: Topics 13–15 · B: Topics 13–15 · C: Topics 13–15 (Ericson and Kurol's split)

Practise this: Question Bank Block T, Q677–721 · Viva V140–V151

IN ONE MINUTE

For every early-treatment question ask one thing: does waiting cost growth I cannot get back?

- Good evidence: crossbite with a displacement, and leeway space maintenance
- Weak evidence: arch expansion for crowding, two-phase Class II treatment
- Localised space loss of $\leq$3 mm can be regained; >4 mm usually means extraction

20.1 The five-step triage

1. Is there a problem at all, or is this normal development? (Transient lower incisor crowding, midline diastema, ugly duckling stage.)
2. Will it **self-correct**?
3. If not, is early intervention **more effective, simpler or safer** than waiting?
4. Can it be done **by me**, or does it need referral?
5. What is the **defined endpoint** for review — a date, not "keep an eye on it"?

20.2 What is worth intercepting — graded by evidence

Problem	Evidence	Action
Posterior crossbite **with a displacement**	**Good**	Treat on diagnosis. Quadhelix or expansion plate beats observation at ages 7–11 (Ugolini 2021, Cochrane, high certainty); quadhelix beats a plate and is ~3 months faster
Leeway space maintenance	**Good**	Lower lingual arch where the space is needed and the arch is otherwise sound
Anterior crossbite with a displacement	Good	Procline the upper incisor if there is room and a positive overbite will result. Corrects in 3–4 weeks with an incline plane; usually stable and needs no retention if a positive overbite is achieved (Wiedel & Bondemark 2015)
Digit habit	Moderate	Interrupt before the permanent incisors are established; the open bite usually closes spontaneously
Ectopic first permanent molar	Moderate	May reverse spontaneously up to ~8 y; otherwise separator, brass wire or distalising spring

Unerupted / impacted maxillary canine	Weak–moderate	Extract the primary canine at 10–13 years with space available. **Only very weak evidence** that this increases eruption, and *no* evidence it reduces the need for surgery (Benson 2021, Cochrane)
Supernumerary preventing eruption	Good (clinical consensus)	Remove the supernumerary, create and maintain space, then observe for eruption; expose and bond if none by 12 months
Early loss of a primary molar	Moderate	Space maintainer where the space matters; band-and-loop or lingual arch
Arch expansion purely to relieve crowding	**Weak**	Expansion yields only about half its width as usable space, and lower arch expansion relapses
Two-phase Class II treatment	**Weak / negative**	Not justified except for trauma risk or significant teasing
Skeletal anterior open bite	Weak	If treated at all, treat early (7–9 years)

20.3 Serial extraction

- A planned sequence in severe crowding with a Class I skeletal pattern and no vertical problem: primary canines (∼8–9 y) → first primary molars (∼9–10 y) → first premolars as they erupt.
- **Advantages**: uses eruptive movement to relieve crowding; may avoid appliances or shorten them; better periodontal and alveolar outcome than late extraction.
- **Disadvantages**: a long period of unattractive spacing; the overbite deepens; the lower incisors retrocline; a fixed appliance is usually needed anyway to finish; commits the patient irreversibly at a young age.
- **Contraindicated** in Class II or III skeletal patterns, deep overbite, open bite, hypodontia and mild crowding.

20.4 The impacted maxillary canine

Item	Detail
Prevalence	∼**2–3%** of Caucasian children; **up to 85% palatal**; bilateral in 17–45%; up to 3× commoner in females; 5× commoner in Europeans than Asians
Eruptive path	The longest in the dentition — **over 20 mm**
Clinical screening	**Palpate the buccal sulcus in every patient from 9–10 years.** If not palpable buccally, assume palatal. A firm, unresorbed primary canine beyond 14 years is a red flag
Localisation	Parallax (horizontal or vertical); "SLOB". Conventional radiography is 39–85% accurate; CBCT where the relationship to the lateral incisor root, or resorption of it, will change management
The real threat	Not the impaction but **resorption of the lateral (and central) incisor root**
Interception	Extract the primary canine at **10–11 y** (RCS guidance: 10–13 y), provided there is space, the canine is distal to the midline of the lateral incisor root, and its angulation to the midsagittal plane is 20–30°. Review at 12 months and abandon if there is no improvement
Ericson & Kurol's split	Overlap *less* than half the lateral incisor root → 91% normalise. Overlap *more* than half → 64% normalise after primary canine extraction
Naumova's trial	9 of 23 erupted after primary canine extraction versus 3 of 22 controls; 7 of 23 versus 11 of 22 still needed surgery. **Alpha angle and sector position** are the most influential predictors

Prognosis for alignment worsens with	**Height** — above the apical third of the incisor roots; **horizontal position** — at or across the lateral incisor midline; **angulation** — above $\sim 55^\circ$ to the midsagittal plane
Exposure technique	**No difference** between open and closed exposure in success or other outcomes (Parkin 2017, Cochrane, low quality); open exposure associated with just over 2 months less time to alignment and lower odds of ankylosis (Cassina 2018). Buccal canines below the mucogingival junction: **apically repositioned flap** (Kokich)
Alternatives	Leave and monitor (if asymptomatic, no resorption, and the patient accepts the primary canine); extract the canine and close or restore the space; **autotransplantation** — success around **88%**, splint semi-rigidly for $\leq$3 weeks then root-treat

20.5 The impacted mandibular second molar

- Prevalence **just over 2%**, mostly mandibular. Warning sign on the panoramic: a third molar follicle sitting *above* the occlusal table of the developing second molar. Evidence of a genetic basis, at least in the mandible (Shapira 2011).
- The escalating ladder:
  1. **Observe** — 6–12 months, with a defined endpoint. No progress means established impaction.
  2. **Disimpact** — a separator, brass wire or open coil placed *below* the contact point.
  3. **Remove the third molar** to relieve the obstruction.
  4. **Surgical uprighting** — expose, remove the third molar, luxate, bond a bracket at surgery, then align (Pogrel 1995; Kravitz 2016).
  5. **Extract the second molar** — the **least successful** outcome, because it commits the patient to unpredictable third molar eruption (Magnusson & Kjellberg 2009).
- **Differential diagnosis**: primary failure of eruption (PTH1R). Orthodontic traction causes ankylosis — do not attempt it.

20.6 The missing or lost maxillary central incisor

- Traumatic loss of a maxillary central affects about **3% of children** — typically unilateral, in the mixed dentition, in boys with an increased overjet (Järvinen 1978).
- **After avulsion, always attempt re-implantation**, even with a poor long-term prognosis: it maintains the space *and* the alveolar bone, and buys years.
- Then the decision: **maintain the space** (denture or retainer with a pontic — but the alveolus resorbs) or **let it close and re-open later** (preserves alveolar bone, but commits the patient to further orthodontics).
- **Definitive options**: unilateral — resin-retained bridge or implant; a lateral incisor rarely substitutes convincingly for a central. Bilateral — move the laterals mesially if size and form allow. Premolar autotransplantation if premolars are being removed anyway.
- **Implants must be deferred until growth is complete**: continued vertical alveolar growth leaves the implant crown in infra-occlusion.

20.7 Dental trauma: the rest period before orthodontic force

Injury	Wait before applying orthodontic force
Uncomplicated crown fracture	3 months

Subluxation	3 months
Lateral luxation — minor	3 months
Complicated crown fracture	Until a hard tissue barrier is seen
Lateral luxation — moderate or severe	12 months
Intrusion	12 months
Extrusion	12 months
Re-implantation after avulsion	12 months
Endodontically treated after trauma	Usually 12 months
Root fracture	12–24 months if dentine and cementum healing; connective tissue healing → treat as a short root with light forces

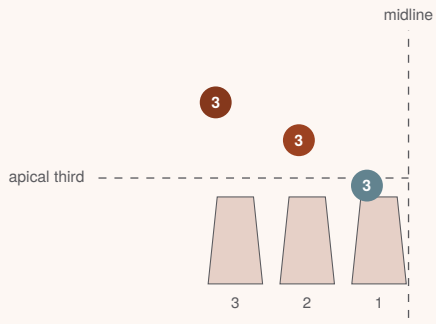
MONITORING A TRAUMATISED TOOTH DURING TREATMENT

Before: vitality testing and periapical radiographs, recorded. **During**: light forces, avoid excessive movement, vitality test three-monthly, repeat radiographs **6–9 months** after starting movement. **If resorption appears**: pause for 3 months (against 3–6 months for OIIRR generally). **The three complications** of moving a previously traumatised tooth: loss of vitality, pulpal calcification, and root resorption where resorption has already occurred.

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

PROGNOSIS FOR ORTHODONTIC ALIGNMENT OF AN IMPACTED CANINE WORSENS AS…



1 HEIGHT

worse above the apical third of the incisor roots

2 HORIZONTAL POSITION

worse at or across the lateral incisor midline

3 ANGULATION

worse above ~55° to the midsagittal plane

INTERCEPTION CRITERIA

10–11 y (RCS 10–13) · distal to the lateral root midline · 20–30° to midsagittal · no crowding · review at 12 months

~3% impacted; up to 85% palatal; eruptive path improves if the canine moves >20 mm.

Compare Fig. 13.12, p. 878; CBCT Fig. 7.27, p. 407.

IMPACTED LOWER SECOND MOLAR — THE ESCALATING LADDER

1. Observe — some resolve spontaneously
2. Separator or brass wire for 1–2 weeks
3. Uprighting spring, or a tube bonded to the 7 with a nickel titanium wire
4. Distal disc of the 6, or extraction of the 8
5. Surgical exposure and uprighting
6. Extraction of the 7 and let the 8 replace it — best before the 8's roots form

Just over 2% prevalence, mostly mandibular. Climb the ladder — do not start at the top.

Compare Ch 12, impacted second molars.

DENTAL TRAUMA: REST PERIOD BEFORE ORTHODONTIC FORCE

Injury	Rest period	Note
Concussion, subluxation	3 months	monitor vitality
Extrusive / lateral luxation	3 months	
Intrusive luxation	3 months	highest resorption risk
Avulsion and re-implantation	3–6 months	expect replacement resorption
Root fracture, healed	3 months	light forces only
Root-filled tooth	no extra delay	no greater resorption risk than a vital tooth

Move root-treated teeth with the same light forces as vital teeth; the evidence does not support the belief that they resorb more.

Compare Ch 12, dental trauma.

MISSING MAXILLARY CENTRAL INCISOR

Avulsion — ALWAYS attempt re-implantation (maintains space AND alveolar bone)

Maintain the space

Denture or retainer with a pontic

— risks alveolar bone loss

Let it close, re-open later

Preserves alveolar bone

— commits to further orthodontics

Definitive options

UNILATERAL → resin-retained bridge or implant (a lateral rarely substitutes convincingly). **BILATERAL** → move the laterals mesially if size and form allow; premolar autotransplant if premolars are coming out anyway.

Implants: defer until growth is complete

Continued vertical alveolar growth leaves the crown in **infra-occlusion** — the implant behaves like an ankylosed tooth while the adjacent dentition continues to erupt.

Compare Figs 12.58–12.59, pp. 821–822.

§21 Class I malocclusion and bimaxillary protrusion

Curriculum: Items 39–44 — treatment of Class I malocclusion in children, adolescents and comprehensive treatment; bimaxillary protrusion — orthodontic management, combined orthodontic/orthognathic treatment, and periodontal concerns and strategies (7 hours of the curriculum). Tested in the written paper, the diagnostic cases and the oral.

Key reading: Companion Ch 13; Proffit Ch 14 and 15; Cobourne cases; Nanda ch. on space closure.

Notes In Orthodontics: Chapter 13 (Class I and the permanent dentition); Topic 3 applied

Supplements: A: the space-analysis unit · B: Class I cases · G: smile aesthetics and space closure

Practise this: Question Bank Block U, Q722–746 · Viva V69–V90

IN ONE MINUTE

In Class I the skeletal base is not the problem, so the problem is space, transverse or vertical. Say which.

- Crowding, spacing, displacement, rotations — plus crossbite and vertical discrepancy
- The lower labial segment is the fixed point you plan around
- In bimaxillary protrusion the decision is set by the lips and the periodontium, not by the incisor angle alone

21.1 Class I: the four presentations

Presentation	Management
Crowding	Quantify in millimetres. Mild 1–4 mm — align, accept, or relieve with IPR or minor expansion. Moderate 5–8 mm — a judgement, usually extraction. Severe ≥9 mm — extraction, almost always first premolars in all four quadrants
Spacing	Generalised spacing (small teeth, hypodontia, macroglossia, tongue thrust) — close and retain permanently, or redistribute and restore. Localised (midline diastema) — establish the cause first
Displacement of individual teeth	Ectopic canines, transpositions, rotations. The plan follows the individual tooth
Vertical or transverse	Deep bite, open bite, crossbite in an otherwise Class I case. See §24 and §25

The midline diastema

- Normal in the mixed dentition (the "ugly duckling" stage) and usually closes as the canines erupt. Reassure and review.
- Persistent causes: small teeth or tooth-size discrepancy, missing or peg lateral incisors, a midline supernumerary (mesiodens), a prominent fleshy fraenum with a blanching test positive, a digit habit, generalised spacing, proclination.
- Investigate with an occlusal or periapical radiograph before ascribing it to the fraenum.
- **Close first, then frenectomy if needed** — a frenectomy performed before closure leaves scar tissue that resists closure. Both circumferential supracrestal fiberotomy and frenectomy are performed *before* appliance removal.
- **A bonded retainer is not optional**: a closed diastema relapses reliably.

21.2 Bimaxillary protrusion

Item	Detail
Definition	Both upper and lower incisors proclined and procumbent, with an increased procumbency of the lips, in a skeletal pattern that is often Class I
Population	Much commoner in patients of African and East Asian descent. **Normal values are population-specific**; do not apply Caucasian cephalometric norms uncritically
The decision	Driven by the **soft tissue**: lip procumbency at rest, lip strain on closure, nasolabial angle, and the patient's own view. An **acute** nasolabial angle means retraction will help; an **obtuse** one means it will harm
Typical plan	Four first premolar extractions with **maximum anchorage** — TADs, headgear or a stabilising arch — and bodily retraction of the labial segments
Soft tissue response	0.7 mm of lip retraction per 1 mm of incisor retraction; nasolabial angle increases 1.6° per 1 mm (Konstantonis 2018)
Periodontal concern	Retraction of proclined incisors in a thin symphysis with a thin gingival biotype risks **dehiscence and recession**. The evidence associates recession more with *retroclination* in high-angle cases with a narrow symphysis and thin biotype than with proclination (Joss-Vassalli 2010; Aziz 2011; Tepedino 2018)
Periodontal strategy	Assess the biotype and symphyseal width *before* planning; consider CBCT of the symphysis; light forces; bodily rather than tipping movement; avoid over-retraction; consider soft tissue grafting before or after; maintain immaculate plaque control; involve a periodontist early
Surgical option	Where the discrepancy is skeletal and severe: anterior segmental osteotomies (Wunderer / Köle) or a two-jaw procedure with setback. Reserve for the case where orthodontic retraction alone cannot reach the facial aim without unacceptable periodontal or dental compromise

EXAM PHRASE TO HAVE READY

"In a bimaxillary protrusion I'm not treating an angle, I'm treating a lip. I'd start from the lips at rest and on smiling and from the nasolabial angle, then check whether the symphysis and the gingival biotype will tolerate the retraction I need. If they won't, that changes the plan before I have chosen a single bracket."

§22 Class II malocclusion

Curriculum: Items 45–52 — interceptive treatment of Class II in children; orthopaedic treatment in adolescents; high-angle and low-angle Class II — orthodontic and combined orthodontic/orthognathic treatment; Class II with TMD (16 hours of the curriculum — the largest single block). Tested in every component.

Key reading: Companion Ch 9, Ch 12 and Ch 13; Proffit Ch 13 and 15; Cobourne cases 12.x; Nanda ch. on distalisation.

Notes In Orthodontics: **Topic 12** (Reducing overjet in a growing child); Chapters 9, 12 and 13; crash notebook *Early Class II treatment: what the trials showed*

Supplements: A: Topic 12 (choose from the vertical) · B: Topic 12 · C: Topic 12 (the randomised trials) · G: distalisation

Practise this: Question Bank Block V, Q747–786 · Viva V152–V161

IN ONE MINUTE

Growth modification, camouflage or surgery. Growth and the soft tissues decide — and the vertical dimension chooses the appliance.

- 15–27% of Caucasians; usually *mandibular retrognathia*, not maxillary excess
- Overjet ≥5 mm falls from 23% of children to 13% of adults — the mandible catches up
- Choose the appliance from the *vertical* dimension, not the anteroposterior discrepancy

22.1 Diagnosis

Feature	Note
Skeletal component	Most often mandibular retrognathia. Establish which jaw is at fault before planning
Division 1	Upper incisors proclined or of average inclination, overjet increased. Often a **lip trap**: the lower lip rests behind the upper incisors, proclining them and retroclining the lowers, and perpetuating the malocclusion
Division 2	Upper central incisors retroclined, overbite increased and often complete to tissue, high lower lip line, reduced lower face height, forward mandibular growth rotation, increased interincisal angle. **The lateral incisors are frequently proclined and mesiolabially rotated**
Vertical pattern	**The variable that chooses the appliance**: cervical pull headgear for the hypodivergent, high pull or occipital for the hyperdivergent
Habits	Digit sucking; assess and intercept
Trauma risk	Increased overjet is a significant risk factor for incisor trauma (Batista 2018; Macey 2020)

22.2 The interceptive question, answered by the trials

Early treatment *did*	Early treatment *did not*
Reduce the overjet (dentoalveolarly)	Improve the final skeletal outcome
Give a transient rise in self-esteem	Improve the final occlusal outcome
Improve profile attractiveness	Shorten overall treatment

Reduce incisor trauma — RR 0.58, a 42% reduction	Reduce total appointments
	The late group finished slightly better

THE PRAGMATIC POLICY

Treat early **only** for children at high risk of trauma — a very large overjet, vertical excess, incompetent lips — or where there is significant teasing. Otherwise a single phase in the late mixed or early permanent dentition. **Two caveats to add**: much incisor trauma occurs before age 9, so even early treatment may come too late; and headgear was marginally more effective than a functional appliance at reducing trauma. The trials to name: Florida and North Carolina (Tulloch, Keeling, Dolce); Manchester twin block (O'Brien 2003, 2009).

22.3 The orthopaedic phase in adolescents

- **Timing**: the circumpubertal period, **CVM stage CS3**. Earlier wastes compliance; later wastes growth.
- **Functional appliance** (twin block, Herbst, Frankel) — see §16. Expect: overjet reduction outpacing overbite reduction, so **lateral open bites are normal** during treatment and are managed by trimming the blocks; upper incisor retroclination and lower incisor proclination; a posterior open bite that must be settled before the appliance is abandoned.
- **Decompensate first**: if the upper labial segment is retroclined (a division 2 pattern being converted to a division 1 before functional appliance therapy), procline the uppers first so that the mandible can be postured forwards.
- **Overcorrect**, then transition carefully — straight into a fixed appliance, or with a period of night-time functional appliance wear. Pancherz recommends **at least 2 years** of activator retention after a Herbst.
- **Headgear** — see §14. Restrains maxillary growth and distalises molars; direction of pull chosen by the vertical pattern; two safety mechanisms mandatory.
- **Distalisation** (pendulum, distal jet, TAD-supported): corrects a *dentoalveolar* Class II only. Distal molar force 150–250 g; molar derotation alone can gain 2–3 mm per side; distalisation always buys reciprocal mesial premolar movement unless a TAD is used.

22.4 Camouflage in the permanent dentition

Vertical pattern	Approach and cautions
Low angle (hypodivergent, deep bite)	Overbite is the dominant problem and it is easy to reduce and hard to hold. Extract to allow retraction (upper first premolars, lower second premolars if lower crowding is mild). Level by **relative intrusion** in a grower (buccal segment extrusion, condylar growth compensates); by **true intrusion** in a non-grower. Cervical headgear is permissible and even helpful. Establish an occlusal stop. Guard against the bite deepening during space closure
High angle (hyperdivergent, increased face height, often reduced overbite)	The difficult one. **Avoid anything extrusive**: no cervical headgear, no Class II elastics if avoidable, no buccal segment extrusion. Use high-pull headgear or TADs; consider molar *intrusion*; consider extraction to allow the mandible to rotate forwards. Anchorage is at risk and the mandible rotates down and back at the slightest provocation

The Class II division 2 problem

- The mechanical difficulty is the **increased interincisal angle** with a **deep, complete overbite**. Retroclined upper centrals must be **torqued labially at the root** — not merely tipped — or the overbite will not be held.
- Sequence: procline the upper labial segment to convert it to a division 1 pattern, reduce the overbite, then retract if extraction space has been created.
- Frequently there is little or no overjet, so extraction space in the upper arch has nowhere to go; **many division 2 cases are treated non-extraction in the upper arch**.
- Bracket prescription: extra palatal root torque on the upper centrals; the mesiolabially rotated laterals need first- and second-order correction.
- Retention: a Hawley with an anterior bite plane, worn long-term, is the classic answer for a deep bite — often into the early twenties.

22.5 Surgery

- Thresholds (Proffit 1992): **overjet >10 mm**, **pogonion to nasion perpendicular ≥18 mm**, **mandibular body length <70 mm**, **anterior face height >125 mm**.
- The usual procedure is a **bilateral sagittal split osteotomy** advancement, with or without a **genioplasty**; a Le Fort I impaction is added where there is vertical maxillary excess and a gummy smile.
- **Mandibular advancement is the least stable of the common movements** in the vertical dimension — see §27's hierarchy of stability.
- Pre-surgical orthodontics **decompensates**: it removes the dental compensation and therefore makes the overjet *worse* before surgery. Say this explicitly, and warn the patient.

22.6 Class II with TMD

- The examinable position is in §26 and does not change here: weak statistical associations exist; there is **no evidence of causation**; orthodontics neither causes nor prevents TMD.
- Practically: screen and **document** joint status before treatment; counsel on natural history; if symptoms appear during treatment, **pause active mechanics** and manage conservatively and reversibly; do not promise that correcting the Class II will cure the TMD, and do not let a TMD diagnosis drive you into an irreversible occlusal change you would not otherwise have made.
- A **stabilisation splint** is a reasonable conservative measure; **occlusal adjustment has little beneficial value** and is irreversible.

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

EARLY CLASS II TREATMENT: WHAT THE TRIALS SHOWED

Early treatment DID

· reduce the overjet (dentoalveolar) · give a transient rise in self-esteem · improve profile attractiveness · **REDUCE INCISOR TRAUMA — RR 0.58, a 42% reduction**

Early treatment did NOT

· improve the final skeletal outcome · improve the final occlusal outcome · shorten overall treatment · reduce total appointments · **the LATE group finished slightly better**

The pragmatic policy

Treat early ONLY for children at high risk of trauma — very large overjet, vertical excess, incompetent lips — or significant teasing. Otherwise a single phase in the late mixed or early permanent dentition.

Caveat: much trauma occurs before age 9, and headgear was marginally better. Trials: Florida · North Carolina (USA) · Manchester twin block (UK). O'Brien et al., 2003, 2009; Tulloch et al., 1997, 2004.

Compare Fig. 12.71, p. 847.

§23 Class III malocclusion

Curriculum: Items 53–56 — interceptive treatment of Class III in children; orthopaedic treatment in adolescents; Class III orthodontic treatment; Class III combined orthodontic/orthognathic treatment (7 hours of the curriculum). Tested in every component.

Key reading: Companion Ch 8, Ch 9 and Ch 13; Proffit Ch 13 and 15; Cobourne cases 17.x.

Notes In Orthodontics: **Topic 17** (Correcting a Class III incisor relationship); Chapters 8, 9 and 13; crash notebook *Class III: the decision pathway*

Supplements: A: Topic 17 (centric relation first) · B: Topic 17 · C: Topic 17 (bone-anchored protraction) · G: the chin cap verdict

Practise this: Question Bank Block W, Q787–816 · Viva V152–V161

IN ONE MINUTE

Unfavourable growth persists. That is what makes Class III harder than Class II — and why monitoring is so often the right answer.

- Residual mandibular growth *helps* in Class II and *harms* in Class III
- Seat the patient in centric relation before you diagnose anything — a pseudo-Class III manipulates back to edge-to-edge
- Facemask before age 10: over 60% avoided surgery versus 36% (Mandall 2016)

23.1 Diagnosis

- Maxillary retrusion, mandibular prognathism, or both. In East Asian populations maxillary retrusion is the commoner component and prevalence is far higher than the ~5–6% of US samples; worldwide Class III is about 6.8% with striking regional variation.
- **Manipulate into centric relation.** A **pseudo-Class III** manipulates back to an edge-to-edge incisor relationship: the true relationship is Class I with an anterior displacement, and it is a completely different prognosis.
- Dental compensation is the rule: **upper incisors proclined, lower incisors retroclined**. The more compensated the case, the less room there is for camouflage.
- Assess the **vertical**: a high-angle, reduced-overbite Class III is the unfavourable pattern.
- Take a **family history** — Class III has the strongest hereditary component of the three.

23.2 The favourable/unfavourable split in the mixed dentition

Favourable — consider early intervention	Unfavourable — monitor into the late teens
Class I or mild Class III skeletal base	Moderate to severe Class III base
Average or reduced lower face height	Increased lower face height
Average or **increased** overbite	**Reduced** overbite
Proclined lower / upright upper incisors (little compensation)	Retroclined lower / proclined upper incisors (heavily compensated)
Anterior displacement on closing	No displacement; Class III molars

23.3 Early intervention

Modality	Detail
Upper removable appliance	For a simple anterior crossbite with room and a positive overbite to come. Z-spring, posterior bite planes. Corrects in 3–4 weeks with an incline plane
Facemask (protraction headgear) ± RME	**12–14 h/day**, elastic force ∼400 g per side, before age 10. Effects: maxillary advancement, downward and backward mandibular rotation, upper incisor proclination — **about half the overjet correction is dental**. **Overcorrect to a +3 to +4 mm overjet**. RME may be used to "disarticulate" the maxilla and improve protraction
Evidence	Mandall 2016: over **60% avoided surgery** versus 36% without early treatment, though most of the initial skeletal effect is lost with growth. Proffit: **25–33% recurrence** of the anterior crossbite 8 years later, mostly needing surgery
Bone-anchored maxillary protraction	Class III elastics to miniplates work through the adolescent spurt at about half the force. The two methods suit different ages: **facemask 8–10 years, miniplate elastics 11–14 years**
Chin cup	Short-term gain is not maintained after growth; **no agreement that it inhibits mandibular growth**. Largely redirects the mandible downwards and backwards

23.4 Camouflage or surgery

Threshold favouring surgery	Value
Wits appraisal	≤ -11 mm
ANB	$\leq -4^\circ$
Reverse overjet	≤ -5 mm
Holdaway angle	$\leq 3.5^\circ$
Gonial angle	$\leq 120^\circ$
Lower anterior face height	$\geq 66\%$
Lower incisor inclination	$\leq 80^\circ$

Discriminant analysis on these variables predicts the need for surgery with over **90% accuracy** (Kerr 1992; Stellzig-Eisenhauer 2002; Kochel 2011).

- **Camouflage** in a mild case: extract upper second premolars and lower first premolars (space further *forward* in the mandible), procline the uppers, retrocline the lowers. Occasionally a single lower incisor extraction. Class III elastics — used cautiously, and never in a high-angle case.
- **Avoid mandibular extractions if surgery is likely**: they consume the decompensation space the surgeon will need.
- **Defer surgery until growth is complete**, confirmed by serial cephalometric superimposition — and remember that Class III mandibular growth persists beyond adolescence, later in males.
- Pre-surgical orthodontics **decompensates**: retrocline the uppers, procline the lowers, and the reverse overjet gets *worse*. Warn the patient in advance and show them a case.

> **VIVA TRAP**
>
> "The parents want the crossbite treated now — the child is 7." The examinable answer separates the anterior crossbite *with a displacement* (treat now — it is causing attrition,

recession on the lower incisor, and asymmetric growth) from the *skeletal* Class III (where early treatment buys time and a 60%-versus-36% chance of avoiding surgery, but most of the skeletal effect is lost, and monitoring into the late teens may still be the right answer).

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

CLASS III: THE DECISION PATHWAY

MIXED DENTITION — is the case FAVOURABLE?

Favourable (Table 8.4)

· Class I or mild Class III base · average or reduced lower face height · average or INCREASED overbite · proclined lower / upright upper incisors · anterior displacement on closing

→ **Early intervention** facemask ± RME, 12–14 h/day, before age 10. >60% avoid surgery versus 36% (Mandall, 2016).

Unfavourable

· moderate to severe Class III base · increased lower face height · REDUCED overbite · retroclined lower / proclined upper incisors · no displacement · Class III molars

→ **Monitor into the late teens** Class III worsens with growth. Avoid mandibular extractions if surgery is likely.

Permanent dentition — camouflage or surgery?

Surgery is more likely if: Wits ≤ -11 mm · ANB $\leq -4°$ · overjet ≤ -5 mm · Holdaway $\leq 3.5°$ · gonial $\leq 120°$ · LAFH $\geq 66\%$ · lower incisor inclination $\leq 80°$

Compare Table 8.4, p. 505; decompensation Fig. 13.37, p. 930.

§24 Transverse problems and facial asymmetry

Curriculum: Items 16, 57–58 — methods of crossbite correction; asymmetry — treatment in children and in adolescents. Tested in the written paper, the diagnostic cases and the oral.

Key reading: Companion Ch 9, Ch 12, Ch 13 and Ch 15; Proffit Ch 13; Cobourne cases 16.x and 19.x.

Notes In Orthodontics: **Topics 16 and 19** (Crossbite correction; facial asymmetry); Chapters 9, 12, 13 and 15; crash notebook *Choosing an expansion approach*, *Facial asymmetry*

Supplements: A: Topics 16 and 19 · B: Topics 16 and 19 · C: Topic 16 (Proffit's dissent on rapid expansion)

Practise this: Question Bank Block X, Q817–841 · Viva V162–V173

IN ONE MINUTE

Classify the crossbite first — dental, functional or skeletal. That decides the appliance. And all humans are slightly asymmetric.

- Shift or no shift: that single finding sets both the urgency and the prognosis
- Posterior before anterior, always
- Transverse is "the first dentofacial dimension to cease growing" — so treat it early

24.1 Classifying a crossbite

Axis	Categories
Site	Anterior or posterior; unilateral or bilateral; buccal (lower buccal cusps buccal to upper) or lingual/scissors bite (upper palatal cusps buccal to lower buccal cusps)
Cause	**Dental** — individual displaced teeth; **skeletal** — a genuine transverse discrepancy of the bases; **functional** — a bilateral discrepancy producing a lateral displacement to a unilateral crossbite
Displacement	With or without a mandibular displacement on closing. **The single most important finding**

WHY THE DISPLACEMENT MATTERS

A unilateral posterior crossbite *with* a displacement is almost always a bilateral constriction that the patient closes out of. It is treated **on diagnosis**, because the displacement causes attrition, is associated with asymmetric mandibular growth and with recession on the tooth taking the load, and because the case will not settle until it is corrected. A crossbite *without* a displacement is treated when convenient.

24.2 Choosing the expansion approach

	Dental or mild skeletal	True transverse skeletal deficiency
Appliance	Removable expansion plate, or **quadhelix**	**Rapid maxillary expansion** (hyrax or Haas)

Activation	Plate: 0.25 mm × 2 per week. Quadhelix: ∼3 months faster than a plate	1–4 turns/day (0.25–1 mm/day) over 2–4 weeks
Force	Light and continuous	Up to **10 kg**
Skeletal component	Minimal	Up to **4 mm at the suture** ≈ 50% of screw expansion
Retention	Overcorrect; retain nights 3–6 months	Retain the appliance in situ 3–6 months
Relapse	—	∼**10% of skeletal change** at 4–8 months
Dental expansion retained	SME **95%**	RME **85%**
Evidence	Ugolini 2021 (Cochrane): high certainty that quadhelix or expansion plate beats observation at 7–11 years; quadhelix beats a plate	Bazargani 2023: limited differences between conventional and hybrid RME at 5 years

- **Growth complete?** Beyond about **5 mm** of required expansion in a non-growing patient, **SARPE** or bone-borne expansion is indicated. Sutural fusion occurs by the late teens, but the correlation between sutural maturation and *chronological* age is poor — assess the suture, not the birthday.
- **Proffit's dissent**: at 4 months, rapid and slow expansion have produced about the same skeletal-to-dental split; the net effect of RME is approximately equal skeletal and dental expansion. Slow expansion at <2 mm/week matches the maximum speed of bone formation. *Rapid activation is not an effective way to minimise tooth movement.*
- Slow expansion **can** split the suture in the early mixed dentition — the constraint is sutural interdigitation with age, not the appliance category.
- RME is **not** recommended as a treatment for nasal obstruction or sleep-disordered breathing on current evidence.

24.3 Anterior crossbite

- Correct **posterior before anterior**. Correct the anterior first and you lose it when you expand.
- Requirements for correction by proclination: adequate space, and a **positive overbite** to result — the overbite is what makes the correction self-retaining.
- An anterior crossbite corrected with a positive overbite is usually stable and requires no retention (Wiedel & Bondemark 2015).
- Look for gingival recession on the lower incisor taking the traumatic load, and record it.

24.4 Facial asymmetry

- **Mild asymmetry is normal.** All humans are slightly asymmetric; clinically significant asymmetric jaw development affects **<0.5%** of the population, and the mandible is involved in **85–90%** of cases. The commonest single cause is a **condylar fracture** in childhood, not a syndrome.
- Distinguish **laterognathy** (true skeletal asymmetry) from **laterocclusion** (a functional shift). Bite on a spatula and ask: is there a *true occlusal cant*?

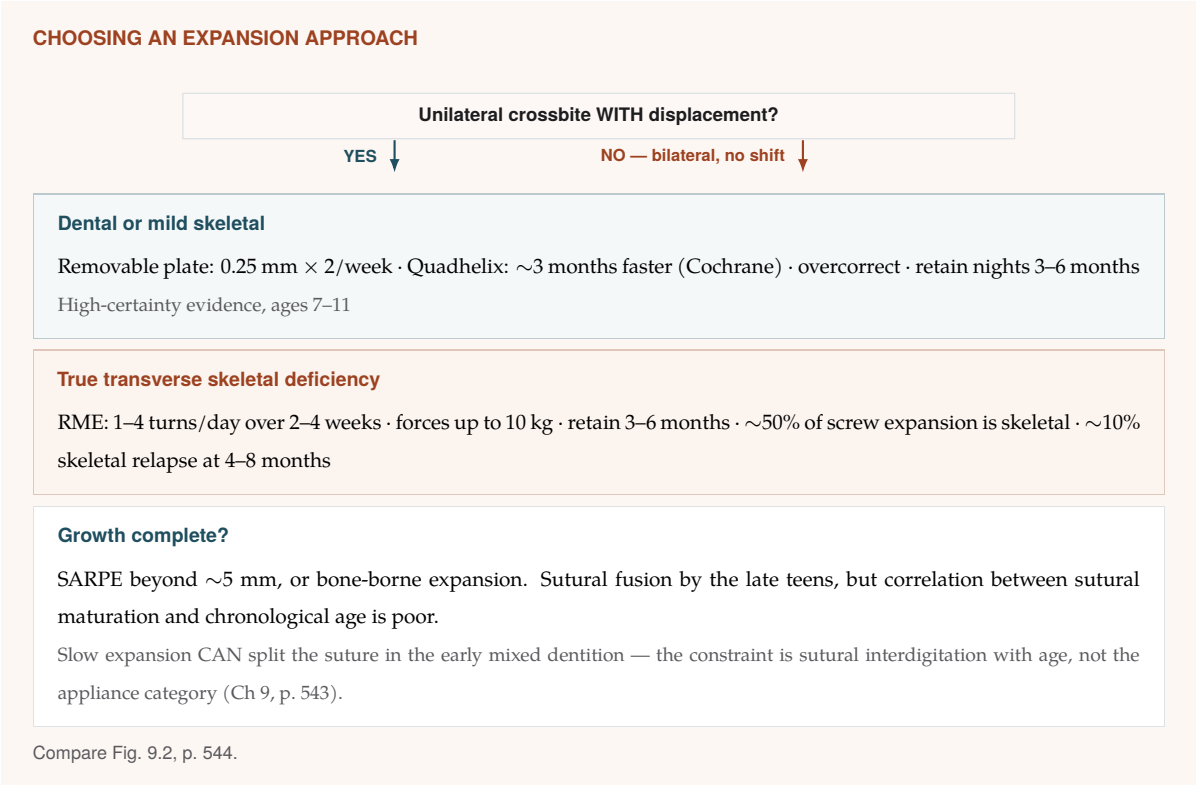
Finding	Interpretation

No cant; deviation of the path of closure RCP → ICP	**Functional** — a crossbite with displacement. Eliminate the shift early; it can drive asymmetric growth
No cant; no deviation; chin deviated to the *unaffected* side; occlusal plane level	**Hemimandibular elongation** (Obwegeser & Makek Type I) — a horizontal vector
True occlusal cant; unilateral open bite; chin and centreline often undisturbed	**Hemimandibular hyperplasia** (Type II) — a vertical vector
Both	Type III, a hybrid

- **Is it progressive?** Accelerating chin deviation, a new cant, or new TMD symptoms in the teens → refer.
- **Investigation**: PA cephalogram for significant asymmetry (identifying the midsagittal plane is the hard part); submentovertex; CBCT. **Scintigraphy** is sensitive but **non-specific**, with false positives. **Serial 3-D facial surface superimposition** is the non-invasive alternative, with good reproducibility using the forehead, mid-nose and infraorbital regions (Häner 2021).
- **Monitor growth before committing to surgery**: operating on an active condylar hyperplasia invites relapse. Slootweg & Müller described four histological subtypes.
- In a growing child with mild asymmetry, **the answer is usually to watch — with a defined review interval and a record you can superimpose**.

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.



Compare Fig. 9.2, p. 544.

FACIAL ASYMMETRY IN A GROWING CHILD

The clinical test

Bite on a spatula. Is the occlusal plane canted? **Cant present** → likely skeletal, and possibly progressive. **No cant** → likely dental or functional.

Is it progressive?

Serial photographs and measurements at 6-monthly intervals; bone scintigraphy where condylar hyperplasia is suspected. Mild asymmetry is present in almost everyone; the task is to identify the minority that is growing worse.

Obwegeser and Makek: hemimandibular elongation (horizontal, chin deviates) versus hemimandibular hyperplasia (vertical, ramus and body enlarge, occlusal cant).

Do not operate on a growing asymmetry

Defer definitive surgery until growth has ceased and the asymmetry is static on serial records — otherwise it recurs.

Compare Ch 15, asymmetry.

§25 Vertical problems: deep bite and open bite

Curriculum: Items 35, 59–60 — overjet and overbite reduction; open bite treatment in children and in adolescents. Tested in the written paper, the diagnostic cases and the oral.

Key reading: Companion Ch 13, pp. 928–945; Proffit Ch 14; Nanda ch. on vertical control; Cobourne cases 11.x.

Notes In Orthodontics: **Topic 11** (Overbite and methods of overbite reduction); Chapter 13, pp. 928–945; crash notebook *Overbite*

Supplements: A: Topic 11 (5 mm of dentoalveolar height) · B: Topic 11 · C: Topic 11 (relative intrusion) · G: vertical control and open bite

Practise this: Question Bank Block Y, Q842–871 · Viva V162–V173

IN ONE MINUTE

Reducing an overbite is easy. Holding it is the difficulty — and the same manoeuvre has opposite consequences in a grower and a non-grower.

- Three mechanisms: intrude incisors · extrude buccal segments · procline the labial segments
- Whichever route, an *occlusal stop* must be established or the overbite relapses
- In open bite, the aetiology dictates everything — habit, skeletal, or tongue

25.1 Deep overbite — the four situations

Situation	Manoeuvre	Why it works, or does not
A. Growing patient	Buccal segment **extrusion** → relative incisor intrusion. Flat anterior bite plane; reverse curve of Spee; cervical headgear	Condylar growth compensates for the increased posterior vertical dimension. **Stable**
B. Non-growing patient	Buccal segment extrusion	The mandible rotates **down and back**, worsening the Class II and increasing face height. **Unstable** — avoid
C. High MMPA or adult	**True incisor intrusion** — utility or bypass arch, segmental mechanics, or miniscrews	The only route where the face height must not increase. **∼60 g total, 15 g per tooth**
D. Retroclined incisors (division 2)	**Proclination** of the labial segments, reducing the interincisal angle	**0.1–0.2 mm** of overbite change per degree. Unstable unless the interincisal angle is genuinely corrected

Technique	Evidence
Segmental intrusion	Achieves ∼1.5 mm maxillary and ∼1.9 mm mandibular incisor intrusion in non-growing patients (Ng 2005)
Segmental vs continuous reverse curve	Segmental gives greater true intrusion; a Ricketts utility arch produces more lower incisor proclination (Dake & Sinclair 1989; Weiland 1996)
Miniscrews	2–3 mm of intrusion; **0.67–0.95 mm more** than segmental mechanics, with less molar extrusion (low-quality evidence) (Sosly 2020; Shakti 2022)

Intrusion arches	Burstone's one-couple bypass intrusion arch; the Connecticut intrusion arch in NiTi requires no helix
Dahl concept	Localised supra-occlusion re-establishes posterior contacts over an average of **6 months**; adverse events rare, minor and transient (Dahl 1975; Poyser 2005)
Stability	Overbite reduction relapses unless the **interincisal angle is corrected and an occlusal stop established** (Houston 1989; Millett 2012)

WRITTEN-PAPER TRAP

The commonest trap in this topic is answering "extrude the buccal segments" without asking whether the patient is growing. In a grower it is the standard answer; in an adult it rotates the mandible down and back, opens the face, worsens the Class II, and relapses.

25.2 Anterior open bite

Aetiology	Recognition	Management
Habit (digit sucking)	Asymmetric, often localised to the digit; associated posterior crossbite; proclined uppers, retroclined lowers	Intercept the habit before the permanent incisors are established; the bite usually closes spontaneously. Reminder therapy, then a palatal crib in a motivated child
Skeletal	Symmetrical, increased lower face height, backward mandibular rotation, high MMPA, obtuse gonial angle, **interlabial gap >3 mm**	Difficult. Vertical control: high-pull headgear, posterior bite blocks, molar intrusion with TADs, or surgery. **If treated early, treat at 7–9 years**
Tongue posture / thrust	Usually *adaptive* to the open bite rather than causative; an endogenous tongue thrust is rare	Do not blame the tongue reflexively. If it is adaptive it resolves; if endogenous, the case is unstable and needs permanent retention
Iatrogenic	Extrusive mechanics in a high-angle patient; Class II elastics; a functional appliance with untrimmed blocks	Prevent it. Recognise the high-angle face before choosing the mechanics

- **1 mm of molar intrusion gives about 3 mm of open bite reduction** (Nanda) — the most efficient lever available, and the reason TADs transformed open bite management. Conversely, 1 mm of molar extrusion *reduces* overlap by 1.5–2.5 mm.
- Open bites relapse in about **20%** of surgical and non-surgical corrections. Say so at consent.
- **Interlabial gap >3 mm** is the key soft tissue sign of a skeletal open bite.
- Extrusion is **contraindicated** where the face is already long.
- **Choose the treatment occlusal plane first**, then design the force system to reach it.

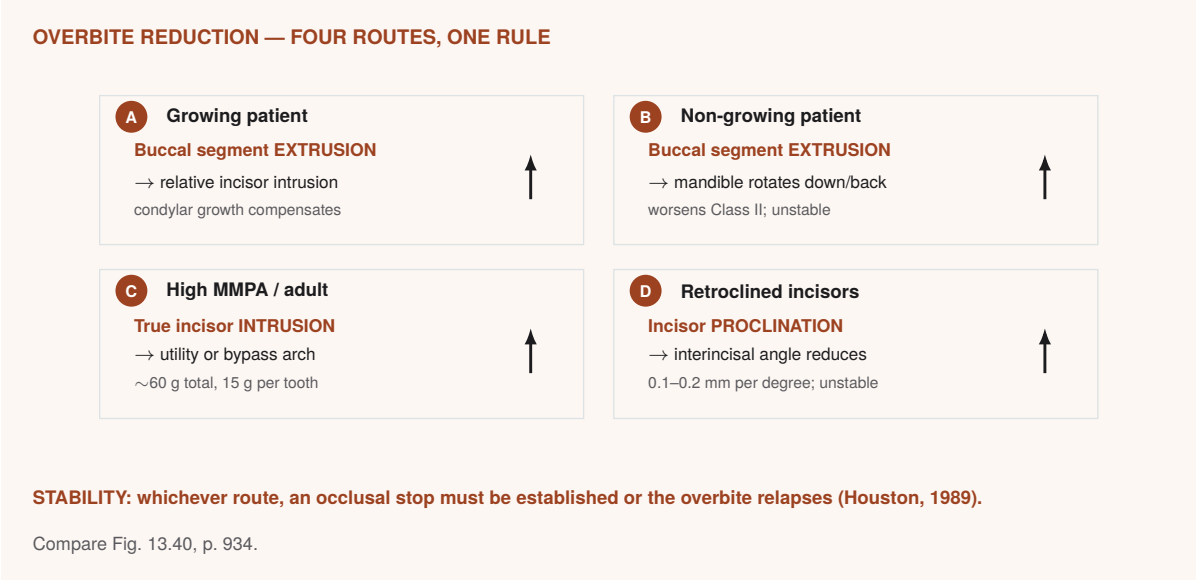
25.3 Vertical control — the rules

- The resting lip decides how much you may intrude. Record incisor show at rest and on smiling — say "Emma" or "Mississippi" for rest position, and "cheese" with the teeth together for the posed smile — and photograph both, **before** planning anything vertical.
- **5 mm of dentoalveolar height separates a deep bite face from an open bite face.** Small changes matter.
- Short face = forward condylar growth = deep bite. Long face = posterior condylar growth = backward rotation = open bite.

- Retention of a deep bite: a Hawley retainer with an anterior bite plane, worn into the early twenties.

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.



Compare Fig. 13.40, p. 934.

§26 TMD, condylar trauma, the airway and myofunctional therapy

Curriculum: Items 24–27 — myofunctional therapy; orthodontics and temporomandibular joint dysfunction; condylar trauma and orthodontic treatment; adenoid hypertrophy and its management (7 hours of the curriculum). Tested in the written paper and heavily in the oral.

Key reading: Companion Ch 2, pp. 102–105; Proffit Ch 1 and 18; Mosby ch. on TMD; Fang Bing and He Hong lectures.

Notes In Orthodontics: **Topic 18** (Malocclusion and temporomandibular disorder); Chapter 2, pp. 102–105; crash notebook *TMD in five dates*

Supplements: A: Topic 18 · B: Topic 18 · C: Topic 18 (Proffit's prevalence argument)

Practise this: Question Bank Block Z, Q872–901 · Viva V174–V183

IN ONE MINUTE

Orthodontics is TMJ-neutral. State it, defend it, and stop there.

- Weak statistical *associations* exist; there is no evidence of *causation*
- Orthodontics neither causes nor prevents TMD — and extraction does not change that
- Symptoms often begin in late adolescence, exactly when we treat. That is the confound

26.1 The history, in five dates

Date	Event
1930s	Costen wrongly attributes ear and joint symptoms to mandibular overclosure
1970s	Roth and the gnathological school: canine guidance, no interferences, RCP = ICP as a treatment goal
Late 1980s	"Functional orthodontists" blame premolar extraction for TMD and for distal condylar displacement
1987	**Brimm v. Malloy** — a Michigan lawsuit; **$850,000** awarded after TMD following four-premolar extraction treatment
1992	**AAO-funded studies (Behrents & White)**: no significant association; TMD cannot be prevented by any intervention; symptoms increase with age and often begin in late adolescence; **orthodontic treatment does not cause TMD**

26.2 The evidence, and the position to state

Source	Finding
Egermark-Eriksson (1983); Mohlin (2007); Thilander & Bjerklin (2012)	Certain occlusal traits show *weak* correlations with TMD signs and symptoms
Manfredini (2017)	Lack of evidence implicating dental occlusion in the pathophysiology of TMD
Luecke & Johnstone (1992)	Conventional incisor retraction does *not* produce distal displacement of the condyles
Macfarlane (2009); Luther (2010)	Lack of evidence that orthodontic intervention reduces TMD signs and symptoms

Luther (2007)	No evidence that an ideal functional occlusion improves stability or prevents TMD
Proffit	TMD is no more prevalent now than before modern orthodontics, and symptoms disappear long before any occlusal change could explain it. Orthodontic treatment for TMD "should not be considered adjunctive treatment"
Kandasamy (2022)	Screen and document before treatment; counsel on natural history; suspend active mechanics if symptoms appear; manage with conservative, reversible therapies

Epidemiology	Figure
At least one TMD *sign*	~75%
At least one TMD *symptom*	~33%
Seeking treatment	A small minority
Peak age of onset	Late adolescence to early adulthood; commoner in females

EXAM PHRASE TO HAVE READY

"The position I'd defend is that orthodontics is TMJ-neutral. There are weak statistical associations between some occlusal traits and TMD signs, but no evidence of causation in either direction, and the AAO studies of 1992 are the ones I'd cite. Practically that means I screen and record joint status before I start, I counsel patients that TMD symptoms commonly appear in late adolescence for reasons unrelated to their braces, and if symptoms appear during treatment I stop active mechanics and manage conservatively rather than reaching for an occlusal explanation."

26.3 Screening, classification and investigation

- **Screen** every patient: maximum opening (normal ≥40 mm), deviation or deflection on opening, joint sounds (click, crepitus), muscle and joint tenderness on palpation, and a history of pain, locking, and parafunction.
- **Classify** broadly: **myofascial pain** (muscles), **internal derangement** (disc displacement with or without reduction), **degenerative joint disease**, and the systemic inflammatory arthropathies (notably **juvenile idiopathic arthritis**, which can destroy the condyle and produce a progressive Class II with an open bite in a child).
- **Imaging**: **MRI for the disc**; **CT or CBCT for bone**; plain films and panoramic for a screening view of condylar form. **CBCT cannot image the disc or muscle.**
- **Management, in order**: reassurance and explanation of the natural history; soft diet, jaw rest, avoidance of wide opening and parafunctional habits; analgesia; physiotherapy and jaw exercises; a **stabilisation splint**; referral for persistent or progressive disease. **Occlusal adjustment has little beneficial value** and is irreversible.

26.4 Condylar trauma and orthodontic treatment

Item	Detail
Why it matters	The condyle is the mandible's principal growth site. A childhood condylar fracture is the **commonest cause of significant mandibular asymmetry**

Presentation	Often missed at the time. Later: reduced opening, deviation to the affected side on opening, ipsilateral posterior open bite acutely, and — as growth proceeds — chin deviation to the affected side, reduced ramus height, occlusal cant
Growth outcome	Highly variable. Many children remodel well and grow normally; a minority develop progressive asymmetry and ankylosis
Management principle in children	**Conservative, functional** management wherever possible: early mobilisation, soft diet, analgesia, physiotherapy. Prolonged immobilisation risks **ankylosis**, which is the outcome to avoid at all costs
Orthodontic role	Monitor growth with serial records and superimposition; a **functional appliance** may be used to encourage condylar growth and maintain symmetry during the growth period; definitive correction of an established asymmetry is orthognathic
Late sequelae	Ankylosis (treat surgically — gap arthroplasty or costochondral graft, with aggressive physiotherapy), asymmetry, TMD

26.5 Adenoid hypertrophy, the airway and sleep-disordered breathing

Item	Detail
The classic association	Chronic nasal obstruction → mouth breathing → low tongue posture, lips apart, head extension → narrow maxilla, increased face height, backward mandibular rotation, anterior open bite. The "adenoid facies"
The honest position	The association is **real but weak, and causation is not established**. Experimental nasal obstruction in primates produced facial change, but human evidence is observational. State the association; do not assert the cause
Adenoid natural history	Adenoid tissue peaks around 3–5 years and involutes through adolescence. Many children "grow out of" the obstruction
Referral	Snoring, witnessed apnoeas, restless sleep, mouth breathing, daytime somnolence or behavioural problems, recurrent otitis media → refer to ENT / paediatrics. **Do not diagnose or treat obstructive sleep apnoea orthodontically without a sleep study**
Paediatric OSA	Prevalence ∼2% of children. **Adenotonsillectomy** is the first-line treatment in most children
RME and the airway	**Not recommended** as a treatment for nasal breathing or OSA on current evidence
Adult OSA	Mandibular advancement appliances are effective in **mild-to-moderate** OSA (Koretsi 2018); CPAP remains the gold standard for moderate-to-severe. Maxillomandibular advancement surgery is the most effective surgical option
Extractions and the airway	**No evidence** that extraction-based treatment reduces the oropharyngeal airway or increases sleep apnoea — described in the literature as "scaremongering and marketing" (Larsen 2015; Orabi 2021)

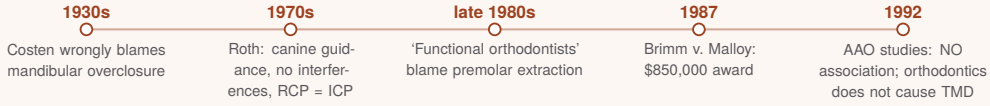
26.6 Myofunctional therapy

Covered in §16.5. The short version for the oral: there is reasonable evidence that *interrupting a habit* before the permanent incisors are established allows spontaneous correction, and reasonable evidence that a lip trap perpetuates a Class II division 1. There is *not* good evidence that muscle exercises alone correct a malocclusion, and there is no evidence at all for the stronger claims made by proprietary "myofunctional" appliance systems.

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

MALOCCLUSION AND TMD — A FRACTIOUS HISTORY



The position to state

Weak statistical associations exist. There is no evidence of **causation**. Orthodontics neither causes nor prevents TMD. Symptoms often begin in late adolescence — exactly when we treat.

In practice (Kandasamy et al., 2022)

Screen and document before treatment · counsel on the natural history · pause mechanics if symptoms appear · manage conservatively with reassurance, soft diet, analgesia and a splint if needed · do not offer orthodontics as a treatment for TMD.

Prevalence: at least one *sign* ∼75%; at least one *symptom* ∼33%.

Compare Where is the evidence 2.3, pp. 104–105.

§27 Adult orthodontics and orthognathic surgery

Curriculum: Items 41, 43, 50, 52, 56 — comprehensive treatment in the adult; combined orthodontic/orthognathic treatment in Class II, Class III and bimaxillary protrusion. Tested in the written paper, the case presentation and the oral.

Key reading: Companion Ch 13 and Ch 15; Proffit Ch 18 and 19; Nanda ch. on interdisciplinary care; Chow / Haers lectures.

Notes In Orthodontics: Chapters 13 and 15 (adult treatment; orthognathic surgery)

Supplements: A: the surgical-planning unit · B: the decompensation unit · C: the hierarchy of stability · G: interdisciplinary finishing

Practise this: Question Bank Block AA, Q902–931 · Viva V184–V193

IN ONE MINUTE

Decompensation deliberately worsens the bite so that the surgery can deliver the face.

- The camouflage-or-surgery decision must be made *before* treatment begins
- Adjunctive treatment serves a larger periodontal or restorative goal; comprehensive treatment does not
- Stability depends on *direction* > *fixation* > *technique*, in that order

27.1 What is different about the adult

Difference	Consequence
No growth	Growth modification is unavailable. Every millimetre must come from tooth movement or from surgery
Reduced periodontium	C_{res} moves apically: the same force produces a larger moment and more tipping. Lighter forces, higher M/F ratios, and consider intrusion of over-erupted teeth to regain attachment level
Pre-existing restorations	Bonding to porcelain and composite; bands where bonding is unreliable; plan the restorative endpoint with the restorative colleague *first*
Missing teeth	Space opening or closing; implant site development; the alveolar bone argument (erupt a tooth through a deficient site, then move it back)
Slower biology	Movement is slower, hyalinisation is more prolonged, and treatment takes longer
Aesthetic expectations	Higher, more specific, and more likely to concern the soft tissues. Record them precisely
Recession	**Probe every adult and record pre-existing recession** before treatment
Root resorption	Somewhat higher risk; take a mid-treatment radiograph
Motivation	Usually excellent, and self-referred. But BDD prevalence is higher in this group

Adjunctive versus comprehensive treatment

- **Adjunctive** — limited tooth movement serving a larger periodontal or restorative goal: uprighting a tipped molar before a bridge, aligning an abutment, intruding an over-erupted tooth, creating space for an implant, correcting a single crossbite. **Often appropriate in general practice.**
- **Comprehensive** — full correction of the malocclusion. Requires the full diagnostic workup and a full appliance.

- **Molar uprighting**: uprighting is easy; knowing when to stop and when not to start is the examination question. Intrusive force **10–15 g per tooth**. **Retain until the lamina dura reappears.** Dolichocephalic patients are poor candidates because the mechanics are extrusive.

27.2 The orthognathic pathway

1. **Joint assessment** — orthodontist and maxillofacial surgeon together, with the patient's own concern recorded first. Clinical photographs, radiographs, casts, and increasingly a CBCT with a merged surface scan.
2. **Diagnosis and planning** — problem list, facial and occlusal aims, cephalometric prediction, model surgery or virtual surgical planning. **The decision to operate is made now, not later.**
3. **Pre-surgical orthodontics** (typically 12–18 months) — align, level, co-ordinate the arches, **decompensate**, and hold the lower centre line on the chin. This deliberately *worsens* the occlusion.
4. **Surgery** — with a surgical wafer, rigid internal fixation.
5. **Post-surgical orthodontics** (typically 4–6 months) — settle the occlusion, detail, finish.
6. **Retention** — and long-term review.

DECOMPENSATION, IN ONE SENTENCE

"Pre-surgical orthodontics is not aligning the arches — it deliberately removes the dental compensation, so the reverse overjet in a Class III or the overjet in a Class II gets *worse* before surgery, because that is what gives the surgeon the movement needed to correct the face." Calling it "aligning the arches" is one of the twelve costly slips.

27.3 The operations

Procedure	What it does
Le Fort I osteotomy	Maxilla: advancement, impaction, down-graft, or segmental. Impaction corrects vertical maxillary excess and a gummy smile, and produces autorotation of the mandible
Bilateral sagittal split osteotomy (BSSO)	Mandible: advancement or setback. Risk to the **inferior alveolar nerve** — altered sensation is common transiently and may be permanent in a minority; quote and consent for it
Vertical subsigmoid osteotomy	Mandibular setback; extra-oral or intra-oral
Genioplasty	Chin: advancement, setback, reduction, augmentation, asymmetry correction. "You cannot orthodontically produce a chin"
Segmental osteotomies	Anterior maxillary (Wunderer) or mandibular (Köle) segments — for bimaxillary protrusion and for closing an open bite
Surgically assisted RPE (SARPE)	Transverse expansion beyond ~5 mm in the non-growing patient
Distraction osteogenesis	Large movements, particularly in syndromic and cleft patients; corticotomy, latency (~5–7 days), distraction (~1 mm/day), consolidation

27.4 The hierarchy of stability

Most stable at the top, least stable at the bottom. **Direction of movement matters more than the method of fixation, which matters more than the surgical technique.** "Very stable" means about a 90% chance of no clinically significant change.

Movement	Stability
Maxilla superiorly (impaction)	Very stable
Mandible forwards (advancement)	Very stable in the low- and average-angle patient
Chin, any direction (genioplasty)	Very stable
Maxilla forwards (advancement)	Stable
Maxilla up + mandible forwards	Stable
Maxilla forwards + mandible back	Stable, but problematic
Mandible back (setback)	Problematic
Maxilla down (down-graft)	**Least stable**
Maxilla wide (surgical expansion)	Least stable

27.5 Consent for orthognathic surgery

Nerve injury (inferior alveolar, lingual, infraorbital); infection; bleeding; relapse; unfavourable split; malocclusion requiring further orthodontics; TMJ symptoms; condylar resorption (particularly in young women with a high-angle pattern and pre-existing joint disease); need for plate removal; prolonged treatment; the psychological adjustment to a changed face; and the possibility that the patient will be dissatisfied even with a technically excellent result. Screen for **body dysmorphic disorder** — up to **17%** of orthognathic patients.

§28 Cleft lip and palate, and craniofacial syndromes

Curriculum: Cleft care is an explicit part of the interdisciplinary component of the Diploma and of the online lecture series (Gillgrass; Chow / Haers). Tested in the written paper and the oral.

Key reading: Companion Ch 16; Proffit Ch 5 and 19; Mosby ch. on craniofacial anomalies.

Notes In Orthodontics: Chapter 16 (Cleft Lip and Palate, and Syndromes); crash notebook *The cleft care timeline*
Supplements: A: the clefts and syndromes unit · B: the cleft unit · C: Proffit Ch 5 and 19

Practise this: Question Bank Block AB, Q932–961 · Viva V184–V193

IN ONE MINUTE

A timeline. Learn who does what, when, and why.

- Lip repair 3–6 months · palate repair 6–12 months · alveolar bone graft 8–10 years
- Graft *before* the canine erupts, when two-thirds of its root is formed — and expand first
- Learn the two epidemiologies as a contrast; they are different developmental failures

28.1 Epidemiology

	Cleft lip ± palate	Isolated cleft palate
Incidence	1 : 700	1 : 2000
Sex ratio	Male 2 : 1	Female 4 : 1
Non-syndromic	~70%	~50%
Laterality	~80% unilateral (left commoner)	—
Higher incidence	East Asian and Native American populations	—
Lower incidence	African populations	—

Aetiology is multifactorial: genetic susceptibility plus environmental factors — maternal smoking, alcohol, folate deficiency, anticonvulsants (phenytoin, sodium valproate), retinoids, corticosteroids, maternal diabetes. **Folic acid supplementation** reduces risk. Recurrence risk rises with the number of affected relatives.

28.2 The care timeline

Age	Intervention	Note
Birth	Diagnosis, feeding advice, hearing assessment, genetics, parental support	Feeding is the first priority. Presurgical orthopaedics (naso-alveolar moulding) is used in some centres; the evidence is contested
3–6 months	**Lip repair**	The "rule of tens": 10 weeks, 10 lb, haemoglobin 10 g/dL
6–12 months	**Palate repair**	Before speech develops. Hearing monitored — glue ear is near-universal; grommets as needed

18 months–5 years	Speech assessment and therapy	Velopharyngeal incompetence may need a pharyngoplasty
Mixed dentition	Prevention, fissure sealing of the first molars, **arch expansion**	Expand *before* you graft: the graft needs access and the arch needs form
8–10 years	**Alveolar bone graft**	**Before the canine erupts**, when about two-thirds of its root is formed. Cancellous bone, usually from the iliac crest. Purpose: provide bone for the canine to erupt into, stabilise the maxillary segments, close the oronasal fistula, support the alar base, and permit later implant placement
Adolescence	Definitive orthodontics	Often complex: hypodontia, supernumeraries, rotations of the central adjacent to the cleft, missing lateral incisor, transverse and sagittal deficiency
Late teens	**Orthognathic surgery** if needed	Maxillary hypoplasia is common; distraction osteogenesis for large advancements. Defer until growth is complete
Throughout	Restorative, psychological, ENT, audiology, speech and language therapy	**Multidisciplinary team care, audited nationally**, is the model — centralisation of cleft care improved outcomes measurably

28.3 Dental features in the cleft patient

- **Missing lateral incisor** on the cleft side is the commonest anomaly; supernumerary and malformed teeth also occur, often *both* sides of the cleft.
- The central incisor adjacent to the cleft is characteristically **rotated and tipped** towards the cleft, and often hypoplastic.
- **Crossbites** — anterior and posterior — from maxillary transverse and sagittal deficiency and from surgical scarring.
- Dental anomalies are **commoner in bilateral cases**.
- Higher caries risk; enamel defects; difficult hygiene. Prevention is central.
- Retention is prolonged, and permanent retention is usual.

28.4 Craniofacial syndromes worth recognising

Syndrome	Features
Crouzon	Craniosynostosis, midface hypoplasia, exophthalmos, relative mandibular prognathism. **FGFR2**. No limb anomalies
Apert	Craniosynostosis, severe midface hypoplasia, **syndactyly**, high arched or cleft palate. **FGFR2** — shares the receptor family with Crouzon
Pierre Robin sequence	Micrognathia, glossoptosis, U-shaped cleft palate, airway obstruction. A *sequence*, not a syndrome: the small mandible causes the rest
Treacher Collins (mandibulofacial dysostosis)	**TCOF1**; autosomal dominant. Downward-sloping palpebral fissures, malar and mandibular hypoplasia, ear anomalies, conductive deafness, cleft palate in ∼30%
Hemifacial microsomia / Goldenhar	Unilateral underdevelopment of the ear, mandible and facial soft tissues; epibulbar dermoid and vertebral anomalies in Goldenhar. Pruzansky classification of the ramus
Cleidocranial dysplasia	**RUNX2**. Absent or hypoplastic clavicles, delayed closure of fontanelles, **multiple supernumerary teeth and failure of eruption of permanent teeth**

Ectodermal dysplasia	Hypodontia or anodontia, conical teeth, sparse hair, hypohidrosis. Prosthetic and implant-led management
Down syndrome (trisomy 21)	Midface hypoplasia, relative mandibular prognathism, macroglossia, hypodontia, delayed eruption, periodontal disease, cardiac anomalies
Turner syndrome (45,X)	Short stature, webbed neck; narrow high palate, mandibular retrognathia, delayed dental development
22q11 deletion (DiGeorge / velocardiofacial)	Cleft palate or velopharyngeal incompetence, cardiac anomalies, characteristic facies, learning difficulties

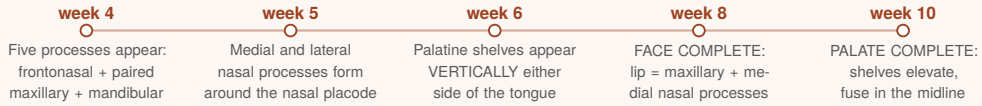
TURNER TOOTH — DO NOT CONFUSE IT WITH TURNER SYNDROME

A **Turner tooth** is a localised enamel hypoplasia of a permanent tooth caused by infection or trauma to its primary predecessor. **Turner syndrome** is 45,X. Examiners do use both words in the same paper.

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

DEVELOPMENT OF THE FACE AND PALATE — AND WHERE CLEFTS ARISE



Cleft lip ± primary palate

Maxillary process fails to fuse with the MEDIAL NASAL process, by week 8

1:700 · male 2:1 · 70% non-syndromic · 80% unilateral

Isolated cleft palate

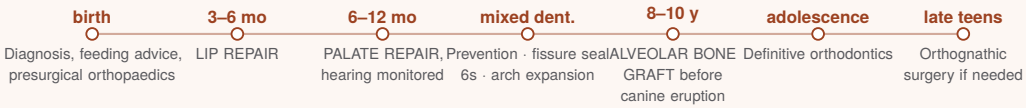
Palatine shelves fail to elevate or fuse, weeks 6–10

1:2000 · female 4:1 · 50% non-syndromic

The sex ratios are opposite because these are *different* developmental failures occurring at *different* times. Ch 3, pp. 121–133; Ch 16, pp. 1113–1118.

Compare Figs 3.8–3.10, pp. 123–133.

THE CLEFT CARE TIMELINE



Epidemiology — learn as a contrast

Cleft lip ± palate	1:700	male 2:1	70% non-syndromic	80% unilateral
Isolated cleft palate	1:2000	female 4:1	50% non-syndromic	

Opposite sex ratios and different syndromic proportions, because these are different developmental failures at different times (lip by 8 weeks; palate weeks 6–10).

The rule of tens for lip repair: 10 weeks, 10 lb, haemoglobin 10 g/dL.

Compare Fig. 16.9, p. 1130.

§29 Risks of orthodontic treatment and their management

Curriculum: Item 65 — risks of orthodontic treatment and their management; and item 10 — medical conditions relevant to orthodontics. Tested in the written paper and the oral.

Key reading: Companion Ch 2, pp. 88–101 and Ch 7, pp. 356–366; Proffit Ch 1.

Notes In Orthodontics: Topics 9 and 10 (Benefits and complications; Medical conditions); Chapter 2, pp. 88–101; Chapter 7, pp. 356–366; crash notebook *Benefits and risks*, *Medical history*

Supplements: A: Topics 9 and 10 · B: Topics 9 and 10 · C: Topics 9 and 10 (demineralisation on QLF)

Practise this: Question Bank Block AC, Q962–991 · Viva V174–V183

IN ONE MINUTE

Elective treatment on a healthy dentition. The benefits are narrower than commonly claimed, and the risks are yours to minimise.

- The four risks to name: demineralisation, root resorption, recession, relapse
- White spot lesions in up to *half* of fixed appliance patients, visible within 6 months
- NICE: antibiotic prophylaxis solely to prevent infective endocarditis is *not* routinely provided for dental procedures

29.1 The benefits — with the right level of confidence

Claimed benefit	Strength of evidence
Dental aesthetics	**Strong**
Trauma prevention (RR 0.58 for early overjet reduction)	**Strong**
Oral-health-related quality of life: emotional and social domains	Moderate
Plaque scores	Moderate
Caries reduction	Weak
Periodontal health	Weak / none
Masticatory efficiency	Weak
Speech	Weak
Long-term psychological health	Little evidence
Nasal breathing (RME)	Not recommended

29.2 The risks, quantified

Risk	Frequency	Management and prevention

Demineralisation / white spot lesions	**Up to half** of fixed appliance patients; visible within **6 months**; commoner in males; commonest on first molars, upper lateral incisors, lower canines	Select patients on plaque control; oral hygiene instruction and reinforcement; daily high-fluoride toothpaste; professional fluoride every **6–8 weeks**; glass ionomer band cement; steel ligatures rather than modules in poor brushers; diet advice; debond early if uncontrolled
Root resorption	Radiographic up to 73%; severe ($\geq \frac{1}{3}$) in **1–5%** of teeth; greatest in the anterior maxilla	Light forces; short treatment; mid-treatment radiograph; pause 3–6 months if significant resorption appears; consent explicitly
Recession	Related more to **retroclination** in high-angle Class III cases with a narrow symphysis and thin biotype than to proclination	Assess biotype and symphysis before planning; light bodily forces; periodontal referral; record pre-existing recession
Relapse	Near-universal	Retention planned and agreed *before* treatment; permanent retention where teeth have been moved out of the neutral zone
Pain	Peaks at **48 hours**, returns to baseline by **7 days**	Warn; analgesics may be taken pre-emptively; paracetamol preferred (prostaglandin inhibitors may slow tooth movement — the effect is small and clinically debated)
Loss of vitality	Uncommon; associated with previous trauma	Vitality testing before treatment; light forces on traumatised teeth
Gingival hyperplasia	Common with poor hygiene; recovers after debond	Hygiene; excise if persistent after debond
Enamel fracture / damage at debond	Especially with ceramic brackets	Correct debonding technique; tungsten carbide clean-up at low speed
Headgear injury	Rare but catastrophic	**Two mandatory safety mechanisms**; written warning; instruct to stop if the bow disengages
Nickel allergy	10–30% F, 1–3% M	Avoid NiTi; stainless steel is safe
Latex allergy	4.3% general, 9.7% healthcare	Latex-free elastics, gloves, dam
Soft tissue trauma	Common, minor	Wax, bracket adjustment, review
Failure to achieve the aim	Real, and must be consented for	Realistic aims; documented plan; recognise your own limits
Discontinuation	High in children and adolescents, especially with removables (Klaus 2017)	Motivation assessment; appropriate appliance choice

29.3 The medical history — what it changes

What changes	Conditions
Force level — use light forces	Sickle cell disease, thalassaemia, childhood malignancy and its treatment, diabetes, previous radiotherapy, short roots
Extractions — avoid if possible	Haemophilia and bleeding disorders, sickle cell, malignancy, **bisphosphonates and antiresorptives**
Appliance choice	Epilepsy → **no removable appliances**. Ceramic or aligner if MRI will be needed. Transplant or phenytoin (gingival hyperplasia) → bonded tubes rather than bands. Asthma → avoid palate-covering removables. Nickel allergy → no NiTi
Timing — delay	≥6 months post-transplant; until a malignancy is in remission; while an eating disorder is active; during pregnancy for anything elective

Expectations	Body dysmorphic disorder in 5–13% of orthodontic patients
Prophylaxis	**NICE: routine antibiotic prophylaxis for dental procedures is not advised** — though NICE conflicts with the AHA and ESC, which still recommend cover for the highest-risk patients (Thornhill 2016). Infective endocarditis mortality ~21%
Rate of movement	**Bisphosphonates slow or inhibit tooth movement and delay eruption** — keep the aims simple and stopping the drug does not help (Zymperdikas 2020). Corticosteroids and NSAIDs also modulate turnover

WRITTEN-PAPER TRAP

"Nickel allergy contraindicates orthodontic treatment" → false. Stainless steel is safe; avoid nickel titanium. "A patient with a heart murmur needs antibiotic cover for band placement" → false under NICE. Both appear regularly.

Schematics

The same figures appear in the crash notebook of Notes In Orthodontics.

MEDICAL HISTORY: WHAT IT CHANGES

Condition	What it changes
Bisphosphonates	tooth movement slowed or arrested; MRONJ risk on extraction
Diabetes, poorly controlled	periodontal breakdown; defer until controlled
Epilepsy	avoid removable appliances that could be inhaled; gingival hyperplasia with phenytoin
Bleeding disorders, anticoagulants	extraction planning and haemostasis; liaise with the physician
Immunosuppression	infection risk; slower healing
Cardiac lesions	antibiotic prophylaxis is NOT routinely indicated (NICE); maintain oral health
Nickel allergy	titanium or gold-plated brackets; nickel-free wires
Latex allergy	latex-free elastics and gloves
Juvenile idiopathic arthritis	condylar resorption; monitor for progressive retrognathia and open bite
Head and neck radiotherapy	avoid orthodontic movement in irradiated bone
Cleft and craniofacial syndromes	team care; timing dictated by the reconstruction

Compare Ch 7, medical history; Handbook **§29.3**.

§30 Orthodontic practice, infection control and governance

Curriculum: Item 63 — orthodontic practice and infection control; and the professional standards underlying every other item. Tested in the written paper and the oral.

Key reading: GDC Standards; local decontamination guidance; Companion Ch 2 and Ch 7; Mosby ch. on practice.

Notes In Orthodontics: Not covered as a separate unit in the Companion; drawn from Chapter 7 (radiographic justification) and professional guidance

Supplements: A: the practice and periodontal-interface units

Practise this: Question Bank Block AD, Q992–1016 · Viva V184–V207

IN ONE MINUTE

The examiners are testing whether you would be safe running an orthodontic list, not whether you can quote a standard.

- Decontamination: single-use where possible; wash, inspect, sterilise, store, track
- Record what you did, why you did it, what you warned about, and what the patient agreed to
- Know your limits and refer — saying so unprompted is the strongest thing you can do in the oral

30.1 Infection control in the orthodontic setting

Area	Practice
Standard precautions	Applied to every patient regardless of known status. Hand hygiene, personal protective equipment, safe sharps handling, safe disposal, surface decontamination between patients
Instrument decontamination	Clean (manual, ultrasonic or washer–disinfector) → inspect → sterilise (vacuum or non-vacuum autoclave, 134–137°C) → store in the sterile pouch → track by batch to the patient. Validate and record cycles daily
Orthodontic-specific items	Pliers and cutters are heat-tolerant but corrode: dry thoroughly, oil hinges after sterilisation. Bands and brackets removed from stock cannot be returned; use a band-sizing kit that is sterilised between patients
Single use	Prophylaxis cups and brushes, saliva ejectors, needles, local anaesthetic cartridges, IPR strips (or sterilise per manufacturer), matrix bands, and anything labelled single-use. **Never reprocess a single-use item**
Impressions and appliances	Rinse, disinfect and label before sending to and on receipt from the laboratory. Document the process
Intra-oral scanners	Disposable or sterilisable tip sleeve; wipe the wand between patients
Water lines	Flush, treat and monitor as per local guidance; *Legionella* risk assessment
Sharps	Wire ends are sharps: use distal-end cutters that retain the fragment, and account for every wire end
Waste	Segregate clinical, sharps, amalgam and domestic waste; consignment notes retained
Aerosols	Debonding and clean-up generate aerosol: high-volume aspiration, eye protection for patient and operator

30.2 Radiation protection

- **Justification, optimisation, limitation.** Every exposure justified for the individual patient by a named practitioner; the dose optimised (rectangular collimation, fastest receptor, correct field size, thyroid protection where appropriate); and the number of exposures limited.
- **No threshold dose.** Stochastic risk means there is no "safe" small exposure — only a justified one.
- Named roles: legal person, referrer, practitioner (justifies), operator (carries out).
- Written justification recorded in the notes; the report on the film recorded too — a radiograph taken is a radiograph that must be reported.
- Quality assurance programme; audit of rejects; staff training and dose monitoring.

30.3 Record keeping and governance

Item	What must be recorded
Assessment	Complaint in the patient's words; history; examination in three planes; IOTN; provisional problem list; records justified and taken
Plan	Aims (facial and occlusal); options discussed **including no treatment**; extraction pattern; anchorage plan; appliance; estimated duration; **retention plan agreed**
Consent	What was discussed, what was warned about, what was given in writing, who was present, and that the patient (and parent) agreed. **A signed form is evidence of consent, not consent itself**
Each visit	What was done, wire sizes and sequence, elastics prescribed, oral hygiene status, breakages, instructions given, next appointment interval
Adverse events	Recorded factually and promptly; decalcification, resorption, appliance failures, missed appointments and the warnings given
Debond	Final records; retention issued; instructions given; review arranged
Audit	Case outcomes (PAR or ICON), failure and breakage rates, decalcification rates, treatment duration, discontinuation rates, radiograph rejects, decontamination cycle records

30.4 Knowing your limits

The Diploma explicitly trains general dentists to practise orthodontics safely and to **collaborate in interdisciplinary care according to treatment needs and context**. Cases that a Diploma holder should be referring rather than treating include: severe skeletal discrepancies and any case where surgery is likely; impacted teeth requiring surgical exposure and complex traction; cleft and craniofacial cases; significant asymmetry; severe hypodontia; the reduced periodontium requiring complex mechanics; suspected primary failure of eruption; and any case where the aims cannot be achieved with the mechanics you are competent to deliver.

EXAM PHRASE TO HAVE READY

"I'd treat this myself; and here is the point at which I would stop and refer." Volunteering the boundary before you are asked is the single most reliable way of demonstrating judgement in an oral examination.

Part 2 — The numbers, classifications and lists

Everything quotable, on twelve pages. If you can reproduce this from memory, the quantitative marks in the written paper are secure and you will never be caught silent in the oral. Read one card a night from four weeks out; read all of them the night before.

Card 1 — Biology, force and anchorage

Tooth movement

Bone: extracellular matrix · mineral · cells	∼30% · ∼60% · ∼10%
Osteocytes as a proportion of bone cells	85–90%
PDL width	0.2 mm (thinnest at the middle third)
Hyalinisation occurs even at	30 g
Initial phase	∼2 days
Lag phase	20–30 days
Historical optimal force (Storey & Smith)	150–200 g
Modelled maximum canine retraction (Ren)	272 cN
Optimal for bodily movement (Theodorou)	50–100 cN
Similar rates across	50–250 cN (0.1–0.47 mm/week)
Normal rate of movement	∼1 mm/month
RANKL peaks at	3 days (baseline from day 6–58)
Force duration threshold	4–8 h/day

Force by movement

Tipping	35–60 g
Bodily movement	50–100 cN (traditionally 100–150 g)
Root uprighting / torque	50–100 g
Rotation	35–60 g
Extrusion	35–60 g
Intrusion	10–20 g per tooth; ∼60 g for four incisors
Space closure (MBT)	∼150 g; 200 g is no better
Molar distalisation	150–250 g

Moment-to-force ratio

Uncontrolled tipping	M/F = 0
Controlled tipping	∼7 : 1
Bodily movement (translation)	∼10 : 1
Root torque	∼12 : 1 and above
Centre of resistance, single-rooted	$\frac{1}{3}$–$\frac{1}{2}$ down the root
Centre of resistance, multirooted	1–2 mm apical to the furcation

Root resorption

Radiographic incidence	up to 73%
Histological incidence	up to 90%
Severe ($\geq \frac{1}{3}$ root length)	1–5% of teeth
Patients losing >5 mm	∼5%
Pause force application	3–6 months
Order of susceptibility	Mx incisors > canines > Md incisors > 6s

Anchorage and headgear

Anchorage support	10–12 h/day · 250–350 g per side
Distalisation / orthopaedic	14 h/day · 450–500 g per side
Facemask	12–14 h/day · ∼400 g per side
Space lost without anchorage support	two-thirds
Mandatory headgear safety devices	2 (self-releasing strap + locking bow)
Reciprocal closure ratio (Proffit)	$\approx 1:1$
With a stabilising lingual/palatal arch	$\approx 2:1$
With headgear + interarch elastics	3 : 1 to 4 : 1

Miniscrews

Overall survival	80–90% (pooled failure 12.3%)
Insertion torque	5–10 Ncm
Median palate failure	1.3%
Paramedian palate	4.8%
Maxillary buccal interradicular	7.5%
Maxillary between 6 and 7	25.9%
Zygomatic buttress	28.1%
Mandibular buccal shelf	65.0%
Higher failure below	8 mm long · 1.2 mm wide
Palatal implant: minimum bone thickness	4 mm; unloaded healing 3 months
Miniplates: cortical depth · loading	2–3 mm · at ∼3 weeks

Card 2 — Appliances, wires and expansion

Archwires

Stainless steel "18–8"	Fe ∼70% · Cr 17–20% · Ni 8–12% · C max 0.08%
Nickel titanium	∼55% Ni, 45% Ti
NiTi modulus relative to steel	∼0.26
Beta titanium stiffness	$\approx \frac{1}{3}$ of stainless steel
Alignment wire diameters	0.012–0.018 in
Levelling (round steel)	0.016–0.020 in
MBT working wire (0.022 slot)	0.019×0.025 in steel

Finishing (0.018 slot / 0.022 slot)	0.017×0.025 / 0.021×0.025
Extra deflection of 0.016×0.022 vs 0.019×0.025	47%
Key identity	strength = stiffness × range
Etch	37% phosphoric acid, 15–30 s
Arch form distribution (Caucasian practice)	tapered 50% · ovoid 42% · square 8%
Arch form change relapse	up to 70%
NiTi expansion of lower intercanine width	up to 5.9 mm

Functional appliances

Twin block bite blocks	∼70° to the occlusal plane
Herbst overjet reduction	6–8 months
Skeletal gain	1–2 mm, transient
Pacha trial failure rate	twin block 37% · Herbst 17%
Overjet reduction achieved (Pacha)	5.8 vs 7.1 mm
Pancherz at 32 years	64% molar · 86% overjet stable
Retention after Herbst	≥2 years with an activator
Best timing	CVM stage CS3 (circumpubertal)
Early overjet reduction, trauma	RR 0.58 — a 42% reduction
Facemask avoiding surgery (Mandall)	60% vs 36%
Facemask crossbite recurrence at 8 y (Proffit)	25–33%
Overcorrect Class III to	+3 to +4 mm overjet

Expansion

Removable plate	0.25 mm × 2 per week
RME activation	1–4 turns/day over 2–4 weeks
RME force	up to 10 kg
Sutural widening	up to 4 mm ≈ 50% of screw expansion
Skeletal relapse at 4–8 months	∼10%
Dental expansion retained	SME 95% · RME 85%
Quadhelix versus plate	∼3 months faster
Cochrane certainty, ages 7–11	High
SARPE indicated beyond	∼5 mm
Expansion yields as usable space	half the expansion

Overbite

Normal overbite	2–4 mm or $\frac{1}{3}$–$\frac{1}{2}$ crown
Ideal overjet	2–4 mm
Intrusion force	∼60 g total · 15 g per tooth
Segmental intrusion achieved	1.5 mm Mx · 1.9 mm Md
Miniscrew intrusion	2–3 mm (0.67–0.95 mm more than segmental)
Proclination effect on overbite	0.1–0.2 mm per degree
Dahl re-establishes contact in	∼6 months
Molar extrusion effect on overlap	1 mm gives 1.5–2.5 mm
Molar intrusion effect on open bite	1 mm gives ∼3 mm

Open bite relapse	∼20%
Interlabial gap suggesting skeletal open bite	>3 mm

Card 3 — Clinical thresholds and epidemiology

Space and extraction

Mild crowding	1–4 mm
Moderate	5–8 mm
Severe	≥9 mm
Proffit's thresholds	<4 mm rarely · 5–9 mm judgement · ≥10 mm almost always
Expansion yields as space	half the expansion
IPR per contact (buccal / anterior)	up to 1 mm / ∼0.5 mm
Space per 1 mm of incisor retraction	2 mm
Space cost of 10° of incisor torque	∼1 mm
Canine distal movement	$<\frac{1}{2}$ unit distalise · $>\frac{1}{2}$ extract · >1 both
Lip response to retraction	0.7 mm per 1 mm
Nasolabial angle change	+1.6° per 1 mm
Lower incisor proclination limit (MBT)	100° to the mandibular plane
Lower incisor advance without lacebacks / with	+1.4 mm / −1.0 mm

Surgery thresholds

Class II: overjet	>10 mm
Class II: pogonion to nasion perpendicular	≥18 mm
Class II: mandibular body length	<70 mm
Class II: anterior face height	>125 mm
Class III: Wits	≤ −11 mm
Class III: ANB	≤ −4°
Class III: reverse overjet	≤ −5 mm
Class III: Holdaway angle	≤ 3.5°
Class III: gonial angle	≤ 120°
Class III: LAFH	≥ 66%
Class III: lower incisor inclination	≤ 80°
Discriminant analysis accuracy	>90%

Impaction

Maxillary canine impacted	∼2–3%
Palatal displacement	up to 85%
Bilateral	17–45%
Sex and ethnicity	3× commoner in females; 5× commoner in Europeans
Canine eruptive path	>20 mm
Palpable by	10–11 years

Conventional radiographic localisation accuracy	39–85%
Interception age	10–11 y (RCS: 10–13 y)
Interception angle	20–30° to midsagittal
Poor prognosis beyond	55°
Ericson & Kurol: overlap $<\frac{1}{2}$ lateral root	91% normalise
Overlap $>\frac{1}{2}$	64% normalise
Autotransplantation success	∼88% (splint ≤3 weeks)
Mandibular second molar impaction	just over 2%
Observation window	6–12 months
First permanent molar impaction	0.75–4.3%; 25× commoner in the maxilla
Transposition	∼0.3%

Development and epidemiology

Incisor liability	7 mm Mx · 5 mm Md
Leeway space per quadrant	1.5 mm Mx · 2.5 mm Md
Increase in intercanine width	∼3 mm Md · ∼4.5 mm Mx
Eruption occurs at	$\frac{2}{3}$–$\frac{3}{4}$ root formation
Flush terminal plane	∼60–75% of children
Tooth agenesis (excluding 8s)	6.4%
Third molar agenesis	up to 30%
Supernumeraries	up to 4% (90% maxillary)
Infra-occluded primary molars	up to 9%
MIH	∼15%
Acceptable natural occlusion	30–40%
Class II (US 12–17 y)	15–22%
Class III (US 12–17 y)	5–6%; worldwide ∼6.8%
Overjet ≥5 mm: children → adults	23% → 13%
RCP = ICP	∼10%
Peak height velocity	girls ∼12 y · boys ∼14 y
CVM	CS1 9.7 · CS2 10.5 · CS3 12.0 · CS4 13.4 y
CLP	1 : 700 · male 2 : 1 · 70% non-syndromic · 80% unilateral
Isolated cleft palate	1 : 2000 · female 4 : 1 · 50% non-syndromic
Alveolar bone graft	8–10 years, $\frac{2}{3}$ root formation
Lip repair · palate repair	3–6 mo · 6–12 mo
Bolton anterior / overall ratio	77.2% / 91.3%
Significant asymmetric jaw development	<0.5%; mandible involved in 85–90%

Card 4 — Risk, evidence and outcome

Risks

White spot lesions	up to half of patients
Appear within	6 months

Commonest sites	6s · upper 2s · lower 3s
Professional fluoride	every 6–8 weeks
Pain peaks at	48 hours, gone by 7 days
Nickel allergy	10–30% F · 1–3% M
Latex allergy	4.3% general · 9.7% healthcare
BDD in orthodontic patients	5–13% (up to 17% orthognathic; 0.7–2.5% general)
Infective endocarditis mortality	~21%
Defer after transplant	≥6 months
Eating disorders, worldwide prevalence	just below 8%

Evidence and outcome

I^2 thresholds	25 low · 50 moderate · 75 high
Power / significance	80–90% / 0.05
PAR: improved / greatly improved	>30% reduction / ≥22 points
Aligners needing no refinement	6%
Aligners converting to fixed	>17%
Invisalign predicted-outcome accuracy	~41%
Acceptable lower incisor relapse (Little)	<3.5 mm
Retainers still worn at 4–5 years	a minority
Vacuum-formed vs Hawley (lower labial segment)	VFR more effective
Fixed vs removable retainer	fixed marginally better for 12 months, then equivalent
Average smile shows	75–100% of the upper incisors
Smile curve parallel to lower lip	~85%
Terminal tooth in the smile is a premolar	~90%

Prevalence of the conditions you will be asked about

TMD: at least one sign	~75%
TMD: at least one symptom	~33%
Paediatric obstructive sleep apnoea	~2%
Dental trauma by age 12	up to 18%
Traumatic loss of a maxillary central	~3% of children

Card 5 — The classifications and lists

Learn these as lists — they are free marks

Classification	Content
Angle	Class I, II div 1, II div 2, III — molar-based
British Standards Institute	Class I, II div 1, II div 2, III — incisor-based
Andrews' six keys	Molar relationship · crown angulation · crown inclination · no rotations · tight contacts · flat curve of Spee

IOTN DHC	Grades 1–5; hierarchical; **MOCDO**
IOTN AC	10 photographs; 1–4 none · 5–7 borderline · 8–10 need
PAR	<30% worse/no different · >30% improved · ≥22 points greatly improved
Ackerman & Proffit	Five characteristics: alignment · profile (appearance) · transverse · anteroposterior · vertical
Björk's structural signs	Seven signs of mandibular rotation
Björk's three rotations	Total (true) · matrix (apparent) · intramatrix. **Total = matrix + intramatrix**
Obwegeser & Makek	Hemimandibular elongation (I) · hemimandibular hyperplasia (II) · hybrid (III)
Veau (clefts)	I soft palate only · II soft and hard palate · III unilateral complete lip and palate · IV bilateral complete
LAHSHAL	The striped-Y style notation recording lip, alveolus, hard palate, soft palate, both sides
Pruzansky	Mandibular ramus grades in hemifacial microsomia
Tanner	Stages of sexual maturation
Demirjian / Nolla	Dental age from tooth formation stages
Baccetti CVM	CS1–CS6
Lekholm & Zarb / Misch	Bone quality — implant crossover
ABO-OGS	Eight finishing criteria
Little's Irregularity Index	Sum of contact point displacements, lower incisors
Bolton	Anterior 77.2% · overall 91.3%

Space analysis — the six requirements

Crowding/spacing · levelling the curve of Spee · arch width change · incisor anteroposterior change · angulation (tip) · inclination (torque).

Space analysis — the seven sources

Tooth reduction or enlargement · extraction · space opening for a prosthesis · molar distal movement · molar mesial movement · differential growth · residue.

The planning sequence — nine steps

Diagnosis · problem list · aims · patient factors · space analysis · anchorage · appliance · phasing · retention.

Removable appliance design — ARAB

Active components · Retention · Anchorage · Baseplate.

The three orders

First = in–out · second = tip (angulation) · third = torque (inclination).

Four causes of an anterior open bite

Habit · skeletal · tongue posture (usually adaptive) · iatrogenic.

Three causes of relapse

Gingival and periodontal fibre recoil (months) · soft tissue pressures (lifelong) · continued growth (lifelong).

Five reasons a canine fails to erupt

Ectopic position · crowding or lack of space · retained or ankylosed primary predecessor · supernumerary or pathology · abnormality of the lateral incisor (the guidance theory) or of the follicle.

Escalating ladder for the impacted lower second molar

Observe · disimpact · remove the third molar · surgically upright · extract the second molar (least successful).

Card 6 — The twelve traps that cost most marks

	The trap	The correction
1	Presenting tooth movement as mechanics	Lead with the PDL — implants and ankylosed teeth do not move
2	Quoting one headgear figure	Force and duration differ by purpose; two safety devices are mandatory
3	Reversing RANKL and OPG	RANKL up on compression; OPG up on tension
4	Reversing the Class II / III extraction pattern	Class II further *back* in the mandible; Class III further *forward*
5	Forgetting the facial aims	Class I canines must not be bought at the cost of the profile
6	Treating overbite without separating growing from non-growing	The same manoeuvre has opposite consequences
7	Calling pre-surgical orthodontics "aligning the arches"	Decompensation deliberately makes the bite *worse*
8	Overclaiming the benefits of treatment	Periodontal, caries, speech and mastication all need qualifiers
9	Saying nickel allergy contraindicates orthodontics	Stainless steel is safe; avoid NiTi
10	Confusing relapse with age-related change	Late lower crowding happens in untreated people too
11	Omitting radiation justification	No threshold dose; justify individually; not for growth prediction
12	Omitting retention from the plan	It is agreed *before* treatment, not mentioned at debond

SIX THINGS TO SAY, AND YOU WILL NOT BE FAR WRONG

1. **Growth status decides everything.** Say whether the patient is growing in your first sentence, and return to it in your last.
2. **Most "skeletal" correction is dentoalveolar** — a few millimetres at best, largely transient.
3. **Anchorage follows Newton's third law.** It cannot be avoided, only planned. Two-thirds of the space is lost without support.
4. **The mandibular arch is the template.** Intercanine width and lower incisor inclination are the least amenable to change.
5. **Prefer evidence to tradition** — and name the study that overturned it.
6. **Retention is agreed before treatment starts**, and late lower incisor crowding is ageing, not relapse.

Part 3 — The diagnostic cases

Two diagnostic exercises are set. You are given records — study casts or a scan, photographs, a panoramic radiograph and usually a lateral cephalogram — and asked to assess, diagnose and plan. The examiners are not testing whether you reach *their* plan. They are testing whether you reach *a defensible* plan, in a reproducible order, and can say what would change it.

The protocol — learn it as a script

SAY THESE HEADINGS OUT LOUD, IN THIS ORDER, EVERY TIME

1. **Who is this?** Age, sex, growth status, and the complaint as I would expect the patient to state it.
2. **Extra-oral** — anteroposterior, vertical, transverse; lips; smile.
3. **Intra-oral** — teeth present, oral health, then each arch, then the occlusion in three planes, then the path of closure.
4. **Radiographic** — what the panoramic shows, then the cephalometric values and what they mean.
5. **Summary in one sentence** — "This is a [age]-year-old [growing/non-growing] patient with a [skeletal class] pattern, [vertical] and [transverse] proportions, and a Class [BSI] incisor relationship on a [Angle] molar relationship, complicated by …"
6. **Ranked problem list.**
7. **Aims** — facial first, then occlusal.
8. **Options** — always at least three, always including no treatment.
9. **The plan**, with the space analysis, the anchorage, the appliance, the phasing and the retention.
10. **What would change my mind**, and **where I would refer**.

The one-sentence summary, worked

"This is a 13-year-old growing girl with a mild skeletal Class II pattern, increased vertical proportions and a symmetrical face, presenting with a Class II division 1 incisor relationship on a half-unit Class II molar relationship, an overjet of 8 mm, an incomplete overbite, 6 mm of upper and 4 mm of lower crowding, and a lip trap."

Everything the examiners need to place the case is in that sentence: age, growth, three planes, incisor class, molar class, the numbers, and the soft tissue driver.

Reading the panoramic — a fixed sweep

Condyles → rami → angles and lower borders → sinuses and nasal floor → dentition right to left, upper then lower → counting teeth, present, missing, unerupted, supernumerary → root form and length → resorption → bone levels → caries and restorations → pathology → third molars: present, position, development. **Say what you looked at and found nothing, not only what you found.**

Reading the cephalogram — six numbers and a sentence

SNA, SNB, ANB (with the Eastman correction if SNA is not 81°), MMPA, upper incisor to maxillary plane, lower incisor to mandibular plane. Then say what they *mean*: which jaw is at fault, what the vertical pattern is, and how compensated the incisors already are. **A heavily**

compensated case has less room for camouflage — that is the sentence that turns numbers into judgement.

The space analysis, spoken

"In the lower arch I have X mm of crowding, plus Y mm to level the curve of Spee, and I intend to hold the incisors where they are — so I need X + Y mm. In the upper arch I have X mm of crowding plus 2 mm of space for every millimetre of overjet reduction, plus about 1 mm for every 10° of torque — so I need Z mm. The upper requirement exceeds the lower, which means I am finishing to a Class II molar relationship, and my extraction pattern must reflect that."

Six diagnostic drills

Work each one against the protocol before reading the model. Time yourself: eight minutes of assessment, four minutes of plan.

Drill 1 — The 12-year-old with an 11 mm overjet

12-year-old boy. Class II division 1. Overjet 11 mm, overbite increased and complete to palatal mucosa. Lips incompetent with a lower lip trap. SNA 82°, SNB 74°, ANB 8°, MMPA 24°. Upper incisors 116°, lower incisors 90°. Mild upper crowding (3 mm), lower arch well aligned. Half-unit Class II molars bilaterally. No previous trauma. Peak height velocity not yet reached (CVM CS2).

MODEL APPROACH

Summary. A 12-year-old growing boy with a moderate skeletal Class II pattern of mandibular retrognathia (SNB 74°), average vertical proportions, a Class II division 1 incisor relationship with an 11 mm overjet and a traumatic overbite, and incompetent lips with a lip trap perpetuating the malocclusion.
Ranked problems. 1. The 11 mm overjet with incompetent lips — a substantial trauma risk (IOTN 5a). 2. The traumatic overbite. 3. The skeletal Class II. 4. Mild upper crowding. 5. Aesthetics and the teasing the family will report if you ask.
Aims. Facially: improve the lip relationship and the profile by advancing the mandible, not by retracting an already-proclined maxillary segment further. Occlusally: reduce the overjet to 2–4 mm, reduce the overbite to one-third coverage with an occlusal stop, Class I incisors and canines, and accept or correct the molar relationship.
Options. (a) No treatment — unacceptable given the trauma risk. (b) Functional appliance now (CS2, close to the window), then a fixed appliance. (c) Fixed appliance alone with headgear and Class II elastics — poorer, because it would camouflage a genuine skeletal problem in a growing patient. (d) Wait and treat surgically — unjustifiable at this ANB.
Plan. Twin block, worn full time, timed to CS3 — but given the overjet is 11 mm and the trauma risk is high, I would not wait long. Expect the overjet to reduce dentoalveolarly, the lateral open bites to appear and be trimmed, and about 6–9 months of active phase. Overcorrect. Transition to a fixed appliance with an upper 2×4 or full appliance; align, level with a flat anterior bite plane effect from the functional phase already achieved; relieve the 3 mm of upper crowding non-extraction if the arch permits. Retain with a vacuum-

formed retainer plus, if the skeletal correction is at risk, night-time functional appliance wear for at least a year.
Consent points. Trauma risk while the overjet persists (RR 0.58 for reduction); the skeletal gain is a few millimetres and largely transient; the possibility of surgery if growth is unfavourable; demineralisation, resorption, relapse.
What would change it. If the boy will not wear the appliance — a fixed functional (Herbst; 17% failure versus 37% for a twin block). If the vertical proportions were increased, I would avoid anything extrusive and change the headgear direction. If he were 16 and non-growing, this becomes camouflage or surgery.

Drill 2 — The unerupted maxillary canine

13-year-old girl. Class I skeletal and incisor relationship. The UR3 is unerupted; the URC is firm and unresorbed. The UL3 is erupted and in alignment. Panoramic shows the UR3 overlapping the distal half of the UR2 root, at approximately 40° to the midsagittal plane, its crown at the level of the middle third of the UR2 root. 2 mm of space is available in the UR3 site. Lower arch well aligned.

MODEL APPROACH

The finding that matters is not the impaction but the risk of **resorption of the UR2 root**. I want a CBCT — justified specifically because the relationship of the canine to the lateral incisor root, and whether that root is resorbing, will change whether I keep the lateral incisor.
Prognosis. Height at the middle third (acceptable), horizontal position overlapping more than half the lateral root (Ericson & Kurol: 64% normalise rather than 91%), angulation 40° (better than 55° but outside the 20–30° interceptive criterion). She is 13 — at the upper end of the RCS interceptive window.
Options. (a) Interceptive extraction of the URC alone — the evidence is weak (Benson 2021: very weak evidence for eruption, none for reducing surgery), and at 13 with only 2 mm of space it is unlikely to succeed on its own. (b) Create space **first** with a fixed appliance, then extract the URC and review for 12 months. (c) Create space, surgically expose and apply traction against a heavy stabilising arch — open or closed exposure; no difference in success (Parkin 2017), open associated with about 2 months less time to alignment (Cassina 2018). (d) Extract the UR3 and either close the space or restore. (e) Autotransplant (∼88% success). (f) Leave and monitor — only if there is no resorption and she accepts the primary canine, with a defined review.
My plan. Space first, then expose and align, with a defined 12-month review and a clear statement to the patient that if the UR2 root is resorbing the decision may change to extracting the lateral incisor and substituting the canine.
Where I would refer. The surgical exposure and the traction mechanics, and any case where the lateral incisor is resorbing.

Drill 3 — The adult Class III

26-year-old woman. Reverse overjet 4 mm. SNA 78°, SNB 83°, ANB −5°, Wits −8 mm, MMPA 32°, LAFH 58%. Upper incisors 118°, lower incisors 82°. Anterior crossbite of all four incisors, no displacement on closing. Class III molars a full unit bilaterally. Mild lower crowding 3 mm; upper arch spaced 2 mm. Concerned about her "bottom teeth

sticking out" and her profile.

MODEL APPROACH

Summary. A non-growing 26-year-old with a moderate–severe skeletal Class III pattern of combined maxillary retrusion and mandibular prognathism, increased vertical proportions, and a Class III incisor relationship with a 4 mm reverse overjet, **already heavily dentally compensated**.

The key judgement. ANB $-5°$, Wits -8 mm, lower incisors already at 82°, reverse overjet 4 mm, LAFH 58%. Against the discriminant thresholds this sits at the surgical end but not unambiguously: Wits is not yet -11 mm, and the reverse overjet is not yet -5 mm. **The decompensation already present is the deciding factor** — there is very little room left to retrocline the lower incisors, and her complaint is about her *profile*, which camouflage cannot change.

Options. (a) No treatment, with an explanation of the natural history. (b) Camouflage: procline the uppers, retrocline the lowers further, possibly with a lower incisor extraction. Would correct the crossbite; would not improve the profile; risks recession on an already-retroclined lower labial segment with a high-angle pattern and probably a thin symphysis — exactly the pattern the evidence associates with recession. (c) **Combined orthodontic and orthognathic treatment** — pre-surgical decompensation (which will make the reverse overjet worse), Le Fort I advancement with or without a mandibular setback, then post-surgical detailing.

My recommendation is (c), with (b) offered honestly and its limitations stated.

Consent for the surgical route. Decompensation makes the bite worse first; 18 months of pre-surgical orthodontics; inferior alveolar nerve injury; relapse (maxillary advancement is stable, setback is problematic); condylar resorption risk in a young woman with a high-angle pattern; the psychological adjustment; and screening for body dysmorphic disorder.

Referral. Joint clinic with a maxillofacial surgeon before any appliance is placed. The camouflage-or-surgery decision is made *now*, not after aligning.

Drill 4 — The unilateral posterior crossbite in a 7-year-old

7-year-old boy. Early mixed dentition. Unilateral right posterior crossbite involving the URD, URE and UR6. On closing there is a clear mandibular displacement to the right of approximately 2 mm; in centric relation the crossbite is bilateral and edge-to-edge. Class I skeletal pattern. Lower incisors mildly crowded (2 mm). No other concerns.

MODEL APPROACH

The diagnosis is not "a unilateral crossbite". It is a bilateral maxillary constriction with a functional lateral displacement — which is why the crossbite appears unilateral in intercuspation and bilateral in centric relation.

Treat on diagnosis. A crossbite with a displacement is the one interceptive intervention with high-certainty Cochrane evidence at ages 7–11. The displacement causes attrition, is associated with asymmetric mandibular growth and with recession on the loaded tooth, and the case will not settle until it is corrected.

Appliance. Quadhelix — it beats an expansion plate on the evidence and is about 3 months faster; it is fixed, so compliance is not a variable in a 7-year-old. Overcorrect. Retain: leave the quadhelix passive in situ for 3–6 months, or retain with a plate at night.

The 2 mm of lower incisor crowding. Transient early crowding of lower incisors is normal in the early mixed dentition. Do nothing about it now; review as the leeway space becomes available.
Review. A defined date, not "keep an eye on it". Re-examine the path of closure in centric relation at each review, and record it.

Drill 5 — Hypodontia of the maxillary lateral incisors

14-year-old girl. Both maxillary lateral incisors are congenitally absent; the ULB and URB have been lost. Class I skeletal pattern, average vertical proportions. Class I molars. Upper arch spaced; lower arch well aligned with no crowding. Overbite average, overjet 3 mm. The canines have erupted adjacent to the central incisors and are of good colour and shape, with a slightly pointed cusp tip. Good oral hygiene, caries-free, motivated.

MODEL APPROACH

The decision is open or close, and **the existing occlusion decides it**, not preference.
Close (**canine substitution**) is favoured when: the case is already Class I or Class II molar, the canine is of good colour and modest prominence, the lower arch needs no space, the smile line is low, and the patient accepts recontouring. It avoids a prosthesis in a young patient and it preserves alveolar bone. It requires: mesial movement of the whole buccal segment, finishing to a **Class II molar relationship**, recontouring and often composite build-up of the canine, gingival recontouring of the canine zenith, and palatal root torque on the canine so that it presents its flatter surface.
Open is favoured when: the occlusion is already Class I with ideal intercuspation and closing would ruin it, the canine is dark, large or very prominent, the smile line is high, or the lower arch also needs space. It commits to a definitive restoration — a resin-retained bridge in the first instance, or an implant **only after growth is complete**, because continued vertical alveolar growth leaves an implant crown in infra-occlusion.
This case. Class I molars with a spaced upper arch and no lower crowding. Closing would require converting sound Class I molars to Class II — achievable, but it must be a deliberate choice. Both are defensible; I would present both, with photographs of each outcome, and involve the restorative colleague *before* starting.
Retention is full-time until the definitive restoration is placed, and a bonded retainer thereafter — a closed space of this kind relapses reliably.
One anomaly is a reason to look for the others: check for other missing teeth, palatally displaced canines, infra-occluded primary molars and peg-shaped teeth.

Drill 6 — The high-angle anterior open bite

15-year-old girl. Anterior open bite of 4 mm from UR3 to UL3. MMPA 38°, LAFH 62%, SNA 80°, SNB 75°, ANB 5°. Interlabial gap 5 mm at rest. Lips incompetent. No history of a digit habit; the tongue is seen between the incisors on swallowing. Class II molars a half unit. Mild crowding in both arches. Concerned that she "cannot bite through a sandwich".

MODEL APPROACH

Aetiology first. This is a **skeletal** open bite: increased MMPA, increased LAFH, backward mandibular rotation, interlabial gap 5 mm. The tongue posture is almost certainly **adaptive** to the open bite rather than causative — an endogenous tongue thrust is rare. Do not blame the tongue and do not promise that muscle exercises will close the bite.
Options. (a) Accept and monitor, with an honest statement about function and stability. (b) Orthodontic camouflage with strict vertical control: high-pull headgear, posterior bite blocks, no extrusive mechanics, no Class II elastics; consider extraction to allow forward mandibular rotation. (c) **Molar intrusion with TADs** — 1 mm of molar intrusion gives about 3 mm of open bite reduction, which is the most efficient lever available. (d) Orthognathic surgery — posterior maxillary impaction with mandibular autorotation, once growth is complete.
Consent. Open bites relapse in about 20% of both surgical and non-surgical corrections. Say the number.
What I would not do. Reach for extrusive mechanics; use Class II elastics to correct the half-unit molar relationship; or promise that a habit appliance will help when there is no habit.
Referral. TAD-supported molar intrusion and any surgical route.

Part 4 — The case presentation

Two treated or almost-completed cases are presented. This is the component in which candidates are most often let down not by their orthodontics but by their *records*. Start collecting properly today, not in October.

The records you must have

Stage	Records
Pre-treatment	Full extra-oral photographs (frontal at rest, frontal smiling, profile, three-quarter); five intra-oral views (frontal in occlusion, both buccal segments, upper and lower occlusal); study casts or scan; panoramic; lateral cephalogram where justified; written problem list, aims, plan and consent
Mid-treatment	Progress photographs at the working-wire stage; a progress radiograph if resorption is a concern or root position is in doubt; a record of any change of plan *and why*
Post-treatment	The full set repeated; a post-treatment panoramic to show root parallelism; a cephalogram if one was taken at the start, for superimposition
Throughout	Contemporaneous clinical notes: wire sizes, elastics, breakages, hygiene, warnings given

THE THREE RECORD FAILURES THAT SINK PRESENTATIONS

1. Photographs taken at different magnifications, with different retraction, in different lighting — so that no honest before-and-after comparison is possible. Standardise now. 2. No pre-treatment cephalogram, so no superimposition and no way of showing what you achieved. 3. No documented *plan*. Without it, every decision you made looks like a reaction rather than a plan, and the examiners will treat it as one.

The presentation structure

1. **The patient and the complaint** — 30 seconds. Age at start, sex, what they asked for.
2. **The diagnosis** — the one-sentence summary from Part 3, then the ranked problem list.
3. **The aims** — facial then occlusal, stated as measurable objectives.
4. **The options considered**, including no treatment, and **why you chose this one**.
5. **The plan as written before you started** — extraction pattern, anchorage, appliance, phasing, retention.
6. **What actually happened** — including what did not go to plan, and what you did about it.
7. **The result** — measured, not asserted: overjet, overbite, molar and canine relationship, centre lines, PAR if you have scored it.
8. **The shortcomings** — name them before the examiners do.
9. **Retention and review** — what was fitted, what the patient was told, what has happened since.
10. **What you would do differently.**

NAMING YOUR OWN SHORTCOMINGS IS THE HIGHEST-SCORING THING YOU CAN DO

Every finished case has them: a residual rotation, a slightly open contact at an extraction site, a centre line 1 mm off, a canine a little extruded, a root not quite parallel. The candidate who says "the UR2 is 1 mm short of full torque expression, and on reflection I should have repositioned the bracket at the working-wire stage rather than adding a bend" is demonstrating exactly the self-audit the examiners are looking for. The candidate who presents a case as flawless invites the examiners to find the flaws, and they will.

Choosing your two cases

- **Choose cases you can defend, not cases that look impressive.** A well-documented, well-finished Class I extraction case with complete records beats a spectacular case with a missing pre-treatment cephalogram.
- **Choose within your competence.** A Diploma candidate presenting a case that should plainly have been referred is inviting the one question they cannot answer.
- **Two contrasting cases** read better than two of the same kind: for example, one growing patient with growth modification and one non-growing extraction case; or one Class II and one Class I with a transverse or vertical problem.
- **Have the numbers.** Start and finish overjet, overbite, molar and canine relationship, treatment duration, number of appointments, number of breakages, and the PAR score if you have it.
- **Rehearse out loud, to time.** Eight minutes of presentation, then questions.

The questions you will be asked about your own case

- Why did you extract those teeth and not others? What did the space analysis say?
- How did you plan the anchorage, and did it hold? How do you know?
- Why that appliance? What would you have used if the patient had refused it?
- How long did it take, and why did it take that long?
- What went wrong, and when did you notice?
- Did anything about the plan change during treatment? Why, and did you re-consent?
- Show me the finish. What is not right?
- What did you tell the patient about retention, and when?
- Has there been any relapse? How would you manage it?
- If this patient walked in tomorrow, would you treat them the same way?

Part 5 — The oral examination

The oral is not a memory test. It is a test of *judgement*, of whether you know your limits, and of whether what you assert is supported. Two hundred practice questions with model answers are in the companion Question Bank; this Part gives you the technique and the four set-pieces to prepare before you walk in.

How the oral is actually scored

What earns marks	What loses them
Structure — answering under headings the examiner can follow	A stream of associations with no shape
Numbers, quoted correctly	Numbers quoted confidently and wrongly. **If unsure, give the range or say you are not certain**
Naming the evidence	Attributing a finding to the wrong author. Cite jointly rather than misattribute
Stating growth status first	Planning as though every patient were the same age
Volunteering the limit of your competence	Being led step by step into a plan you would not actually carry out
Saying "I don't know, but this is how I would find out"	Guessing, and then defending the guess
Balance — the case for and against	One-sided advocacy
Finishing in the clinic — what you would actually *do*	Ending on theory

THE THREE SENTENCES THAT RESCUE ANY ANSWER

"**The first thing I would establish is whether this patient is still growing**, because that changes everything that follows."
"**I would treat this myself up to the point where … , and beyond that I would refer.**"
"**The evidence there is weaker than the confidence with which it is usually stated** — what I'd say to the patient is …"

Set-piece 1 — Present this case

Sixty seconds, unprompted, in a fixed order. Rehearse it until it is automatic, because everything after it is easier if you have taken control of the frame.

"This is a [age]-year-old [growing / non-growing] [boy/girl/man/woman]. Facially, [skeletal class] with [average / increased / reduced] vertical proportions and [a symmetrical face / a chin point deviated to the …]. The lips are [competent / incompetent] with [an interlabial gap of …/ a lip trap]. Intra-orally, [teeth present / absent], oral health is […], with [X] mm of crowding in the upper arch and [Y] mm in the lower. The incisor relationship is Class […] with an overjet of […] mm and an overbite that is […]. Molars are […] bilaterally. [There is / is no] displacement on closing. My working diagnosis is

[…]. What I would want to establish before planning is […], and the decision in this case turns on […]."

Set-piece 2 — Growth modification, camouflage or surgery

You will be asked this, in one form or another, in every oral.

THE STRUCTURE

"There are three routes, and which is available depends first on growth. **If the patient is growing and the discrepancy is skeletal**, growth modification is on the table — but I would be honest that the skeletal gain is a few millimetres at best and largely transient, that it is far more effective in Class II than Class III, and that it depends entirely on compliance and on favourable growth. **If the patient is not growing and the discrepancy is mild to moderate**, camouflage — accepting the skeletal pattern and positioning the teeth to disguise it. The limits are the alveolar envelope, the soft tissue response of about 0.7 mm of lip retraction per millimetre of incisor retraction, and the degree of dental compensation already present. **If the discrepancy is severe, or the facial aims cannot be met by moving teeth**, surgery. For Class II the thresholds I would quote are an overjet over 10 mm, pogonion to nasion perpendicular of 18 mm or more, a mandibular body under 70 mm, or an anterior face height over 125 mm. For Class III, a Wits of -11 mm or worse, an ANB of -4° or worse, a reverse overjet of -5 mm or worse, and a lower incisor inclination of 80° or less. And the decision has to be made *before* treatment starts, because pre-surgical orthodontics decompensates and makes the bite worse."

Set-piece 3 — Consent for fixed appliance treatment

THE STRUCTURE

"Consent is a process rather than a form. I would cover: what I propose and why; the alternatives, **including doing nothing** and what happens if we do nothing; the material risks — **demineralisation**, which affects up to half of fixed appliance patients and can be visible within six months, and which is the reason I will not start until plaque control is adequate; **root resorption**, which happens to some degree in most people and exceeds a third of the root in one to five per cent of teeth; **recession**, particularly where I am moving incisors out of the alveolar envelope; **relapse**, which is near-universal, and the fact that **retention is lifelong if the result is to be held**; **pain**, which peaks at 48 hours and settles by a week; the time commitment and appointment frequency; the cost, including replacement retainers; and the possibility that I will not achieve the aim, and what I would do then. I would give it in writing, allow time, check understanding by asking the patient to tell me back what we agreed, and record what I said. In a child I would obtain consent from a person with parental responsibility, but I would satisfy myself that the child assents, because a plan the child will not wear is not a plan."

Set-piece 4 — The evidence, and where it contradicts tradition

HAVE FIVE READY, WITH THE NAME

"Several things I was taught have not survived the evidence, and I think it matters to say so. **Early Class II treatment** does not improve the final result — O'Brien's Manchester trial in 2009 found no long-term difference, with the late group finishing slightly better; the one exception is incisor trauma, where early overjet reduction gives a relative risk of 0.58. **Self-ligation is not faster** — Scott and Fleming found no difference in rate, treatment time, visits or occlusal outcome. **Third molars do not cause late lower incisor crowding** — it happens in untreated people, and NICE advises against prophylactic removal for that indication. **Orthodontics neither causes nor cures TMD** — the AAO studies of 1992. And **interceptive primary canine extraction** has only very weak evidence for improving eruption and none at all for reducing the need for surgery — the 2021 Cochrane review."

Four worked examiner question flows

These show the *shape* of an oral: a chain of questions that gets harder, ending where your limits are. Full model answers to two hundred questions are in the Question Bank.

Flow A — from a single fact to the whole of biomechanics

Q What force would you use to retract a canine?

"Around 150 grams for space closure — that is the MBT figure and the evidence is that 200 grams is no better. But the more useful answer is that force alone doesn't decide the movement; the moment-to-force ratio does. If I want bodily retraction I need a ratio of about 10 to 1, which is why I do it on a rectangular working wire rather than a round one."

Q Where is the centre of resistance of that canine?

"About a third to halfway down the root. It isn't accessible, which is the fundamental problem — I have to apply the force at the bracket, so a moment is always produced, and the whole of appliance design is about managing that."

Q This patient has lost 4 mm of bone around that canine. What changes?

"The centre of resistance moves apically, so the same force at the same bracket now acts over a greater distance and produces a bigger moment — more tipping for the same force. I would use lighter forces and a higher moment-to-force ratio, and I would want the periodontal disease treated and stable before I placed anything."

Q How much lighter?

"I wouldn't quote a precise figure, because there isn't good evidence for one. The principle is that force should be reduced roughly in proportion to the reduction in root surface area within bone. What I would do practically is start lighter than I would in an intact periodontium, review at four weeks, and check mobility and the response rather than working from a number."

Flow B — from anchorage to knowing your limits

Q You've planned four first premolar extractions with maximum anchorage. How will you get it?

"First by recognising that anchorage is spent before active mechanics begin — so lacebacks during levelling, and I would keep the anchor teeth out of the initial aligning wire, which is Melsen and Verna's point. Then by reinforcement: including the second molars, a transpalatal arch, headgear, or skeletal anchorage. If I genuinely need one hundred per cent of the space for the anterior teeth, only bone will do."

Q Would you place the miniscrew yourself?

"In the maxillary buccal interradicular region with adequate space and good imaging, that is within a general dentist's scope with appropriate training — the pooled failure rate there is about seven and a half per cent. The palate is the most reliable site, at around one per cent. But the mandibular buccal shelf fails in about sixty-five per cent of cases, and that is not somewhere I would be placing screws. If the case depends on a screw in a difficult site, I would refer the whole case rather than the screw."

Q Your screw is mobile at four weeks. What now?

"Remove it. A mobile screw will not become stable and will keep the inflammation going. I would check why — root contact is the commonest cause, so I would look at the radiograph again — then re-site it at least a few millimetres away, or change the anchorage plan. And I would not have designed a case in which the failure of one screw leaves me with no fallback."

Flow C — from an interceptive decision to the evidence

Q A 9-year-old has a unilateral posterior crossbite with a 2 mm displacement. What do you do?

"Treat it on diagnosis. The displacement tells me this is a bilateral constriction the patient is closing out of, and it is one of the few interceptive interventions with high-certainty evidence — the 2021 Cochrane review found quadhelix or an expansion plate beats observation at ages seven to eleven, and the quadhelix is better and about three months faster. I would overcorrect and retain for three to six months."

Q And if there were no displacement?

"Then it is a true unilateral discrepancy, the urgency disappears, and I would treat it when it is convenient to do so within the overall plan — usually as part of comprehensive treatment."

Q The parents have read that expansion will improve their child's breathing. What do you say?

"I would say honestly that rapid maxillary expansion is not recommended as a treatment for nasal breathing or sleep apnoea on the current evidence. If they are worried about the breathing, that is a real concern and I would refer to ENT or paediatrics for proper assessment — including a sleep study if there are symptoms — rather than treating it with an orthodontic appliance."

Flow D — from a complication to governance

Q You debond a 15-year-old and there are white spot lesions on six teeth. What went wrong?

"The honest answer is that I started, or continued, treatment in a mouth where plaque control was not adequate. Demineralisation affects up to half of fixed appliance patients and can be visible within six months, so it is foreseeable, and in elective treatment that makes it my responsibility."

Q What should you have done?

"Not started until hygiene was demonstrably adequate, with a documented programme and review; used glass ionomer for bands and steel ligatures rather than modules; prescribed high-fluoride toothpaste and applied professional fluoride every six to eight weeks; reinforced at every visit and recorded the hygiene score; and debonded early if it was not improving rather than pressing on."

Q What do you do now?

"Tell the patient and the parent plainly what has happened, record it, stop any further active treatment, put intensive prevention in place — fluoride, diet advice, hygiene — and review over several months, because many early lesions arrest and remineralise superficially. I would not reach for restorative treatment or microabrasion early. And I would audit my own case series, because if it has happened once it is probably happening more widely."

Part 6 — The literature

Which book, which paper, what number. Learn the *finding*, not just the name — and cite jointly rather than mis-attribute.

The books, and what each is for

Text	Use it for
Cobourne & DiBiase, *Handbook of Orthodontics*, 3e (2025)	The spine. British practice, the numbers, the evidence appraisal, the "where is the evidence" boxes
Proffit et al., *Contemporary Orthodontics*, 6e	The differential voice. Force numbers, the soft tissue paradigm, the anchorage ratios, the envelope of discrepancy, the hierarchy of stability
Cobourne et al., *Clinical Cases in Orthodontics*	102 worked cases — the best preparation for the diagnostic exercises
Mosby's Orthodontic Review	Breadth and American prevalence data; lasers, CBCT dosimetry, the periodontal interface
Bennett & McLaughlin (1993)	Lacebacks, bendbacks, tiebacks, light continuous force
McLaughlin, Bennett & Trevisi (2001)	The MBT system: bracket selection and positioning, archwire sequence, force levels, arch form
Nanda, *Biomechanics and Esthetic Strategies*	Force system design, vertical control, skeletal anchorage, smile aesthetics

The papers, by theme

Biology and mechanics

Sandstedt; Oppenheim	Independent light-microscopy studies founding the pressure–tension theory
Reitan (1967)	Hyalinisation occurs even at 30 g, and more with tipping than bodily movement
Baumrind (1969)	Bone-bending hypothesis — little evidence forces bend bone meaningfully
Storey & Smith (1952)	Historical optimal force 150–200 g
Quinn & Yoshikawa (1985)	Positive dose–response at low forces to a plateau
Ren et al. (2003, 2004)	No universal ideal force; modelled maximum canine retraction 272 cN
Theodorou et al. (2019)	50–250 cN give similar rates; 50–100 cN optimal for bodily movement
Kanzaki et al. (2004)	RANKL increased in compressed PDL cells; OPG gene transfer reduces movement
Yang et al. (2018)	RANKL deletion in transgenic mice blocks tooth movement
Henneman et al. (2008)	Four-step mechanotransduction model
Danz et al. (2021)	RANKL peaks at 3 days, baseline from day 6–58
Levander et al. (1994)	Pause force for 3–6 months where significant resorption is seen
El-Angbawi et al. (2023)	Photobiomodulation does not shorten overall treatment time
Jepsen (1963)	Root surface areas — the basis of relative anchorage values

Weltman et al. (2010)	Heavy force is the only substantively evidenced cause of resorption

Diagnosis and records

Houston (1983)	"Cephalometrics is definitely not an exact science"
Baumrind & Frantz (1971a)	Every landmark has a non-circular envelope of error in x and y
Baumrind & Frantz (1971b)	Landmark error is cumulative and can create artificial correlation
Ricketts (1957, 1972); Björk (1969)	Growth prediction, since discredited
Baumrind (1984)	Growth prediction no better than chance
Graf et al. (2022)	No superimposition method proved completely accurate
Baumrind et al. (1976); Ghafari et al. (1987)	All superimposition methods carry appreciable error; none is more reliable
Posselt (1952)	Only $\sim$10% have RCP coincident with ICP; 0–2 mm discrepancy is normal

Planning, extraction and anchorage

Kirschen, O'Higgins & Lee (2000)	Royal London Hospital space analysis
Al-Abdallah et al. (2008); Ahmed & Sharma (2023)	Formal space analysis gives a more disciplined approach and better anticipation of anchorage
Proffit et al. (1992)	Cephalometric thresholds for Class II surgery versus camouflage
Konstantonis et al. (2018)	1 mm of incisor retraction gives 0.7 mm of lip retraction, +1.6° nasolabial angle
Ackerman & Proffit (1997); Holdaway (1983)	The soft tissues define the limits of treatment
Rushing (1995); Paquette (1992); Iared (2017)	Minimal soft tissue difference between extraction and non-extraction
Bowman & Johnston (2000)	Lip retraction must be considerable before a profile is judged unattractive
Meyer (2014); Christou (2019); Cheng (2016)	Extractions do not produce larger buccal corridors or a less attractive smile
Larsen (2015); Orabi (2021)	No evidence extraction narrows the airway — "scaremongering and marketing"
O'Higgins & Lee (2000)	Expansion creates space equal to about half the amount of expansion
Haque et al. (2017)	Two-thirds of the space is lost unless anchorage is supported
Diar-Bakirly et al. (2017)	A transpalatal arch provides very little sagittal anchorage
Rebellato et al. (1997)	A lingual arch provides little anchorage and can procline lower incisors
Melsen & Verna (1999)	In maximum anchorage cases, anchor teeth should not be in the initial aligning wire
Samuels (1996)	Two types of headgear injury — catapult and nocturnal disengagement
Giannopoulou et al. (2016)	The maxillary arch is particularly susceptible to anchorage loss

Appliances

Angle (1928)	The edgewise slot; horizontal orientation gives greater control
Begg (1956); Cadman (1975)	Differential force; the three stages of Begg treatment
Kesling (1988, 1991)	Tip-Edge — limited tipping, then full prescription at finishing
Andrews (1972)	Six keys of occlusion, from 120 untreated ideal occlusions
Andrews (1979)	The pre-adjusted prescription built into each bracket
Scott (2008); Fleming (2009, 2010); DiBiase (2011); Pringle (2009)	No difference with self-ligation in rate, time, visits or outcome
Penning et al. (2017)	Customised appliances: comparable outcome, no time saving, more breakages
Kapila & Sachdeva (1989)	No archwire material offers all the ideal properties together
Wang et al. (2018), Cochrane	Coaxial superelastic NiTi better than single strand at 12 weeks
Papageorgiou et al. (2014)	Austenitic-active slightly better than martensitic-stabilised NiTi
Kravitz et al. (2009)	Invisalign achieves the predicted outcome with ∼41% accuracy

Growth modification and timing

Tulloch (1997, 2004); Keeling (1998); Dolce (2007)	Florida and North Carolina trials: no long-term difference
O'Brien et al. (2003)	Early twin block reduces the overjet dentoalveolarly; transient self-esteem gain
O'Brien et al. (2009)	No long-term difference; more attendances; late group finished slightly better
Cobourne (2022); Kallunki (2021)	RR 0.58 for new incisor trauma; headgear marginally more effective
Batista (2018); Macey (2020); Arraj (2019)	Increased overjet is a significant risk factor for incisor trauma
Pancherz (1982)	Herbst overjet reduction in 6–8 months
Pancherz et al. (2014)	At 32 years: 64% molar and 86% overjet correction stable
Pacha et al. (2023, 2024)	Twin block 37% failure vs Herbst 17%; 5.8 vs 7.1 mm overjet reduction
Koretsi et al. (2015)	Functional appliances have a positive effect on the Class II soft tissue profile
Voudouris et al. (2003)	Implanted electrodes show reduced lateral pterygoid activity on protrusion
Santana et al. (2020)	Little difference between maximal and incremental advancement
Mandall et al. (2016)	Facemask: >60% avoided surgery vs 36%; most skeletal effect lost with growth
Baccetti, Franchi & McNamara	Cervical vertebral maturation staging

Specific clinical problems

Houston (1989); Millett et al. (2012)	Overbite reduction relapses unless the interincisal angle is corrected and an occlusal stop established
Eberhart et al. (1990)	0.1–0.2 mm of overbite change per degree of incisal angular change
Ng et al. (2005)	Segmental mechanics achieve ∼1.5 mm Mx and ∼1.9 mm Md intrusion
Dake & Sinclair (1989); Weiland (1996)	Segmental gives greater intrusion; Ricketts utility arch proclines lower incisors
Sosly (2020); Shakti (2022)	Miniscrews give 0.67–0.95 mm more intrusion than segmental mechanics

Burstone (1977); Nanda et al. (1998)	One-couple bypass intrusion arch; the Connecticut intrusion arch
Dahl et al. (1975); Poyser et al. (2005)	Localised supra-occlusion re-establishes contacts over ∼6 months
Järvinen (1978)	Traumatic loss of a maxillary central affects ∼3% of children
Ferguson (1990)	Maxillary canine fails to erupt in ∼3%; up to 85% palatal
Peck et al. (1994)	Bilateral 17–45%; 3× commoner in females; 5× commoner in Europeans
Ericson & Kurol (1986, 1988)	Palpable by 10–11 y; interceptive extraction between 10 and 13 with space
Naoumova et al. (2015, 2018)	Alpha angle and sector position are the most influential predictors
Benson et al. (2021), Cochrane	Very weak evidence that primary canine extraction increases eruption; none that it reduces surgery
Parkin et al. (2017), Cochrane	No difference between open and closed exposure
Cassina et al. (2018)	Open exposure: ∼2 months less to alignment, lower odds of ankylosis
Kokich (2004)	Apically repositioned flap for buccal canines below the mucogingival junction
Patel et al. (2009)	Autotransplantation of canines ∼88% success
Shapira et al. (2011)	Genetic basis for mandibular second molar impaction
Pogrel (1995); Kravitz et al. (2016)	Surgical uprighting with a bracket bonded at surgery
Magnusson & Kjellberg (2009)	Extraction of the second molar gives the least successful outcome
Decker et al. (2008)	PTH1R loss-of-function in primary failure of eruption
Ugolini et al. (2021), Cochrane	Quadhelix or plate beats observation at 7–11 y; quadhelix ∼3 months faster
Liu et al. (2015)	RME achieves up to 4 mm at the suture, ∼50% of screw expansion
Lo Giudice et al. (2020)	∼10% of skeletal change relapses at 4–8 months
Bucci (2016); Rutili (2021, 2022); Srivastava (2020)	SME retains 95% vs RME 85% of initial expansion
Bazargani et al. (2023)	Limited differences between conventional and hybrid RME at 5 years
Wiedel & Bondemark (2015)	Anterior crossbite corrected with a positive overbite is usually stable
Kerr (1992); Stellzig-Eisenhauer (2002); Kochel (2011)	Class III surgery thresholds; discriminant analysis >90% accurate
Obwegeser & Makek (1986)	Three types of condylar hyperplasia asymmetry
Slootweg & Müller (1986)	Four histological subtypes of condylar hyperplasia
Häner et al. (2021)	Serial 3-D facial superimposition — good reproducibility
Sinclair & Little (1983)	Mandibular arch width and length decrease with age in untreated subjects
Behrents (1985); Solow (1980)	Facial growth continues throughout life and is rotational
Southard et al. (1990)	Anterior component of occlusal force — proposed, but the forces are too brief
Thilander (2009); Abdulraheem (2020)	Shorter arch perimeter and increased crowding in untreated adults
Little (1975)	Under 3.5 mm of lower incisor irregularity as acceptable relapse
Rowland et al. (2007)	Vacuum-formed retainers more effective than Hawley in the lower labial segment
Forde (2017, 2018); Martin (2023)	Fixed retainers marginally better for 12 months, then equivalent
Al-Moghrabi (2018); Krämer (2022)	By 4–5 years the majority have stopped wearing retainers

Benefits, risks and appraisal

Macey et al. (2020)	Weak evidence that crowding is a caries risk or that treatment reduces caries
Bollen (2008); Papageorgiou (2018)	Lack of evidence that treatment improves periodontal health
Gorelick (1982); Tufekci (2011); Øgaard (2008)	White spot lesions in up to half of patients, within 6 months
Benson et al. (2019)	Daily high-fluoride toothpaste plus professional foam every 6–8 weeks
Sameshima & Sinclair (2004)	Severe root resorption in 1–5% of treated teeth, greatest anterior maxilla
Monk et al. (2017)	Pain peaks at 48 hours, baseline by 7 days
Joss-Vassalli (2010); Aziz (2011); Tepedino (2018)	Recession relates more to retroclination in high-angle Class III cases
Koretsi et al. (2018)	Mandibular advancement appliances effective in mild-to-moderate OSA
Seehra et al. (2011)	Increased overjet increases susceptibility to teasing and bullying
Kenealy et al. (2007)	Little objective evidence of long-term psychological benefit
Javidi et al. (2017)	Treatment improves the emotional and social domains of OHRQoL
Klaus et al. (2017)	High discontinuation rates, especially with removable appliances
NICE	Antibiotic prophylaxis not routinely provided for dental procedures; no prophylactic third molar removal
Thornhill et al. (2016)	NICE conflicts with AHA and ESC for the highest-risk patients
Chakravarthi et al. (2012)	Nickel allergy 10–30% F, 1–3% M
Wu et al. (2016)	Latex allergy 4.3% general, 9.7% healthcare
Zymperdikas et al. (2020)	Bisphosphonates slow or inhibit tooth movement and delay eruption
Dons et al. (2022)	BDD in 5–13% of orthodontic and up to 17% of orthognathic patients
Egermark-Eriksson (1983); Mohlin (2007); Thilander & Bjerklin (2012)	Weak correlations only between occlusal traits and TMD
Manfredini et al. (2017)	Lack of evidence implicating occlusion in TMD pathophysiology
Brimm v. Malloy (1987)	$850,000 award after TMD following four-premolar extraction treatment
Behrents & White (1992)	AAO studies: orthodontic treatment does not cause TMD
Luecke & Johnstone (1992)	Incisor retraction does not displace the condyles distally
Macfarlane (2009); Luther (2010)	No evidence orthodontics reduces TMD signs and symptoms
Luther (2007)	No evidence an ideal functional occlusion improves stability or prevents TMD
Kandasamy et al. (2022)	Screen, counsel, suspend mechanics, manage conservatively
Richardson (2002)	Third molars are not a primary cause of late lower incisor crowding

Part 7 — Examination technique

The written paper

- Two hours across sixty-five curriculum items. **Roughly a minute per question.** Answer every question — there is no negative marking in the great majority of dental MCQ papers, and an unanswered question is a guaranteed zero.
- **Read the last line of the stem first.** Find the discriminating word: *most likely*, *first*, *single best*, *least appropriate*, *initial*, *definitive*.
- **Form your answer before reading the options.** Then look for it. If it is not there, you have misread the stem — reread it rather than settling for the nearest option.
- **Age and growth status are never decoration.** If the stem gives an age, the answer changes at that age.
- **Absolutes are usually wrong.** "Always", "never", "all" and "only" are rarely the right answer in a subject with this evidence base. "Usually", "may" and "most" more often are.
- **Flag and move on.** Do not spend three minutes on one question. Mark it, answer your best guess now in case you never return, and come back.
- **Change an answer only for a reason you can articulate.** A remembered fact is a reason; a feeling is not.
- In the last five minutes, **check that every question has an answer recorded**, and that your answers are on the right lines of the sheet.

The diagnostic cases and case presentation

- **Use the protocol out loud.** The examiner is following a marking structure; make it easy to give you the marks.
- **Quantify.** "Some crowding" earns nothing; "6 mm of crowding, of which about half will be relieved by levelling" earns marks.
- **Offer three options and say why you rejected two.** A single plan asserted is worth less than a plan chosen.
- **Say what would change your mind**, and **where you would refer**.
- **Name your own case's shortcomings before the examiner does.**

The oral

- **Take one breath before answering.** A structured five-second pause reads as consideration; an immediate stream reads as recitation.
- **Signpost.** "There are three things I'd want to say about that: first …"
- **Finish in the clinic.** End every answer with what you would actually do.
- **If you do not know, say so — then say how you would find out.** That answer scores; a guess defended does not.
- **Do not be led.** If an examiner pushes you towards a plan you would not carry out, say politely where your limit is and why.

- **Recover from a mistake cleanly.** "I said 150 grams for intrusion — that is wrong; intrusion is 10 to 20 grams per tooth, because the force concentrates at the apex." Correcting yourself costs one mark; leaving it costs the examiner's confidence for the rest of the viva.

The countdown to 5 November

Dated, and built around one idea: the written paper is the engine. Revising for it is simultaneously revising for the oral and the diagnostic cases, because all three test the same knowledge. What the paper cannot do for you is produce records from patients, or teach you to speak — so the case requests go out before the revision starts, and the speaking starts in week one.

Before it starts — one session

Job	Detail
The case audit	Pick the three write-up cases and the two presentation cases. For each, list what is missing: pre-treatment records, radiographs, photographs, models or scans, progress notes, debond records, PAR scoring
The requests	Send them the same afternoon — patient recalls, radiograph retrieval, laboratory work, trainer sign-off. A recall is a three-to-four week loop you do not control
The deadline	Ask the Programme Management Office when the three written-up cases are due. The regulations say "at the end of the programme", which is not necessarily 5 November. **If that date is earlier, it outranks this whole table**
The baseline	Sit Question Bank Blocks **H** and **AD** cold — 55 questions on ethics, consent and governance. The likeliest thin spot, the cheapest marks, and a number to measure the next seven weeks against

The seven weeks

The spine is the curriculum, not the book. This volume's thirty sections and the Question Bank's thirty blocks run one-to-one — §1 is Block A, §30 is Block AD — so the weeks advance through them in order, and the Companion column names the units whose main *treatment sits in that week's sections. Note that the Companion has* ***two separate series****: the twenty essay* ***topics*** *of its Part II and the sixteen* ***chapters*** *of its Part III. They are not the same thing — Chapter 1 is Evidence-Based Orthodontics, Topic 1 is the biological basis of tooth movement. Each chapter and each topic appears exactly once below.*

Week	Dates	Here	Read in the Companion	Sit, and alongside
1	14–20 Sep	§1–§5	Chapters 2–6 · Topic 1	Blocks A–E · Viva V1–V20 aloud
2	21–27 Sep	§6–§10	Chapters 1, 7 · Topic 2 · Part V.1	Blocks F–J · V21–V48 · first Part 3 drill
3	28 Sep–4 Oct	§11–§15	Chapters 8, 11 · Topics 3–5, 7	Blocks K–O · V49–V80 · write up case 1
4	5–11 Oct	§16–§20	Chapters 9, 10, 12–14 · Topics 6, 8, 13–15, 20	Blocks P–T · V81–V111 · write up case 2
5	12–18 Oct	§21–§25	Topics 11, 12, 16, 17, 19	Blocks U–Y · V112–V151 · write up case 3

Week	Dates	Here	Read in the Companion	Sit, and alongside
6	19–25 Oct	§26–§30	Chapters 15, 16 · Topics 9, 10, 18 · Part V.2–V.4	Blocks Z–AD · V152–V207
7	26 Oct–1 Nov	Part 2 cards nightly; Part 5 set-pieces until automatic	Only what the mark sheet sends you back to	Re-sit every block under 70% · mock oral · present both cases

Week 5 lists no chapters because §21–§25 are Chapter 13 territory, read in week 4 — that week is topic-led. Part V is new in the revised Companion and has no chapter or topic of its own; it is the only home for §8 and §30.

The last three days

When	What
Mon 2 Nov	One timed full-length sitting — two or three blocks back to back, marked together. This trains stamina, which is the part of a two-hour paper nobody practises
Tue 3 Nov	The Companion's crash notebook: work the schematics, reproducing each from memory before looking. Part 6 literature, skimmed by theme
Wed 4 Nov	Part 2 number cards only. Part 7 technique. Nothing new. Sleep
The morning	Card 6 — the twelve traps and the six things to say. Then stop

WHERE THIS PLAN GOES WRONG

- **Speaking gets crowded out.** Question blocks are measurable and satisfying, so they expand to fill the time. The viva tracker at the back of the Question Bank exists to stop this: three ticks per section, and the first pass does not count unless it was aloud and to the end.
- **The cases slip to October.** They are only eight to twelve hours of writing, but they depend on records you may not have. That is why the requests go out in the first session and the writing sits in weeks 3 to 5, by which point your knowledge is better and the discussions will be sharper.
- **Re-reading replaces testing.** Reading the Companion cover to cover feels like work and is the least efficient thing available to you. Let the mark sheet decide what you re-read.
- **Week 7 gets used for new material.** It is for re-sitting failures and rehearsing out loud. Anything not learned by 25 October is not going to be learned.

THE LAST HOUR

Structure earns breadth. Numbers earn depth. Weighing the evidence earns judgement. Say whether the patient is growing in your first sentence; say where you would refer before you are asked; and end every answer with what you would actually do.

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By section number, unless a Part is named. Bold indicates the main treatment of a topic.

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